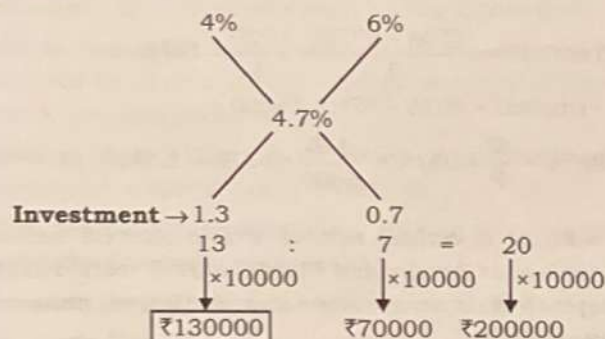


Simple Interest

Based on Alligation

Ex. A total of ₹2,00,000 is divided into two parts for investing in different banks on simple interest. One part yields 4% p.a. while the other part yields 6% p.a. If the total interest at the end of one year is equivalent to 4.7% p.a. on the whole amount, the amount (in ₹) invested at 4% p.a. is:

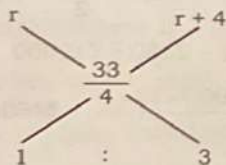
HINTS



Ex. A person invested one-fourth of the sum of ₹25,000 at a certain rate of simple interest and the rest at 4% p.a. higher rate. If the total interest received for 2 years is ₹4,125. What is the rate at which the second sum was invested?

HINTS

$$\text{Overall interest\%} = \frac{4125}{25000 \times 2} \times 100\% = \frac{33}{4}\%$$



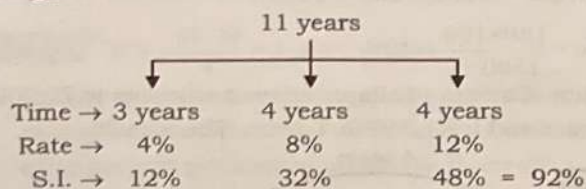
$$\Rightarrow \frac{33}{4} = \frac{r + 3r + 12}{4} \Rightarrow r = 5.25\%$$

∴ Rate at which 2nd sum was invested
= 5.25 + 4 = 9.25%

Concept of total SI

Ex. A certain sum is lent at 4% p.a. for 3 years, 8% p.a. for next 4 years and 12% p.a. beyond 7 years. If for a period of 11 years, the simple interest obtained is ₹27,600, then the sum is (in ₹) :

HINTS



ATQ,

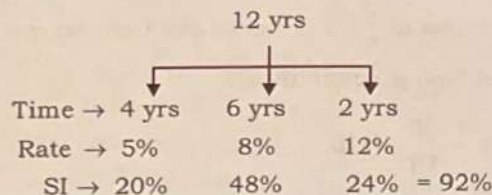
$$92\% \rightarrow ₹27600$$

$$100\% \rightarrow \frac{27600}{92} \times 100 = ₹30000$$

∴ Sum = ₹30,000

Ex. Ramesh borrowed some money at the rate of 5% per annum for the first four years, 8% per annum for the next six years and 12% per annum for the period beyond ten years. If the total interest paid by him at the end of twelve years is ₹9,016, then find the money borrowed by Ramesh.

HINTS



ATQ,

$$\therefore 92\% \rightarrow ₹9016$$

$$\Rightarrow 100\% \rightarrow \frac{9016}{92} \times 100 = ₹9800$$

∴ Money borrowed by Ramesh = ₹9800

Concept of Equal SI

If a sum 'P' is divided in 3 parts i.e., P_1, P_2, P_3 and then each part is lent at respective rate of r_1, r_2, r_3 for t_1, t_2, t_3 years, respectively at simple interest and received SI from all the 3 parts are equal, then:

$$P_1 : P_2 : P_3 = \frac{1}{r_1 t_1} : \frac{1}{r_2 t_2} : \frac{1}{r_3 t_3}$$

Ex. A person invested a total amount of ₹9,000 in three parts at 3%, 4% and 6% per annum on simple interest. At the end of a year, he received equal interest in all three cases. The amount invested at 6% is:

HINTS

Let the sum be invested in the ratio = $P_1 : P_2 : P_3$.

ATQ,

SI on all the 3 amounts invested is equal.

$$P_1 : P_2 : P_3 = \frac{1}{r_1 t_1} : \frac{1}{r_2 t_2} : \frac{1}{r_3 t_3}$$

$$\Rightarrow P_1 : P_2 : P_3 = \frac{1}{3} : \frac{1}{4} : \frac{1}{6} = 4 : 3 : 2$$

$$\therefore \text{Amount invested at 6\%} = 9000 \times \frac{2}{9} = ₹2000$$

Ex. In how much time will ₹5,000 at 4% per annum simple interest produce the same interest as ₹8,000 in $2\frac{1}{2}$ years at 8% per annum simple interest?

HINTS

ATQ,

$$\frac{5000 \times t \times 4}{100} = 8000 \times \frac{5}{2} \times \frac{8}{100}$$

$$\Rightarrow 20t = 20 \times 8$$

$$\Rightarrow t = 8 \text{ years}$$

Concept of Equal Amount

If a sum 'P' is divided in 3 parts i.e., P_1, P_2, P_3 and then each part is lent at r_1, r_2, r_3 rates for t_1, t_2, t_3 years respectively at simple interest and amount received from all the 3 parts are equal, then:

$$P_1 : P_2 : P_3 = \frac{1}{100 + r_1 t_1} : \frac{1}{100 + r_2 t_2} : \frac{1}{100 + r_3 t_3}$$

Ex. If the amount obtained by A by investing ₹9,100 for 3 years at a rate of 10% p.a. on simple interest is equal to the amount obtained by B by investing a certain sum of money for five years at a rate of 8% p.a. on simple interest, then 90% of the sum invested by B (in ₹) is:

HINTS

$$P_1 : P_2 = \frac{1}{100 + r_1 t_1} : \frac{1}{100 + r_2 t_2}$$

$$P_1 : P_2 = \frac{1}{100 + 10 \times 3} : \frac{1}{100 + 8 \times 5}$$

$$\Rightarrow P_1 : P_2 = \frac{1}{130} : \frac{1}{140} \Rightarrow P_1 : P_2 = 14 : 13$$

$$\frac{9100}{P_2} = \frac{14}{13} \Rightarrow P_2 = \frac{9100 \times 13}{14}$$

$$\therefore 90\% \text{ of } P_2 = \frac{9100 \times 13}{14} \times \frac{90}{100} = ₹7605$$

Alternatively

130% A = 140% of B

$$\Rightarrow 13 \times 9100 = 14 \times B \Rightarrow B = 8450$$

$$\therefore \frac{90}{100} \times B = \frac{90}{100} \times 8450 = ₹7605$$

Ex. ₹21750 is invested by a person in the bank account of his two sons, whose ages are 9 years and 13 years, in such a way that they will get equal amount at an age of 21 years at the rate of 4.5% per annum. Find the share of the younger son.

HINTS For younger son:

Time period for investment = 21 - 9 = 12 years

For elder son:

Time period for investment = 21 - 13 = 8 years

Let the sum invested for younger and elder child is 'x' and 'y'.

ATQ,

$$(100 + 12 \times 4.5)\% \text{ of } x = (100 + 8 \times 4.5)\% \text{ of } y$$

$$\Rightarrow 154\% \text{ of } x = 136\% \text{ of } y$$

$$\Rightarrow \frac{x}{y} = \frac{136}{154} = \frac{68}{77}$$

$$\therefore \text{Share of younger child} = 21750 \times \frac{68}{77 + 68} = ₹10200$$

Miscellaneous

Ex. A moneylender claims that he lends money at a simple rate of interest of 8% per annum. But he cleverly includes the interest amount in the principal when he calculates it in every six months. The effective rate of interest becomes:

HINTS $R = 8\% \text{ p.a.}$

Then, Rate = 4% (6 monthly)

$$\text{Effective rate} = 4 + 4 + \frac{4 \times 4}{100} = 8.16\%$$

Ex. A sum of ₹5,000 is divided into two parts such that the simple interest on the first part for $4\frac{1}{5}$ years at $6\frac{2}{3}\%$ p.a. is double the simple interest on the second part for $2\frac{3}{4}$ years at 4% p.a. What is the difference between the two parts?

HINTS Let, two parts be x and y.

ATQ,

$$x \times \frac{20}{3} \times \frac{21}{5} = 2 \times y \times 4 \times \frac{11}{4}$$

$$\Rightarrow 14x = 11y$$

$$\Rightarrow \frac{x}{y} = \frac{11}{14} \rightarrow -3 \text{ unit}$$

$$\begin{array}{l} \times 200 \\ 25 \text{ unit} \rightarrow ₹5000 \\ 3 \text{ unit} \rightarrow ₹600 \\ \times 200 \end{array}$$

$$\therefore \text{Difference between 2 parts} = ₹600$$

Ex. A sum of ₹10,000 is taken as a loan by Rajesh at a rate of 15% p.a. simple interest for 2 years. But Rajesh could not repay it at the agreed time and asked for an extension of two more years. So, the lender included the interest amount for the period as principal for the next two years at the same rate of interest. The total amount paid by Rajesh at the end of 4 years is:

HINTS Amount after 2 years

$$= \frac{10000 \times 15 \times 2}{100} + 10000 = 13000$$

Now amount becomes principal for next 2 years.

Amount after 4 years

$$= 13000 + \frac{13000 \times 15 \times 2}{100}$$

$$= 13000 + 3900 = 16900$$

Alternatively

$$P = 10000$$

$$\text{SI after 2 years} = 15 \times 2 = 30\%$$

Now

$$\text{Amount after 4 years} = 10000 \times \frac{13}{10} \times \frac{13}{10} = 16900$$

Installment

IS If sum is to be deposited in equal installment, then:

$$\text{Equal installment} = \frac{A \times 200}{T[200 + (T-1) \times r]}$$

Where, A = amount, T = Time, r = Rate of interest

Ex. If P = ₹30,000, R = 8% per annum, T = 4 years, then find the value of each installment.

HINTS Amount = $30,000 + 30000 \times \frac{8 \times 4}{100} = ₹39600$

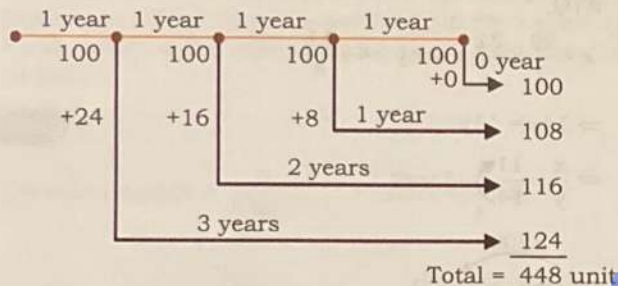
$$\therefore \text{Value of each installment} = \frac{A \times 200}{T[200 + (T-1) \times r]}$$

$$\frac{39600 \times 200}{4[200 + 3 \times 8]} = \frac{39600 \times 200}{4 \times 224} = ₹8839.28 \approx ₹8840$$

Alternatively

$$\text{Due debt} = 30,000 + 30000 \times \frac{8 \times 4}{100} = ₹39600$$

Let value of each installment = 100



$$\therefore 448 \text{ unit} \rightarrow ₹39,600$$

$$100 \text{ unit} \rightarrow \frac{39600}{448} \times 100 = ₹8840$$

$\therefore$ Value of each installment = ₹8840

Ex. What annual installment will discharge a debt of ₹1740 due in 5 years at 8% simple interest?

HINTS Due debt = ₹1740

Let, value of each installment = 100 unit

$$\text{Due debt} = 100 + 108 + 116 + 124 + 132 = 580 \text{ unit}$$

$$\therefore \text{Value of each installment} = \frac{1740}{580} \times 100 = ₹300$$

Ex. What five-monthly payment will discharge a debt of ₹94500 in $1\frac{1}{4}$ years at 12% per annum simple interest?

HINTS $R = \frac{12}{12} \times 5 = 5\%$, $T = \frac{15}{5} = 3$ (5-monthly cycles)

$$\text{Due debt} = ₹94500$$

Let, value of each installment = 100 unit

$$\text{Due debt} = 100 + 105 + 110 = 315 \text{ unit}$$

$$\therefore \text{Value of each installment} = \frac{94500}{315} \times 100 = ₹30000$$

Ex. A man borrowed a sum of ₹7000 from bank at simple interest. After 3 years, he paid ₹3000 to the bank and at the end of 5 years, he paid ₹5450 and clear all his dues. Find the rate of interest.

HINTS Borrowed sum = ₹7000

$$\text{Total amount paid} = 3000 + 5450 = ₹8450$$

$$\text{Interest paid} = 8450 - 7000 = ₹1450$$

$$\text{We know, SI} = \frac{P \times R \times T}{100}$$

Here, he will pay the interest for two principal i.e.

₹7000 for 3 years and ₹4000 for 2 years.

$$\Rightarrow \frac{7000 \times R \times 3}{100} + \frac{4000 \times R \times 2}{100} = 1450$$

$$\Rightarrow 290R = 1450 \Rightarrow R = 5\%$$

Ex. A camera is sold for ₹2500 as cash down payment and ₹2100 after 3 months. If the rate of interest charged is 20% p.a., then find the cash price of the camera.

HINTS Cash down payment = ₹2500

$$\text{Installment paid after 3 months} = ₹2100$$

Rate of interest = 20% p.a.

So, Principal amount for ₹2100:

$$\Rightarrow P \left(100 + 20 \times \frac{3}{12} \right) \% = 2100$$

$$\Rightarrow P \times 105\% = 2100$$

$$\Rightarrow P = \frac{2100}{105} \times 100 = ₹2000$$

$$\therefore \text{Cash price of camera} = 2500 + 2000 = ₹4500$$

Ex. A computer is sold for ₹30,000 cash or ₹18000 cash down payment and 6 monthly installments of ₹2150 each. Find the rate of interest charged under the installment plan.

HINTS Cash price of the computer = ₹30000

$$\text{Cash down payment} = ₹18000$$

$$\text{Payment in 6 installments} = (6 \times 2150) = ₹12900$$

$$\text{Total amount paid} = (18000 + 12900) = ₹30900$$

$$\therefore \text{Interest paid} = (30900 - 30000) = ₹900$$

$$\text{Principal for 1st month} = (30000 - 18000) = ₹12000$$

$$\text{Principal for 2nd month} = (12000 - 2150) = ₹9850$$

$$\text{Principal for 3rd month} = (9850 - 2150) = ₹7700$$

$$\text{Principal for 4th month} = (7700 - 2150) = ₹5550$$

$$\text{Principal for 5th month} = (5550 - 2150) = ₹3400$$

$$\text{Principal for 6th month} = (3400 - 2150) = ₹1250$$

$$\therefore \text{Total Principal for one month} = (12000 + 9850 + 7700 + 5550 + 3400 + 1250) = ₹39750$$

$$\therefore 39750 \times \frac{r}{100} \times \frac{1}{12} = 900$$

$$\Rightarrow r = \frac{900 \times 12 \times 100}{39750} = \frac{1440}{53} = 27.17\%$$

$$\therefore \text{Rate of interest} = 27.17\% \text{ per annum}$$

COMPOUND INTEREST

- **Compound interest** is the interest earned on an amount borrowed or lent, which is calculated on both the initial principal amount and the accumulated interest of previous periods.
- The compound interest for an amount depends on both Principal and interest gained over periods. This is the main difference between compound and simple interest.
- For finding Compound Interest we will use the following formula

$$A = P \left(1 + \frac{r}{100} \right)^t, \text{ C.I.} = A - P$$

$$\text{C.I.} = P \left[\left(1 + \frac{r}{100} \right)^t - 1 \right]$$

Where,

A = Amount

P = Principal

r = Rate

CI = Compound Interest

t = Number of years

☞ If in the question stated that interest is compounded half-yearly, Quarterly, Monthly or n-Monthly then you have to find new rate and time cycle to calculate CI with the help of following Table:

	Rate	Time (n)
Annually	r%	t years
Half-yearly	$\frac{r}{2}\%$	$t \times 2$ cycle
Quarterly	$\frac{r}{4}\%$	$t \times 4$ cycle
Monthly	$\frac{r}{12}\%$	$t \times 12$ cycle

☞ There are various ways to find compound interest

- Using Basic Formula
- Using Tree Method
- Using Ratio Method
- Using Successive Method
- Golden ratio method



It is important to remember Basic Formulae, but we rarely use it to solve questions. Tree Method and Ratio Method are generally used when it is easy to convert the given interest rate into fraction like, 5%, 10%, 15%, 20%, 25%, 30%

or $8\frac{1}{3}\%$, $12\frac{1}{2}\%$, $14\frac{2}{7}\%$, $16\frac{2}{3}\%$

Whereas Successive Method is used when it is difficult to convert Rate% into Fraction like, 3%, 6%, 7%, 9%, 13%, 18% etc.

Table for Successive Percentage

Remember the following table for solving questions quickly

Table 01: For 2 cycles/years

Rate	CI	CI - SI
2%	4.04%	0.04%
3%	6.09%	0.09%
4%	8.16%	0.16%
5%	10.25%	0.25%
6%	12.36%	0.36%
7%	14.49%	0.49%
8%	16.64%	0.64%
9%	18.81%	0.81%
10%	21%	1%
11%	23.21%	1.21%
12%	25.44%	1.44%
15%	32.25%	2.25%
20%	44%	4%
25%	56.25%	6.25%
30%	69%	9%

Table 02: For 3 cycles/years

Rate	CI	CI - SI
2%	6.12%	0.12%
3%	9.27%	0.27%
4%	12.48%	0.48%
5%	15.76%	0.76%
6%	19.1%	1.1%
7%	22.5%	1.5%
8%	25.97%	0.97%
9%	29.5%	2.5%
10%	33.1%	3.1%
11%	36.76%	3.36%
12%	40.49%	4.49%
15%	52.09%	7.09%
20%	72.80%	12.80%
25%	95.31%	20.31%
30%	119.7%	29.7%

How to Calculate Successive Percentage Change

For 2 years

- Successive percentage change = $\left[a + b + \frac{ab}{100} \right] \%$
- If $a = b$ then Successive percentage change = $\left[2a + \frac{a^2}{100} \right] \%$

So, whenever we have to find successive percentage increment of same value –

Step I: Double the value

Step II: Square the value, put decimal after two digits from the right & add it to the result of step I.

Ex. 2% & 2%

Step I: $2 \times 2 = 4\%$

Step II: $2^2 = 4 \rightarrow 0.04\%$

Now, $4 + 0.04 = 4.04\%$

Ex. 5% & 5%

Step I: $2 \times 5 = 10\%$

Step II: $5^2 = 25 \rightarrow 0.25\%$

Now, $10 + 0.25 = 10.25\%$

Ex. 10% & 10%

Step I: $2 \times 10 = 20\%$

Step II: $10^2 = 100 \rightarrow 1.00\%$

Now, $20 + 1.00 = 21\%$

For 3 years

- Successive percentage change = $\left[a + b + c + \frac{(ab + bc + ca)}{100} + \frac{abc}{10000} \right] \%$
- If $a = b = c$ then Successive percentage change = $3a + \frac{3a^2}{100} + \frac{a^3}{10000} = 3a \left| 3a^2 \right| a^3$

Ex. 2%, 2% & 2% = $3 \times 2 \mid 3 \times 2^2 \mid 2^3 = 6.1208\%$

Ex. 3%, 3% & 3% = $3 \times 3 \mid 3 \times 3^2 \mid 3^3 = 9.2727\%$

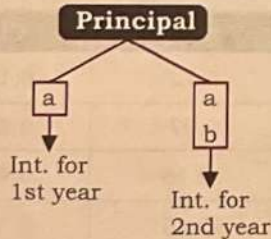
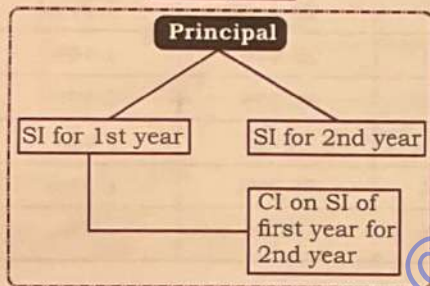
Ex. 5%, 5% & 5% = $3 \times 5 \mid 3 \times 5^2 \mid 5^3 = 15 \mid 75 \mid 125 = 15.7625\%$

Ex. 6%, 6% & 6% = $18 \mid 108 \mid 216 = 19.1016\%$

Ex. 8%, 8% & 8% = $24 \mid 192 \mid 512 = 25.9712\%$

Tree Method Structure for 2 years/Cycles and 3 years/Cycles

For 2 years



1st year interest = a
 2nd year interest = $a + b$
 Diff. between 2 years CI and SI = b

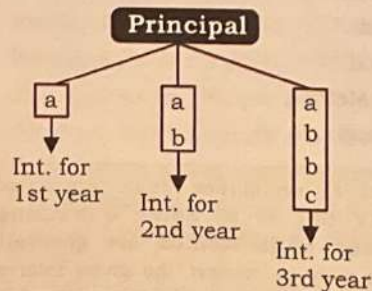
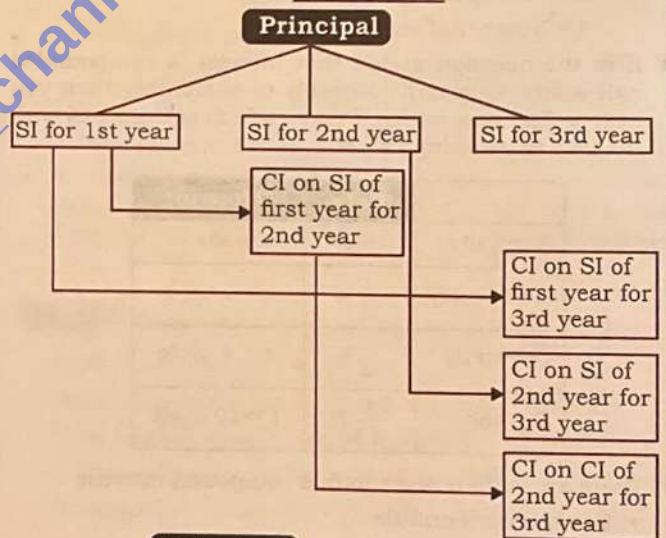
- Keep in mind That simple and compound interest remain the same for the first year.
- Difference of CI and SI for 2 cycle/years can be found

by using direct formula = $P \left(\frac{R}{100} \right)^2$

- Difference of CI and SI for 3 cycle/years can be found

by using direct formula = $P \left(\frac{R}{100} \right)^2 \left(\frac{300 + R}{100} \right)$

For 3 years



1st year interest = a
 2nd year interest = $a + b$
 3rd year interest = $a + 2b + c$
 3 years SI = $3a$
 3 years CI = $3a + 3b + c$
 Diff. between 3 years CI and SI = $3b + c$

For 2 Cycle/Years

Ex. A sum of ₹2500 is invested at 20% per annum compound interest (compounded annually) for 2 years. Find:

- Compound interest
- Amount
- 2nd year CI

HINTS**(Successive Method)**

Equivalent successive % change for 2 cycle of 20% = 44%

Given, Principal (100%) = ₹2500

$$(i) \text{ CI} = 2500 \times \frac{44}{100} = ₹1100$$

$$(ii) \text{ Amount (144\%)} = 2500 \times \frac{144}{100} = ₹3600$$

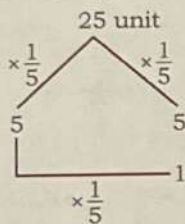
$$(iii) \text{ 2nd year CI} = 20 + 4 = 24\%$$

$$\Rightarrow 2500 \times \frac{24}{100} = ₹600$$

Alternatively**(By Tree Method)**

$$20\% = \frac{1}{5}$$

For 2 cycle, let Principal $(5)^2 = 25$ units



$$2 \text{ year CI} = 5 + 5 + 1 = 11 \text{ units}$$

Here We have, 25 unit = ₹2500

$$(i) \text{ CI(11unit)} = \frac{2500}{25} \times 11 = ₹1100$$

$$(ii) \text{ Amount after 2 year} = P + \text{CI} = 25 + 11 = 36 \text{ unit}$$

$$\Rightarrow \frac{2500}{25} \times 36 = ₹3600$$

$$\text{or } A = P + \text{CI} = 2500 + 1100 = ₹3600$$

$$(iii) \text{ 2nd year CI} = 5 + 1 = 6 \text{ unit}$$

$$\Rightarrow \frac{2500}{25} \times 6 = ₹600$$

Alternatively**(By Ratio Method)**

$$20\% = \frac{1}{5}$$

Principal	:	Amount
5^2	:	6^2
25	:	36
CI		
11		

Given, 25 unit = ₹2500

$$(i) \text{ CI (11 unit)} = ₹1100$$

$$(ii) \text{ Amount (36 unit)} = ₹3600$$

Ex. On what amount, the interest compounded semi-annually at 12% per annum for one year is ₹1545?

HINTS CI = 1545, R = 12% per annum

$$T = 1 \text{ year} = 12 \text{ months}$$

Since interest is compounded annually

$$\therefore T' = \frac{12}{6} = 2 \text{ cycle}$$

$$\& R' = \frac{12}{12} \times 6\% = 6\%$$

(Successive Method)

Equivalent successive percentage for 2 cycle of 6% = 12.36%

$$\text{We have CI (12.36\%)} = 1545$$

$$\therefore \text{Principal (100\%)} = \frac{1545 \times 100}{12.36}$$

$$= ₹12,500$$

Ex. A certain amount of money was lent for a period of 1 year 9 months at a rate of 10% per annum compounded annually. If the compound interest is ₹1460, find the amount of money lent.

HINTS Rate of interest for 1st year = 10%

$$\begin{aligned} \text{Rate of interest for remaining 9 months} &= \frac{10 \times 9}{12} \% \\ &= 7.5\% \end{aligned}$$

Equivalent rate of interest for 1 year 9 months

$$= \left[10 + 7.5 + \frac{10 \times 7.5}{100} \right] \%$$

$$= (17.50 + 0.75)\%$$

$$= 18.25\%$$

$$\therefore \text{Principal} = \frac{1460}{18.25} \times 100 = ₹8000$$

Ex. What is the difference between the compound interest on ₹6,400 for 1 year at 5% per annum C.I. compounded yearly and half-yearly?

HINTS Time = 1 year = 2 cycles of 6 months

$$R = 5\% \text{ yearly}$$

$$R = \frac{5}{2} \% = 2.5\% \text{ (half-yearly)}$$

$$\begin{aligned} \text{Effective rate} &= \left(2.5 + 2.5 + \frac{2.5 \times 2.5}{100} \right) \% \\ &= 5.0625\% \end{aligned}$$

$$\begin{aligned} \text{Difference} &= 0.0625\% \times 6400 \\ &= ₹4 \end{aligned}$$

For 03 Cycle/Years

Ex. A sum of ₹1000 is invested at 20% per annum compound interest (compounded annually) for 3 years. Find:

- Compound interest
- Amount
- 3rd year CI

HINTS (Successive Method)

Equivalent successive% change for 3 cycle of 20% = 72.8%

(i) We have principal (100%) = ₹1000

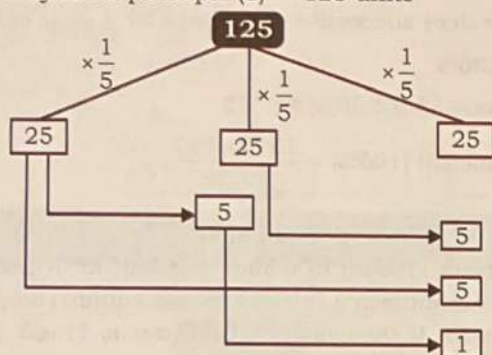
∴ CI (72.8%) = ₹728

(ii) Amount (172.8%) = ₹1728

Alternatively (By Tree Method)

$$R = 20\% = \frac{1}{5}$$

For 3 cycle let principal $(5)^3 = 125$ units



Total CI = $3 \times 25 + 3 \times 5 + 1 = 91$ units

We have 125 unit = ₹1000

(i) CI (91 unit) = $\frac{1000}{125} \times 91 = ₹728$

(ii) Amount = Principal + CI = $1000 + 728 = ₹1728$

(iii) 3rd year CI = $25 + 2 \times 5 + 1 = 36$ unit $\Rightarrow \frac{1000}{125} \times 36 = ₹288$

Alternatively (By Ratio Method)

Principal		Amount
5^3	:	6^3
125	:	216
CI		
	:	91

Principal (125 unit) = ₹1000

(i) CI = $91 \times 8 = ₹728$

(ii) Amount (216 unit) = $216 \times 8 = ₹1728$

Ex. What is the compound interest on ₹25,000 for 3 years at 10% per annum compounded annually?

HINTS Equivalent rate for 3 years at 10% = 33.1%

CI = 33.1% of 25000 = ₹8275

Alternatively

	P	A
1 st year	→ 10	: 11
2 nd year	→ 10	: 11
3 rd year	→ 10	: 11
	1000	: 1331
	×25	331unit ×25
	↓	→ ₹8275
	₹25000	

Golden Ratio Method

For 2 years → 2 : 1

For 3 years → 3 : 3 : 1

For 4 years → 4 : 6 : 4 : 1

For 5 years → 5 : 10 : 10 : 5 : 1

For 6 years → 6 : 15 : 20 : 15 : 6 : 1

Ex. Find the amount of ₹75,000 for one year at the rate of 16% per annum, if the interest is compounded quarterly (rounded off to two decimal places).

HINTS

For 4 Years

		CI
4	$75000 \times \frac{4}{100} = 3000$	12000
6	$3000 \times \frac{4}{100} = 120$	720
4	$120 \times \frac{4}{100} = 4.8$	19.2
1	$4.8 \times \frac{4}{100} = 0.192$	~0.2
		12739.4

Total amount = $75000 + 12739.4 = ₹87739.4$

Ex. A sum of ₹1,00,000 was taken from a bank at the rate of 7% p.a. to be compounded annually for 5 years. Calculate the compound interest. (Closest to a ₹)

HINTS

For 5 Years

		CI
5	$100000 \times \frac{7}{100} = 7000$	35000
10	$7000 \times \frac{7}{100} = 490$	4900
10	$490 \times \frac{7}{100} = 34.3$	343
5	$34.3 \times \frac{7}{100} = 2.401$	12
1	$2.401 \times \frac{7}{100} = 0.168$	~0.1
		40255

Ex. Find the amount (integral value only) if a sum of ₹6,500 is being borrowed at 10% interest per annum for 2 years if interest is compounded half-yearly.

HINTS

For 4 years

		CI
4	$6500 \times \frac{5}{100} = 325$	1300
6	$325 \times \frac{5}{100} = 16.25$	97.5
4	$16.25 \times \frac{5}{100} = 0.8125$	3.25
1		~1400

Amount = $6500 + 1400 = 7900$

Alternatively

P	A
20	: 21
20	: 21
20	: 21
20	: 21
20	: 21
160000	: 194481
160000 unit	= 6500
194481 unit	= 7900

Compound Interest

When Time is given in Fraction

Ex. The compound interest on a sum of ₹20,000 at 15% p.a. for $2\frac{2}{3}$ years, when interest is compounded yearly:

HINTS $P = ₹20000$

$T = 2\frac{2}{3}$ years, $R = 15\%$ per annum

Since interest is compounded annually.

∴ Rate of interest for first 2 complete years = 15%

& Rate of interest for remaining $\frac{2}{3}$ year = $15 \times \frac{2}{3} = 10\%$

(By Ratio Method)

P	:	A
20	:	23
20	:	23
10	:	11
4000		5819
$\times 5$		
20000		
	CI	
	1819	
	9095	



Wherever, we have the different rates of interest for different year it is better to apply ratio method.

Alternatively

(By Successive Method)

Equivalent Successive percentage for 2 cycle of 15% = 32.25%

Again,

equivalent successive percentage of 32.25% & 10%
 $= \left[32.25 + 10 + \frac{32.25 \times 10}{100} \right] \%$

$= (42.25 + 3.225) \% = 45.475\%$

Given,

Principal (100%) = ₹20,000

45.475% = ₹9095

Ex. What is the compound interest (in ₹) on a sum of ₹8192 for $1\frac{1}{2}$ years at 15% per annum, if interest is compounded 5-monthly?

HINTS Rate = $\frac{15}{12} \times 5\% = \frac{25}{4} \% = \frac{1}{16}$

Time = $1\frac{1}{2} = \frac{5}{4} \times 12 = 15$ months = 3 cycle

P	A
1 st cycle → 16	: 17
2 nd cycle → 16	: 17
3 rd cycle → 16	: 17
4096	: 4913
$\times 2$	
8192	
	817 unit $\times 2 \rightarrow ₹1634$

Based on Difference of CI & SI for 2 & 3 Cycle

Ex. The difference between the simple interest and the compound interest on a certain amount at 9% per annum for two years is ₹162. What is the principal?

HINTS Difference of CI and SI for 2 years

$$= P \left(\frac{R}{100} \right)^2 \Rightarrow 162 = P \left(\frac{9}{100} \right)^2 \Rightarrow P = \frac{162 \times 10000}{81} = ₹20000$$

Alternatively

Successive rate for 2 years = $9 + 9 + \frac{9 \times 9}{100} = 18.81\%$

ATQ,

Diff. = $(18.81 - 18) = 0.81\% \rightarrow ₹162$

$100\% \rightarrow 20,000 \Rightarrow \text{Principal} = ₹20,000$

Ex. Find the difference between the compound interest and simple interest when a sum of ₹15,625 is invested for 3 years at 4% per annum.

HINTS Difference between CI & SI for 3 years =

$$P \left(\frac{R}{100} \right)^3 \left(3 + \frac{R}{100} \right) = 15625 \left(\frac{4}{100} \right)^3 \left(3 + \frac{4}{100} \right)$$

$$= 15625 \times \frac{16}{10000} \times \frac{304}{100} = ₹76$$

Ex. What is the difference between the compound interest and simple interest on a sum of ₹12,000 for $1\frac{1}{4}$ years at 12% per annum, when interest is compounded 5-monthly?

HINTS $P = ₹12000$

$T = 1\frac{1}{4}$ year = 15 months, $R = 12\%$ per annum

Since interest compounded 5-monthly, then

$T' = \frac{15}{5} = 3$ cycles & $R' = \frac{12}{12} \times 5 = 5\%$

$$D = P \left(\frac{R}{100} \right)^2 \left(\frac{300 + R}{100} \right)$$

$$= 12000 \times \frac{25}{100 \times 100} \times \frac{305}{100} = ₹91.50$$

Alternatively

(By Successive Method)

Equivalent successive percentage for 3 cycle of 5% = 15.7625%

Diff. = $(15.7625 - 15)\% = 0.7625\%$

Given, 100% = ₹12000

$$\text{Diff. } (0.7625\%) = \frac{12000}{100} \times 0.7625$$

$$= ₹91.5$$

Compound Interest

Combination of both CI & SI

Ex. A certain sum amounts to ₹14,641 in 4 years at 10% p.a. when interest is compounded annually. What will be the amount of the same sum (in ₹) in $3\frac{2}{5}$ years at the same rate at simple interest?

HINTS CI for 4 years at 10% = 46.41%

$$\text{Amount} = 146.41\%$$

$$\Rightarrow 146.41\% = 14641$$

$$1\% = 100$$

$$\text{Principal (100\%)} = ₹10000$$

$$\begin{aligned}\therefore \text{Amount} &= 10000 + 10000 \times \frac{17}{5} \times \frac{10}{100} \\ &= 10000 + 3400 \\ &= ₹13400\end{aligned}$$

Ex. Varun and Madhur invested ₹25,000 each in different schemes. Varun earned simple interest at 11% per annum, whereas Madhur earned compound interest at 10% per annum compounded annually. Who received more interest after 2 years and how much?

HINTS Eff. rate for Varun = 22%

$$\text{Eff. rate for Madhur} = 21\%$$

$$\text{Varun received more interest} = (22 - 21)\% = 1\% \text{ extra}$$

$$\Rightarrow \frac{1}{100} \times 25000 = ₹250$$

To Find Rate

Ex. A sum of ₹6,400 invested on the basis of yearly compounding of interest, grows to ₹7,056 in two years. What is the percentage rate of interest?

HINTS

P	:	A
6400	:	7056
Simplified form		
400	:	441
for 2 years, take square root		
20	:	21

$$\text{Rate of interest} = \frac{1}{20} \times 100\% = 5\%$$

Ex. Hema invested ₹1728 at the rate of compound interest (Compounded annually) for 3 years. She got ₹4,096 after the specified period. Find the rate of interest per annum Compounded annually.

HINTS P = 1728, A = 4096

P	:	A
(1728)	:	(4096)
For 3 years, take cube root		
12	:	16

$$R = \frac{4}{12} \times 100\% = 33\frac{1}{3}\%$$

Ex. The compound interest on a sum of ₹8,000 becomes ₹9,261 in 18 months. Find the rate of interest if interest is compounded half-yearly.

HINTS

P	:	A
8000	:	9261
For 3 cycles, take cube root		
20	:	21

Rate of interest = $\frac{1}{20} \times 100 = 5\%$
Annual rate of interest = 10%

To Find Time

Ex. In how many years will a sum of ₹320 amount to ₹405 if interest is compounded at 12.5% per annum?

HINTS

Principal	:	Amount
320	:	405
Simplified forms		
64	:	81
Which is a perfect square		
8	:	9

Square root

$$\therefore \text{Time} = 2 \text{ years}$$

Ex. The compound interest on ₹4,000 at the rate of 5% p.a. is ₹630.50, then the time period is :

HINTS

Principal	:	Amount
4000	:	4630.5
Simplified form		
8000	:	9261
Which is a perfect cube		
20	:	21

cube root

$$\therefore \text{Time} = 3 \text{ years}$$

Ex. The compound interest on ₹10,000 at 20% per annum is ₹4,641. when interest is compounded half-yearly, then for how many years was the sum invested?

HINTS Rate half yearly = 10%

$$\text{Amount} = 10,000 + 4641 = 14641$$

$$\therefore A = P \left(1 + \frac{r}{100}\right)^n$$

$$\Rightarrow 14,641 = 10000 \left(\frac{11}{10}\right)^n \Rightarrow \frac{14641}{10000} = \left(\frac{11}{10}\right)^n \Rightarrow \left(\frac{11}{10}\right)^4 = \left(\frac{11}{10}\right)^n$$

$$\therefore n = 4 \Rightarrow \text{Required time} = \frac{4}{2} = 2 \text{ years}$$

Alternatively

$$\text{Half year rate} = \frac{20\%}{2} = 10\%$$

P	:	A
10000	:	14641
↓		↓
10 ⁴	:	11 ⁴

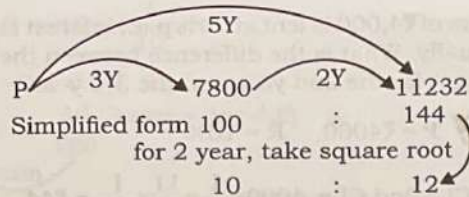
$$\therefore \text{Required years} = \frac{4}{2} = 2 \text{ yrs}$$



If the simplified ratio of principal and amount is perfect square then the time is 2 years (cycle) and if it is perfect cube then the time is 3 years (cycle).

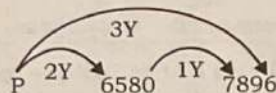
Munni Method

Ex. A sum invested at compound interest amounts to ₹ 7,800 in 3 years and ₹ 11,232 in 5 years. What is the rate percent?

HINTS

$$\text{Rate of interest} = \frac{2}{10} \times 100\% = 20\%$$

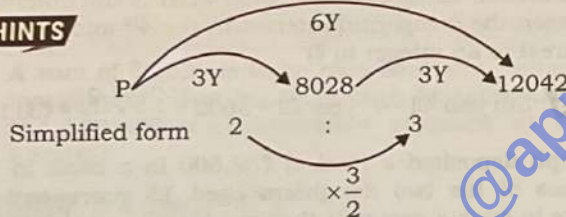
Ex. A sum amounts to ₹ 6580 in 2 years and amounts to ₹ 7896 in 3 years on compound interest (compounding annually) at the same rate of interest. What is the annual rate of interest?

HINTS

$$\text{Interest for 1 year} = (7896 - 6580) = ₹ 1316$$

$$\text{Rate of interest} = \frac{1316}{6580} \times 100\% = 20\%$$

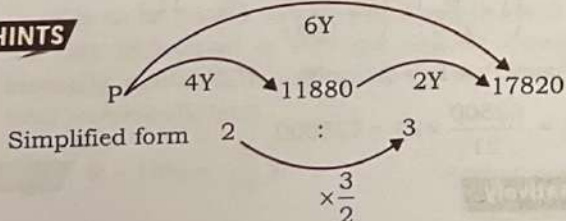
Ex. A sum amounts to ₹ 8,028 in 3 years and to ₹ 12,042 in 6 years at a certain rate percent per annum, when the interest is compounded yearly. The sum is:

HINTS

Since in 3Y amount becomes $\frac{3}{2}$ times of itself.

$$\Rightarrow P \times \frac{3}{2} = 8028 \Rightarrow P = ₹ 5352$$

Ex. A sum of money becomes ₹ 11,880 after 4 years and ₹ 17,820 after 6 years on compound interest, the interest is compounded annually. What is the half of the sum (in ₹)?

HINTS

Since in 2Y amount becomes $\frac{3}{2}$ times of itself.

$$\Rightarrow P \times \frac{3}{2} \times \frac{3}{2} = 11880 \Rightarrow P = ₹ 5280$$

$$\therefore \text{Half of sum} = \frac{5280}{2} = ₹ 2640$$

Principal Becomes 'n' Times after 't' Years

A certain sum becomes x times in n_1 year and y times in n_2 years when rate of interest is compounded, then

$$\frac{1}{x^{n_1}} = \frac{1}{y^{n_2}}$$

Ex. A sum of money becomes 3 times in 10 years at the rate of compound interest (compounded annually). In how many years will it become 243 times?

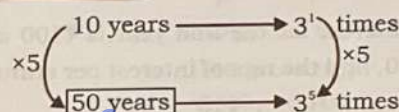
HINTS Using formula,

$$n_1 = 10, x = 3, y = 243, n_2 = ?$$

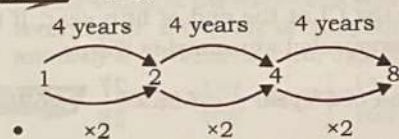
$$\Rightarrow \frac{1}{3^{10}} = \frac{1}{3^{n_2}}$$

$$\Rightarrow \frac{1}{10} = \frac{5}{n_2}$$

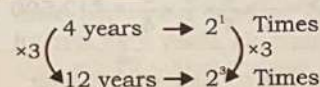
$$\Rightarrow n_2 = 50 \text{ years}$$

Alternatively

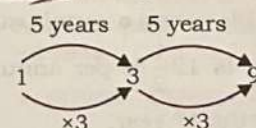
Ex. A sum of money at compound interest doubles itself in 4 years. In how many years does the sum become 8 times of itself at the same rate of interest?

HINTS ATQ,

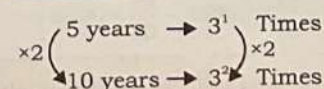
$$\text{Total time} = 4 + 4 + 4 = 12 \text{ years}$$

Alternatively

Ex. A certain amount of money becomes thrice in 5 years at compound interest. How many years it will take to become 9 times?

HINTS ATQ,

$$\text{Total time} = 5 + 5 = 10 \text{ years}$$

Alternatively

Step-1: Convert Times (Multiplication) to Power

Step-2: The multiplication factor in power will be multiplied by the same multiplication factor in time.

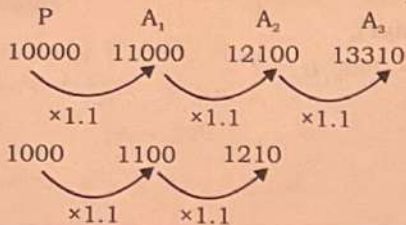
Miscellaneous

Amount & CI_1 , CI_2 , CI_3 and so on will be in Geometric Progression(GP).

Let's Observe the following Example,

Assume, Principal = ₹10000, Rate = 10% and T = 3 Years

$R = 10\% = \frac{10}{100}$, Multiplying Factor = $\frac{11}{10}$ or 1.1



We can find the Rate or Principal by taking the ratio of any two amounts or CI.

Ex. If the compound interest for the 2nd year is ₹100 and for the 3rd year ₹110, find the rate of interest per annum.

HINTS Simplified form $100 : 110$
 $10 : 11$

$$R = \frac{1}{10} \times 100\% = 10\%$$

Ex. Compound interest on a certain sum at the end of fourth year is ₹625. Find the CI at the end of fifth year, If the rate of interest(compounded annually) is 8% pa.

HINTS C.I at the end of 5th year = $625 \times \frac{27}{25} = ₹675$

Ex. A sum of ₹25600 is invested at compound interest at the rate of 25% per annum. Find the compound interest for the 4th year.

HINTS 4th year CI = $25600 \times \frac{1}{4} \times \frac{5}{4} \times \frac{5}{4} \times \frac{5}{4} = ₹12,500$



Multiply by $\frac{1}{4}$ in principal we get first year CI, after multiplying by $\frac{5}{4}$ we get second year CI and again multiplying by $\frac{5}{4}$ and $\frac{5}{4}$ we get third and fourth year CI.

Ex. The compound interest for the 1st year on a certain sum is ₹6400. If the rate of interest is $12\frac{1}{2}\%$ per annum. Find the compound interest for the 4th year.

HINTS 4th year CI = $6400 \times \frac{9}{8} \times \frac{9}{8} \times \frac{9}{8}$
 $= \frac{72900}{8} = ₹9112.5$



Since 6400 is the CI of the first year, hence it multiply by $\frac{9}{8}$ we get second year and again multiply by $\frac{9}{8}$ and $\frac{9}{8}$ we get third and fourth year CI.

Ex. The compound interest for the 3rd year on certain sum is ₹1,25,000. If the rate of interest is 25% per annum. Find the compound interest for the 1st year.

HINTS 1st years CI = $125000 \times \frac{4}{5} \times \frac{4}{5} = ₹80,000$

Ex. A sum of ₹4,000 is lent at 10% p.a., interest compounded annually. What is the difference between the compound interest for the 2nd year and the 3rd year?

HINTS $P = ₹4000$, $R = 10\%$

$$3rd\ CI - 2nd\ CI = 4000 \times \frac{1}{10} \times \frac{11}{10} \times \frac{1}{10} = ₹44$$



Multiplying by $\frac{1}{10}$ in Principal we get first year CI, After multiplying by $\frac{11}{10}$ we get CI of the second year, and after multiplying by $\frac{1}{10}$ we get the difference between the CI of the third year and the second year.

Ex. A sum of ₹18000 is lent at 10% p.a. compound interest, compounded annually. What is the difference between the compound interest for 3rd year and 4th year?

HINTS 4th year CI - 3th year CI

$$= 18000 \times \frac{1}{10} \times \frac{11}{10} \times \frac{11}{10} \times \frac{1}{10} = \frac{198 \times 11}{10} = ₹217.8$$

Ex. If the compound interest in the third year at 8% p.a. on a certain sum is ₹3600, then what is the difference between the compound interest in the 4th and 5th year? (nearest to an integer in ₹)

HINTS 5th year CI - 4th year CI = $3600 \times \frac{27}{25} \times \frac{2}{25} = ₹311.04$

Ex. Joseph deposited a total of ₹52,500 in a bank in the names of his two daughters aged 15 years and 16 years in such a way that they would get equal amounts when they become 18 years old. If the bank gives 10% compound interest compounded annually then what is the amount (in ₹) that Joseph had deposited in the name of his younger daughter?

HINTS $P_1 \left(1 + \frac{R}{100}\right)^3 = P_2 \left(1 + \frac{R}{100}\right)^2$
 $\Rightarrow \frac{P_1}{P_2} = \frac{1}{\left(1 + \frac{R}{100}\right)} = \frac{1}{\left(1 + \frac{10}{100}\right)} \Rightarrow \frac{P_1}{P_2} = \frac{10}{11}$

We have, $P_1 + P_2 = 21$ units $\rightarrow ₹52500$

$$\Rightarrow P_1 = \frac{52500}{21} \times 10 = ₹25000$$

Alternatively

$$R = 10\% = \frac{10}{100}$$

Younger Elder

$$10 \quad 11 \text{ (Power = age diff)}$$

$$(10 + 11 = 21) \text{ unit} = ₹52500$$

$$\text{Younger (10 unit)} = ₹25000$$



Ex. Divide ₹66,300 between A and B in such a way that the amount that A receives after 8 years is equal to the amount that B receives after 10 years; with compound interest being compounded annually at a rate of 10% per annum.

HINTS $R = 10\% = \frac{1}{10}$

A	B
11	10 (Power = age diff)
121	100
sum	

221 unit = ₹66300

A (121 unit) = ₹36300

B (100 unit) = ₹30000

Installment

One of the Regular payments that you make for something until you have paid the full amount.

Concepts related to Installment of Compound Interest Let's understand through a scenario.

Method for two installments:

$$R = 10\% = \frac{1}{10}$$

	P	I
1 st year →	10 _{,11}	11 _{,11}
2 nd year →	100	121
	210	242

32 units = CI

Ex. A sum of ₹4,620 is to be paid back in 2 equal annual installment. How much is each installment (in ₹) if the interest is compounded annually at 10% per annum?

HINTS $R = 10\% = \frac{1}{10}$

P	I
10 _{,11}	11 _{,11}
100	121
210	242
↓ ×22	↓ ×22
4620	2662

Ex. Loan is to be repaid in two equal yearly instalment. If the rate of interest is 10% per annum, compounded annually, and each installment is ₹ 6,897, then find the total interest charged.

HINTS $R = 10\% = \frac{1}{10}$

P	I
10 _{,11}	11 _{,11}
100	121
210	242
CI	
32 units	
↓	
₹1824	

Method for three installment:

$$R = 10\% = \frac{1}{10}$$

	P	I
1 st year →	10 _{,121}	11 _{,121}
2 nd year →	100 _{,11}	121 _{,11}
3 rd year →	1000	1331
	3310	3993

683 units = CI

Ex. A mobile phone is available for ₹79,860 by cash payment or by paying cash of ₹60,000 as down payment and the remaining amount in three equal annual installments. If the shopkeeper charges interest at the rate of 10% per annum compounded annually, then the amount of each installment (in ₹) will be:

HINTS $R = 10\% = \frac{1}{10}$

Due debt = 79860 - 60000 = ₹19860

P	I
10 _{,121}	11 _{,121}
100 _{,11}	121 _{,11}
1000	1331
3310	3993
↓ ×6	↓ ×6
19860	7986

Ex. A man borrowed a certain sum and agrees to repay it by paying ₹4000 at the end of first year and ₹7700 at the end of second year. If the rate of compound interest compounded annually is 10% per annum, then find the sum (in ₹).

HINTS $\left[(\text{Sum}) \times \frac{110}{100} - 4000 \right] \times \frac{110}{100} - 7700 = 0$

$$\Rightarrow \text{Sum} \times \frac{11}{10} - 4000 = 7000 \Rightarrow \text{Sum} \times \frac{11}{10} = 11000$$

$$\Rightarrow \text{Sum} = ₹10000$$

Ex. Amit borrows a sum of ₹8,000 at 10% p.a. compound interest for 4 years. He repays ₹2,800 at the end of the first year and ₹2,600 at the end of the second year. To clear the loan, how much should he pay after 4 years, interest being compounded yearly?

HINTS Given, principal = ₹8000

Amount left after 1st year = 8000 × (1.1) - 2800 = ₹6000

Amount left after 2nd year = 6000 × (1.1) - 2600 = ₹4000

Amount paid at the end of 4th year = 4000 × (1.21) = ₹4840

$$\Rightarrow [37268 - 13268] \times \frac{11}{10} = x$$

Alternatively

Amount left after 2 years

$$= \left(8000 \times \frac{11}{10} - 2800 \right) \times \frac{11}{10} - 2600$$

$$= 6000 \times \frac{11}{10} - 2600 = 4000$$

Amount paid at the end of 4th year

$$= 4000 \times \frac{11}{10} \times \frac{11}{10} = 4840$$

CHAPTER

19

Ratio & Proportion

Ratio

- $\frac{a}{b}$ is called ratio if 'a' and 'b' are of same type.
- $\frac{a}{b}$ is the ratio of 'a' & 'b' and is written as a:b and read as "a is to b", where 'a' is called '**antecedent**' and 'b' is called '**consequent**'.

✎ (i) If $ad = bc$, then $\frac{a}{b} = \frac{c}{d}$.

(ii) If $ad > bc$, then $\frac{a}{b} > \frac{c}{d}$.

(iii) If $ad < bc$, then $\frac{a}{b} < \frac{c}{d}$.

(iv) If $\frac{a}{b} = \frac{c}{d}$, then:

(a) $\frac{b}{a} = \frac{d}{c}$

(b) $\frac{a}{c} = \frac{b}{d}$

(c) $\frac{a+b}{b} = \frac{c+d}{d}$

(d) $\frac{a-b}{b} = \frac{c-d}{d}$

(e) $\frac{a+b}{a-b} = \frac{c+d}{c-d}$

(f) For $\frac{a}{b} = \frac{c}{d} = k$

$a = bk, c = dk$ (k method)

(g) For continued proportion, $\frac{a}{b} = \frac{b}{c} = k$

$b = ck$ and $a = bk$

$a = (ck)k = ck^2$

Mixed Ratio

If $x : y$ and $P : Q$ are two ratios, then $Px : Qy$ is called **mixed ratio or compound ratio**.

Duplicate Ratio

The square of a certain ratio is called its duplicate ratio.

Duplicate ratio of $\frac{a}{b} = \frac{a^2}{b^2}$

Triplicate ratio

The cube of a certain ratio is called its triplicate ratio.

Triplicate ratio of $\frac{a}{b} = \frac{a^3}{b^3}$

Inverse ratio

The reciprocal of a ratio is called its inverse ratio. If there is a ratio 'a : b', then its inverse ratio will be 'b : a'.

Sub-duplicate ratio

The square root of a certain ratio is called its sub-duplicate ratio.

Sub-duplicate ratio of $\frac{a}{b} = \sqrt{\frac{a}{b}}$

Sub-triplicate ratio

The cube root of a certain ratio is called its sub-triplicate ratio.

Sub-triplicate ratio of $\frac{a}{b} = \sqrt[3]{\frac{a}{b}}$

Ex. The compound ratio of 3 : 4 and 6 : 7 is 36 : x. Find 'x'.

HINTS Compound ratio = $\frac{3}{4} \times \frac{6}{7}$

$\Rightarrow \frac{36}{x} = \frac{3 \times 6}{4 \times 7} \Rightarrow x = 56$

Ex. Find the inverse ratio of 5 : 8.

HINTS Inverse ratio = 8 : 5

Ex. Find the duplicate ratio of 14 : 17.

HINTS Duplicate ratio = $\frac{(14)^2}{(17)^2} = \frac{196}{289}$

Ex. Find the sub-duplicate ratio of $\frac{1}{0.225} : \frac{1}{0.256}$.

HINTS $\frac{1}{0.225} : \frac{1}{0.256} = \frac{1000}{225} : \frac{1000}{256} = 256 : 225$

$\therefore$ Sub-duplicate ratio = $\sqrt{256} : \sqrt{225} = 16 : 15$

Ex. Find the triplicate ratio of $\frac{1}{0.49} : \frac{1}{0.343}$.

HINTS $\frac{1}{0.49} : \frac{1}{0.343} = \frac{100}{49} : \frac{1000}{343}$
 $= 7 : 10$

$\therefore$ Triplicate ratio = $(7)^3 : (10)^3$
 $= 343 : 1000$

Ex. Find the sub-triplicate ratio of 512 : 729.

HINTS Sub-triplicate ratio = $\sqrt[3]{512} : \sqrt[3]{729}$
 $= 8 : 9$



(i) If we multiply any ratio with a certain number, then the ratio is not changed.

$\Rightarrow \frac{a}{b} = \frac{ak}{bk}$

(ii) If we divide any ratio with a certain number, then the ratio is not changed $\Rightarrow \frac{a}{b} = \frac{a}{k} \times \frac{k}{b}$

Based on Simple Ratio

Ex. If $a : b = 2 : 3$ and $b : c = 6 : 8$, then find $a : b : c$.

HINTS

$$\begin{array}{ccc}
 a & : & b & : & c \\
 2 & : & 3 & : & \\
 & \swarrow & \searrow & & \\
 & 6 & : & 8 & \\
 \hline
 12 & : & 18 & : & 24 \\
 \hline
 2 & : & 3 & : & 4
 \end{array}$$

Ex. If $a : b = 3 : 4$, $b : c = 2 : 3$ and $c : d = 5 : 6$, then find $a : b : c : d$.

HINTS

$$\begin{array}{cccc}
 a & : & b & : & c & : & d \\
 3 & : & 4 & \rightarrow & 4 & : & 4 \\
 2 \leftarrow & 2 & : & 3 & \rightarrow & 3 \\
 5 & : & 5 & \leftarrow & 5 & : & 6 \\
 \hline
 30 & : & 40 & : & 60 & : & 72 \\
 \hline
 15 & : & 20 & : & 30 & : & 36
 \end{array}$$

Ex. If $P : Q = 3 : 4$ and $Q : R = 4 : 7$, then find the value of $(P + Q) : (Q + R) : (R : P)$.

HINTS $P : Q = 3 : 4$
 $Q : R = 4 : 7$
 $P : Q : R = 3 : 4 : 7$
 $\therefore (P + Q) : (Q + R) : (R : P) = 7 : 11 : 10$

Ex. If $a : (b + c) = 5 : 7$, $b : (c + a) = 7 : 9$, then find $c : (a + b)$.

HINTS $\frac{a}{b+c} = \frac{5_{\times 4}}{7_{\times 4}} \xrightarrow{\text{Total}} 12_{\times 4}$

$$\frac{b}{a+c} = \frac{7_{\times 3}}{9_{\times 3}} \xrightarrow{\text{Total}} 16_{\times 3}$$

$$\therefore a = 20, b = 21 \text{ and } a + c = 27 \Rightarrow c = 7$$

$$\therefore \frac{c}{a+b} = \frac{7}{20+21} = \frac{7}{41}$$

Ex. If $\frac{p}{q} = \frac{r}{s} = \frac{t}{u} = \frac{2}{5}$, then what is the value of

$$(4p + 3r + 7t) : (4q + 3s + 7u)?$$

HINTS $\frac{p}{q} = \frac{r}{s} = \frac{t}{u} = \frac{2}{5}$

$$\therefore \frac{4p+3r+7t}{4q+3s+7u} = \frac{2(4+3+7)}{5(4+3+7)} = \frac{2}{5} \quad \left\{ \begin{array}{l} p=r=t=2 \\ \text{and } q=s=u=5 \end{array} \right\}$$

Ex. Three numbers are in the ratio $\frac{1}{2} : \frac{2}{3} : \frac{3}{4}$. The difference between the greatest and the smallest number is 27. The smallest number is:

HINTS LCM of (2, 3, 4) = 12

$$\begin{array}{ccc}
 \text{I} & : & \text{II} & : & \text{III} \\
 6 & : & 8 & : & 9 \\
 & \swarrow & \searrow & & \\
 & \text{diff.} & & & \\
 & \downarrow \times 9 & 3 \text{ unit} \rightarrow 27 & & \\
 & \boxed{54} & & &
 \end{array}$$

$$\therefore \text{Smallest number} = 54$$

Ex. A, B and C divide an amount of ₹10,500 amongst themselves in the ratio 5 : 7 : 9, respectively. If each one gets ₹500 more, then what will be the ratio of the amounts with A, B and C?

HINTS $A : B : C$
 $5 : 7 : 9 = 21 \text{ unit} \rightarrow ₹10500$
 $1 \text{ unit} \rightarrow ₹500$

$\therefore$ Each gets ₹500 more, so their new ratio will increase by 1 then:

$$\text{Required ratio} = 6 : 8 : 10 = 3 : 4 : 5$$

Ex. By mistake, instead of dividing ₹702 among Ram, Ramesh and Naresh in the ratio $\frac{1}{3} : \frac{1}{4} : \frac{1}{6}$, it was divided in the ratio of 3 : 4 : 6. Who gained the most and by how much?

HINTS

	Ram	: Ramesh	: Naresh
Mistaken $\rightarrow$	3	: 4	: 6
Actual $\rightarrow$	$\frac{1}{3}$	: $\frac{1}{4}$	: $\frac{1}{6}$
	4	: 3	: 2

Clearly, Naresh gained the most.

$$\text{Naresh's actual share} = \frac{2}{9} \times 702 = ₹156$$

$$\text{Naresh's mistaken share} = \frac{6}{13} \times 702 = ₹324$$

$$\therefore \text{Profit} = 324 - 156 = ₹168$$

Based on Coins

Ex. A bag contains 1 rupee, 50 paise and 25 paise coins in the ratio 3 : 4 : 6. If the total amount is ₹143, then the number of 50 paise coins is:

HINTS

Denomination $\rightarrow$	₹1	₹0.50	₹0.25
No. of coins $\rightarrow$	3	: 4	: 6
Amount $\rightarrow$	3	$\frac{2}{\times 22}$	$\frac{1.5}{\times 22}$
		44	143

$$\therefore \text{Number of 50 paise coins} = \frac{44}{0.5} = 88$$

Ex. In a bag, there are coins of ₹5, ₹10 and ₹20 denominations. The total number of coins in the bag is 240. If the number of coins of ₹5, ₹10 and ₹20 denominations are in the ratio of 2 : 3 : 5, then what is the total amount of money in the bag?

HINTS

Denomination $\rightarrow$	₹5	₹10	₹20
No. of coins $\rightarrow$	2	: 3	: 5
Amount $\rightarrow$	10	: 30	: 100

$$\text{When there are 10 coins, then value} = ₹140$$

$$\text{When there are 240 coins, then value} = 14 \times 240 = ₹3,360$$

$$\therefore \text{Total amount of money in the bag} = ₹3,360$$

Alternatively

$$\begin{array}{ccc}
 ₹5 & : & ₹10 & : & ₹20 \\
 2 & : & 3 & : & 5 \\
 \downarrow \times 24 & & \downarrow \times 24 & & \downarrow \times 24 \\
 48 & : & 72 & : & 120
 \end{array}$$

$$\therefore \text{Amount} = (48 \times 5) + (72 \times 10) + (120 \times 20) = 240 + 720 + 2400 = ₹3,360$$

$$\therefore \text{Total amount of money in the bag} = ₹3,360$$

Ex. Aman has a certain number of notes of denomination ₹20 and ₹10 which amount to ₹390. If the number of notes of each kind is interchanged, then the new amount is ₹90 less than before. Find the number of notes of ₹20 denomination.

HINTS Denomination → ₹20 ₹10
No. of notes → x y

ATQ,

$$20x + 10y = 390 \quad \dots\dots (i)$$

$$10x + 20y = 300 \quad \dots\dots (ii)$$

Adding equation (i) and (ii),

$$x + y = 23 \quad \dots\dots (iii)$$

Subtract Eqⁿ (ii) from Eqⁿ (i),

$$x - y = 9 \quad \dots\dots (iv)$$

From eqⁿ (iii) & (iv),

$$x = 16$$

∴ No. of notes of ₹20 denomination = 16

Based on Addition or Subtraction

Ex. Two numbers are in the ratio 3 : 4. On increasing each of them by 30, the ratio becomes 9 : 10. The numbers are:

HINTS $3 : 4$ $\xrightarrow{6\text{unit} \rightarrow 30}$
 $9 : 10$
 $1 \text{ unit} \rightarrow 5$

∴ Required numbers = 15, 20

Ex. The number of students in three classes are in the ratio 3 : 4 : 5. If 20 students are increased in every class, then this ratio becomes 4 : 5 : 6. What was the original number of students in these classes taken together?

HINTS Initial → $3 : 4 : 5$
 $\downarrow +20 \quad \downarrow +20 \quad \downarrow +20$
Final → $4 : 5 : 6$

$1 \text{ unit} \rightarrow 20$
 $12 \text{ unit} \rightarrow 12 \times 20 = 240$
∴ Number of students initially = 240

Ex. The number of students in three sections of Grade 10 in a school are in the proportion 3 : 5 : 8. If 15, 30 and 15 more students are admitted in the three sections, respectively, the new proportion becomes 4 : 7 : 9. The total number of students before the new admissions is:

HINTS $3 : 5 : 8 \rightarrow 16 \times 15 = 240$
 $\downarrow +15 \quad \downarrow +30 \quad \downarrow +15$
 $4 : 7 : 9$

Based on Income & Expenditure

Ex. The ratio of the monthly incomes of A and B is 11 : 13 and the ratio of their expenditures is 9 : 11. If both of them manage to save ₹4000 per month, then find the difference in their income (in ₹).

HINTS

	A	:	B
Income →	11	:	13
Exp. →	9	:	11
Savings →	4000	:	4000

∴ (121 - 117) unit → ₹44,000 - ₹36,000
4 unit → ₹8,000
2 unit → ₹4,000
∴ Required difference = ₹4,000

Ex. The income of A and B are in the ratio 5 : 3. The expenses of A, B and C are in the ratio 8 : 5 : 2. If C spends ₹2000 and B saves ₹700, then how much amount does A save?

HINTS

	A	:	B	:	C
Exp. →	8	:	5	:	2
	$\downarrow \times 1000$		$\downarrow \times 1000$		$\downarrow \times 1000$
	₹8000		₹5000		₹2000

B's income = ₹5,000 + ₹700 = ₹5,700

Income of A = $\frac{5700}{3} \times 5 = ₹9,500$

∴ A's savings = ₹9,500 - ₹8,000 = ₹1,500

Ex. The ratio of last year income of A, B and C is 3 : 4 : 5, while the ratio of their last year income to current year income is 4 : 5, 2 : 3 and 3 : 4 respectively. If their total current year income is ₹98,500, then find out the present income of (B + C).

HINTS Ratio of last year income of A, B and C = 3 : 4 : 5
Ratio of present year income of A, B and C
 $= \frac{3 \times 5}{4} : \frac{3 \times 4}{2} : \frac{5 \times 4}{3} = \frac{15}{4} : 6 : \frac{20}{3}$
 $\left(\frac{15}{4} + 6 + \frac{20}{3} \right) \text{ unit} \rightarrow ₹98,500$
 $\frac{197}{12} \text{ unit} \rightarrow ₹98,500$
 $1 \text{ unit} \rightarrow ₹6,000$

$\left(6 + \frac{20}{3} \right) \text{ unit} = \frac{38}{3} \text{ unit} \rightarrow \frac{38}{3} \times 6000 = ₹76,000$

∴ Present income of (B + C) = ₹76,000

Ex. The expenditure and savings of a person are in the ratio of 4 : 3. His savings increase by $\frac{1}{5}$ of his income and income remains the same. What is the new ratio of his expenditure and savings?

HINTS

	Exp.	Savings	Income
Initial →	4	3	7
	$\downarrow$	$\downarrow$	$\downarrow$
Final →	2.6	$3 + \frac{7}{5} = 4.4$	7

∴ New ratio of expenditure and savings = 2.6 : 4.4 = 13 : 22

Ex. The salaries of P and Q together amount to ₹9,60,000. P and Q saves 64% and 44% respectively of their individual salaries. If P's expenditure is 1.5 times Q's expenditure, then what is the ratio of P's salary to Q's salary?

	Income	:	Expenditure	:	Savings
P →	100x	:	36x	:	64x
Q →	100y	:	56y	:	44y

$$\text{ATQ, } \frac{36x}{56y} = \frac{3}{2} \Rightarrow \frac{x}{y} = \frac{7}{3}$$

$$\therefore x : y = 7 : 3$$

Based on Age

Ex. The ratio of the ages of two friends is 4 : 3 which will become 6 : 5 after 4 years. What will be the sum of their age (in years) after 22 years from now?

HINTS	Present age → 4 : 3
	diff. 2 unit → 4 years
	After 4 years → 6 : 5
	1 unit → 2 years

Present age of first friend = $4 \times 2 = 8$ years

Present age of second friend = $3 \times 2 = 6$ years

Sum of their age after 22 years = $8 + 22 + 6 + 22 = 58$ years

Ex. Ratio of the present age of a mother to that of the daughter is 7 : 1. After five years the ratio will become 4 : 1. What is the difference (in years) in their present age?

HINTS	M	D	M	D
	At present → 7	: 1	7	: 1
	After 5 years → 4×2	: 1×2	8	: 2

$\therefore 1 \text{ unit} \rightarrow 5 \text{ years}$

6 unit → 30 years

$\therefore$ Difference in their present age = 30 years

"Difference between the age of two persons will always be same."

Ex. Eight years ago, the ratio of ages of A and B was 5 : 4. The ratio of their present ages is 6 : 5. What will be the sum (in years) of the age of A and B after 7 years from now?

HINTS	A	B
	8 years before → 5	: 4
		diff. 1 unit → 8 years
	At present → 6	: 5

A's present age = $6 \times 8 = 48$ years

B's present age = $5 \times 8 = 40$ years

$\therefore$ Sum of the age of A and B after 7 years from now = $(40 + 48) + 2 \times 7 = 102$ years

Ex. 5 years ago, the ratio of the age of A to that of B was 4 : 5. Five years hence, the ratio of the age of A to that of B will be 6 : 7. If, at present, C is 10 years younger than B, then what will be the ratio of the present age of A to that of C?

HINTS	A	B
	5 years ago → 4	: 5
		diff. 2 unit → 10 years
	5 years hence → 6	: 7
		1 unit → 5 years

Present age of A = $6 \times 5 - 5 = 25$ years

Present age of B = $7 \times 5 - 5 = 30$ years

Present age of C = Present age of B - 10 years
= $30 - 10 = 20$ years

$\therefore$ Ratio of present age of A and C = $25 : 20 = 5 : 4$

Ex. The ratio of the ages of A and B, 8 years ago, was 5 : 7. The ratio of their ages, 8 years from now, will be 9 : 11. If the present age of C is 13 years less than that of B and the present age of D is 8 years less than that of A, then the sum of the present age of C and D, in years, is:

HINTS	A	B
	8 years ago → 5	: 7
		diff. 4 unit → 16 years
	8 years from now → 9	: 11
		1 unit → 4 years

Present age of A = $5 \times 4 + 8 = 28$ years

Present age of B = $7 \times 4 + 8 = 36$ years

Present age of C = B's present age - 13 = 23 years

Present age of D = A's present age - 8 = 20 years

$\therefore$ Sum of the present age of C and D = $23 + 20 = 43$ years

Ex. In a family, the combined age of husband, wife and son is 90 years. The husband is the oldest. The difference in the ages of the husband and wife is $\frac{1}{3}$ rd of the age of

their son. After 10 years, the age of the son would be half the age of his father. What is the present age of his mother (in years)?

HINTS Given,

$$H + W + S = 90 \quad \dots (i)$$

$$H - W = \frac{1}{3} S \quad \dots (ii)$$

$$\text{Also, } S + 10 = \frac{1}{2} (H + 10)$$

$$\Rightarrow H = 2S + 10 \quad \dots (iii)$$

From equation (ii) & (iii),

$$6S + 30 - 3W = S$$

$$\Rightarrow 3W = 5S + 30 \Rightarrow W = \frac{5S}{3} + 10 \quad \dots (iv)$$

Put eqⁿ (iii) and eqⁿ (iv) in eqⁿ (i),

$$\Rightarrow (2S + 10) + \left(\frac{5S}{3} + 10 \right) + S = 90 \Rightarrow 14S = 210 \Rightarrow S = 15$$

$$\therefore W = 5 \times \frac{15}{3} + 10 = 35 \text{ years}$$

Ex. Anil and Beena are friends and the difference between their ages is 5 years. Anil's father Dinesh is three times as old as Anil and Beena is twice as old as her sister Charu. The ages of Dinesh and Charu differ by 45 years. If Beena is older than Anil, then find the sum of the present age of both Dinesh and Charu.

HINTS Let, present age of Anil = x

Dinesh	Anil	Beena	Charu
↓	↓	↓	↓
Age → 3x	x	x+5	$\frac{x+5}{2}$

ATQ,

$$3x - \frac{x+5}{2} = 45$$

$$\Rightarrow 6x - x - 5 = 90 \Rightarrow x = 19$$

$\therefore$ Sum of the present age of both Dinesh and Charu

$$= 3x + \frac{x+5}{2} = \frac{7x+5}{2} = \frac{7 \times 19 + 5}{2} = \frac{138}{2} = 69 \text{ years}$$

Proportion

When two ratio are equal, i.e., $\frac{a}{b} = \frac{c}{d}$, then a, b, c, d are said to be in proportion & a, b, c & d are said to be 1st, 2nd, 3rd and 4th proportion respectively.

When 3 numbers are given

$$x : a :: b : c$$

First proportion

$$x : a :: b : c \Rightarrow x = \frac{ab}{c}$$

Second proportion

$$a : x :: b : c \Rightarrow x = \frac{ac}{b}$$

Third proportion

$$a : b :: x : c \Rightarrow x = \frac{ac}{b}$$

Fourth proportion

$$a : b :: c : x \Rightarrow x = \frac{bc}{a}$$

Ex. The first proportional of three numbers 6, 12 and 24 is:

HINTS First proportion = $\frac{6 \times 12}{24} = 3$

Ex. The second proportional of three numbers 28, 4 and 5 is :

HINTS Second proportion = $\frac{28 \times 5}{4} = 35$

Ex. The third proportional of three numbers 2, 3 and 15 is :

HINTS Third proportion = $\frac{2 \times 15}{3} = 10$

Ex. The fourth proportion of 12, 14 and 24 is :

HINTS Fourth proportion = $\frac{14 \times 24}{12} = 28$

When 2 numbers are given

$$x : a :: a : b$$

First proportion

$$x : a :: a : b \Rightarrow x = \frac{a^2}{b}$$

Mean proportion

$$a : x :: x : b \Rightarrow x = \sqrt{ab}$$

Third proportion

$$a : b :: b : x \Rightarrow x = \frac{b^2}{a}$$

Ex. The first proportional of two numbers 3 and 5 is :

HINTS First proportion = $\frac{(3)^2}{5} = \frac{9}{5}$

Ex. Find the mean proportion of 3 and 27.

HINTS Mean proportion = $\sqrt{3 \times 27} = 9$

Ex. The third proportional to 81 and 27 is:

HINTS Third proportion = $\frac{27 \times 27}{81} = 9$

Proportion After Addition

Let, 'x' be a number which is added to a, b, c and d to make them proportional, then:

$$x = \frac{bc - ad}{(a + d) - (b + c)}$$

Ex. When 'x' is added to each of 2, 3, 30 and 35, then the numbers obtained in this order are in proportion. What is the mean proportional between $(x + 7)$ & $(x - 2)$?

HINTS
$$x = \frac{bc - ad}{(a + d) - (b + c)} = \frac{3 \times 30 - 2 \times 35}{(2 + 35) - (3 + 30)}$$
$$= \frac{90 - 70}{37 - 33} = \frac{20}{4} = 5$$

$$\therefore \text{Mean proportional of } (x + 7) \text{ and } (x - 2) = \sqrt{12 \times 3}$$
$$= \sqrt{36} = 6$$

Alternatively

$$\frac{2+x}{3+x} = \frac{30+x}{35+x}$$

By assuming, $x = 5$

$$\frac{2+5}{3+5} = \frac{30+5}{35+5} \Rightarrow \frac{7}{8} = \frac{35}{40} \Rightarrow \frac{7}{8} = \frac{7}{8} \text{ (satisfy)}$$

$$\therefore \text{Mean proportional of } (x + 7) \text{ and } (x - 2) = \sqrt{12 \times 3}$$
$$= \sqrt{36} = 6$$

Alternatively

$$\text{Diff.} = 1 \left(\frac{2+x}{3+x} = \frac{30+x}{35+x} \right) \text{Diff.} = 5$$

$$\Rightarrow 5 \times (2 + x) = 1 \times (30 + x)$$

$$\Rightarrow 10 + 5x = 30 + x \Rightarrow 4x = 20 \Rightarrow x = 5$$

$$\therefore \text{Mean proportional of } (x + 7) \text{ and } (x - 2) = \sqrt{12 \times 3}$$
$$= \sqrt{36} = 6$$

Ex. When 'x' is added to each of the numbers 7, 11, 18 and 23 then the number so obtained are in proportion. What is the mean proportional between $(x - 1)$ and $(2x - 10)$?

HINTS
$$x = \frac{11 \times 18 - 7 \times 23}{(7 + 23) - (11 + 18)} = \frac{11 \times 18 - 7 \times 23}{(7 + 23) - (11 + 18)} = \frac{198 - 161}{30 - 29} = 37$$
$$\therefore \text{Mean proportional between } (x - 1) \text{ and } (2x - 10)$$
$$= \sqrt{(37 - 1) \times (2 \times 37 - 10)} = \sqrt{36 \times 64} = 6 \times 8 = 48$$

Ratio & Proportion

Alternatively

$$\text{Diff.} = 4 \left(\frac{7+x}{11+x} = \frac{18+x}{23+x} \right) \text{Diff.} = 5$$

$$\Rightarrow 5(7+x) = 4(18+x) \Rightarrow 35 + 5x = 72 + 4x \Rightarrow x = 37$$

$$\therefore \text{Mean proportional between } (x-1) \text{ and } (2x-10)$$

$$= \sqrt{(37-1) \times (2 \times 37 - 10)} = \sqrt{36 \times 64} = 6 \times 8 = 48$$

Alternatively

$$\text{Diff.} = 4 \left(\frac{7}{11} = \frac{18}{23} \right) \text{Diff.} = 5 \left(\frac{11}{12} = \frac{44}{48} \right) \text{Diff.} = 7 \left(\frac{44}{48} = \frac{11}{12} \right)$$

$$\therefore \text{Mean proportional between } (x-1) \text{ and } (2x-10)$$

$$= \sqrt{(37-1) \times (2 \times 37 - 10)} = \sqrt{36 \times 64} = 6 \times 8 = 48$$

Proportion After Subtraction

Let, 'x' be a number which is subtracted from a, b, c and d to make them proportional, then:

$$x = \frac{ad - bc}{(a+d) - (b+c)}$$

Ex. If 'x' is subtracted from each of 24, 40, 33 and 57, the numbers, so obtained are in proportion. The ratio of $(5x+12)$ to $(4x+15)$ is:

HINTS

$$x = \frac{ad - bc}{(a+d) - (b+c)} = \frac{24 \times 57 - 40 \times 33}{(24+57) - (40+33)}$$

$$= \frac{1368 - 1320}{81 - 73} = \frac{48}{8} = 6$$

$$\therefore \frac{5x+12}{4x+15} = \frac{5 \times 6 + 12}{4 \times 6 + 15} = \frac{42}{39} = \frac{14}{13}$$

Alternatively

$$\text{Diff.} = 16 \left(\frac{24-x}{40-x} = \frac{33-x}{57-x} \right) \text{Diff.} = 24$$

$$\Rightarrow 3(24-x) = 2(33-x) \Rightarrow 72 - 3x = 66 - 2x \Rightarrow x = 6$$

$$\therefore \frac{5x+12}{4x+15} = \frac{5 \times 6 + 12}{4 \times 6 + 15} = \frac{42}{39} = \frac{14}{13}$$

Alternatively

$$\frac{24-x}{40-x} = \frac{33-x}{57-x}$$

By assuming, $x = 6$

$$\frac{24-6}{40-6} = \frac{33-6}{57-6} \Rightarrow \frac{18}{34} = \frac{27}{51} \Rightarrow \frac{9}{17} = \frac{9}{17} \text{ (satisfy)}$$

$$\therefore \frac{5x+12}{4x+15} = \frac{5 \times 6 + 12}{4 \times 6 + 15} = \frac{42}{39} = \frac{14}{13}$$

Alternatively

$$\text{Diff.} = 9 \left(\frac{24}{40} = \frac{33}{57} \right) \text{Diff.} = 17 \left(\frac{9}{17} = \frac{18}{34} \right) \text{Diff.} = 24 \left(\frac{18}{34} = \frac{9}{17} \right)$$

$$\therefore \frac{5x+12}{4x+15} = \frac{5 \times 6 + 12}{4 \times 6 + 15} = \frac{42}{39} = \frac{14}{13}$$

Ex. When 'x' is subtracted from each of 24, 33, 20 and 26, the number so obtained in this order are in proportion. What is the mean proportional between $(2x+1)$ and $(5x-11)$?

HINTS

$$x = \frac{ad - bc}{(a+d) - (b+c)} = \frac{24 \times 26 - 33 \times 20}{(24+26) - (33+20)}$$

$$= \frac{624 - 660}{50 - 53} = \frac{36}{3} = 12$$

$$\therefore \text{Mean proportion between } (2x+1) \text{ and } (5x-11)$$

$$= \sqrt{25 \times 49} = 5 \times 7 = 35$$

Alternatively

$$\text{Diff.} = 9 \left(\frac{24-x}{33-x} = \frac{20-x}{26-x} \right) \text{Diff.} = 6$$

$$\Rightarrow (24-x) \times 2 = (20-x) \times 3$$

$$\Rightarrow 48 - 2x = 60 - 3x \Rightarrow x = 12$$

$$\therefore \text{Mean proportion between } (2x+1) \text{ and } (5x-11)$$

$$= \sqrt{25 \times 49} = 5 \times 7 = 35$$

Alternatively

$$\text{Diff.} = 4 \left(\frac{24}{33} = \frac{20}{26} \right) \text{Diff.} = 6 \left(\frac{4}{7} = \frac{12}{21} \right) \text{Diff.} = 12 \left(\frac{12}{21} = \frac{4}{7} \right)$$

$$\therefore \text{Mean proportion between } (2x+1) \text{ and } (5x-11)$$

$$= \sqrt{25 \times 49} = 5 \times 7 = 35$$

Componendo & Dividendo Rule

$$\text{Let, } \frac{a}{b} = \frac{c}{d}$$

If we add 1 to both sides, then:

$$\frac{a}{b} + 1 = \frac{c}{d} + 1$$

$$\Rightarrow \frac{a+b}{b} = \frac{c+d}{d} \quad \dots\dots (i)$$

If we subtract 1 from both sides, then:

$$\frac{a}{b} - 1 = \frac{c}{d} - 1$$

$$\Rightarrow \frac{a-b}{b} = \frac{c-d}{d} \quad \dots\dots (ii)$$

Divide equation (i) by (ii),

$$\frac{a+b}{a-b} = \frac{c+d}{c-d}$$

Ex. If $\frac{5x+3y}{5x-3y} = \frac{7}{3}$, then find $x:y$.

HINTS

$$\frac{5x+3y}{5x-3y} = \frac{7}{3}$$

Using Componendo and Dividendo rule,

$$\frac{5x+3y+5x-3y}{5x+3y-5x+3y} = \frac{7+3}{7-3}$$

$$\Rightarrow \frac{10x}{6y} = \frac{10}{4} \Rightarrow \frac{5x}{3y} = \frac{5}{2}$$

$$\therefore x:y = 3:2$$

Directly and Inversely Proportional

Directly Proportional :- If $x = ky$, where k is a constant, then x is directly proportional to y .

$$x \propto y$$

Inversely Proportional :- If $x = \frac{k}{y}$, where k is a constant, then x is inversely proportional to y .

$$x \propto \frac{1}{y}$$

Ex. A varies directly with B. When $A = 52$, $B = 91$. Find the value of A when $B = 126$.

HINTS Given,

A is directly proportional to B

$$\Rightarrow \frac{A_1}{A_2} = \frac{B_1}{B_2} \Rightarrow \frac{52}{A_2} = \frac{91}{126} \Rightarrow A_2 = 72$$

Ex. M is inversely proportional to N. If M is 20, then N is 70. If $N = 50$, then what is the value of M?

HINTS $M \propto \frac{1}{N}$

$$\Rightarrow M_1 N_1 = M_2 N_2 \Rightarrow 20 \times 70 = M_2 \times 50 \Rightarrow M_2 = 28$$

Ex. 'A' varies directly as the positive square root of 'B', and inversely as the cube of 'C'. If $A = 15$, when $B = 27$ and $C = 2$, then find B when $A = 9$ and $C = 2$.

HINTS $A \propto \frac{\sqrt{B}}{C^3}$

$$\Rightarrow \frac{A_1}{A_2} = \frac{\sqrt{B_1}}{\sqrt{B_2}} \times \frac{C_2^3}{C_1^3} \Rightarrow \frac{15}{9} = \frac{\sqrt{27}}{\sqrt{B_2}} \times \frac{2^3}{2^3}$$

$$\Rightarrow \frac{27}{B_2} = \left(\frac{15}{9}\right)^2 = \frac{25}{9} \Rightarrow B_2 = \frac{243}{25}$$

Ex. The cost of a diamond varies directly as the square of its weight. Once this diamond broke into four pieces with weights in the ratio $4 : 3 : 2 : 1$. When the pieces were sold, the merchant got ₹63,000 less than that of the price of whole diamond. Find the original price of the diamond.

HINTS Cost $\propto$ (Weight)²

$$\text{Cost of whole diamond} = k \times (10)^2 = 100k$$

$$\text{Cost of four pieces} = (16 + 9 + 4 + 1)k = 30k$$

ATQ,

$$100k - 30k = ₹63,000$$

$$\Rightarrow k = ₹900$$

$$\therefore \text{Original price of the diamond} = 900 \times 100 = ₹90,000$$

Ex. The age of a tree depends on the number of rings in the trunk. The value of such a tree is proportional to its age. If a tree with 40 rings and 50 years of age has a value of ₹50,000, then a tree with 30 rings has _____ years of age and its value is ₹ _____.

HINTS Age $\propto$ Number of rings

Value of tree $\propto$ Age

ATQ,

$$\frac{40}{30} = \frac{50}{A} = \frac{50000}{V}$$

$$\therefore A = \frac{30 \times 50}{40}$$

$$= 37.5 \text{ years and } V = \frac{37.5 \times 50000}{50}$$

$$= ₹ 37,500$$

CHAPTER

20

PARTNERSHIP

Partnership is a legal relationship between two or more people who agree to run a business together and share its profit and loss.

- **Simple partnership:** If all partners invest same capital for the different time or the different capital for the same time.
- **Compound partnership:** If all partners invest different capital for the different time period.

Type of Partners

Active Partners

Sleeping Partners

- **Active partner:** Invest money as well as takes part in business activity for which he is paid salary from the profit.
- **Sleeping partners:** Who only invest money and does not take part in business activities.

Basic Terms

I	Investment
P	Profit
C	Capital
T	Time

Relation between Profit, Capital and Time

(i) Profit = Investment × Time

Ex. X, Y and Z invested ₹4,000, ₹6,000 and ₹8,000 respectively in a business. They invested the money for 3, 2 and 4 years respectively. If the total profit is ₹56,000, then what is the share of Y in the profit?

HINTS

	X	Y	Z	
Investment	→ 4000	→ 6000	→ 8000	
Time	→ 3	→ 2	→ 4	
Profit	→ 12	→ 12	→ 32	
	3	3	8	= 14
		↓ ×4000	↓ ×4000	
		₹12000	₹56000	

(ii) Investment = $\frac{\text{Profit}}{\text{Time}}$

Ex. Three partners shared the profit in a business in the ratio 8 : 7 : 5. They invested their capitals for 7 months, 8 months and 14 months respectively. What was the ratio of their capitals?

HINTS

	A	B	C	
Profit	→ 8	→ 7	→ 5	
Time	→ 7	→ 8	→ 14	
Investment	→ $\left(\frac{8}{7} : \frac{7}{8} : \frac{5}{14}\right) \times 56$			
	= 64	49	20	

(iii) Time = $\frac{\text{Profit}}{\text{Investment}}$

Ex. The ratio of investment of A to B in a business is 14 : 15 and the ratio of their respective profits at the end of a year is 2 : 5. If A invested the money for 3 months, then for how much time (in months) B invested his money?

HINTS

	A	B
Investment	→ 14	→ 15
Time	→ 3	→ x
Profit	→ 2	→ 5

$$\frac{3}{x} = \frac{2}{14} \times \frac{15}{5} \Rightarrow x = 7 \text{ months}$$

When Someone Joined or Left

Ex. A and B enter into a partnership with ₹25,000 and ₹50,000 respectively, in a company for a year. After 7 months, C gets into a partnership with them with ₹30,000 and A withdraws his contribution after 9 months. What would be C's share in the profit, assuming a total profit of ₹3,900 at the end of the year?

HINTS

Investment × Time = Profit

	A	B	C	
Investment	→ 5	→ 10	→ 6	
Time	→ 9	→ 12	→ 5	
Profit	→ 45	→ 120	→ 30	
	3	8	2	= 13
			↓ ×300	
			₹3900	

∴ C's share = 2 × 300 = ₹600

Ex. A, B and C started a business with the investment of ₹100000, ₹140000 and ₹200000 respectively. After 3 months, C left the business. 7 months after C left the business, B also left the business. B and C took their investments with them. At the end of the year, C received his share of profit as ₹1155. What is the total share of profits of A and B?

HINTS

	A	B	C	
Investment	→ 10	→ 14	→ 20	
Time	→ 12	→ 10	→ 3	
Profit	→ 120	→ 140	→ 60	
	6	7	3	
	↓ sum		↓ ×385	
	13		1155	
	↓ ×385			
	5005			

Partnership

Increase/Decrease in Capital

Ex. Three partners X, Y and Z started their business by investing ₹40,000, ₹38,000 and ₹30,000 respectively. After 6 months, X and Z made additional investments of ₹20,000 and ₹15,000 respectively, whereas Y withdrew ₹8,000. Find the share of Y (in ₹) in the total profit of ₹38,880 made at the end of the year.

HINTS X, Y and Z started their business by investing ₹40,000, ₹38,000 and ₹30,000 respectively.

	X	Y	Z
For First 6 months	40,000	38,000	30,000
For Next 6 months	60,000	30,000	45,000
Profit	100	68	75

$$\text{Share of Y} = \frac{38880}{243} \times 68 = ₹10880$$

Alternatively

$$\begin{array}{l} \text{Investment} \rightarrow \begin{array}{l} X : Y : Z \\ 40_{\times 6} : 38_{\times 6} : 30_{\times 6} \\ 60_{\times 6} : 30_{\times 6} : 45_{\times 6} \\ \hline 600 : 408 : 450 \end{array} \\ \text{Profit} \rightarrow 100 : 68 : 75 = 243 \\ \quad \quad \quad \downarrow \times 160 \quad \downarrow \times 160 \\ \quad \quad \quad \boxed{10880} \quad 38880 \end{array}$$

Ex. A, B and C started a business with their capitals in the ratio 2 : 3 : 5. A increased his capital by 50% after 4 months, B increased his capital by $33\frac{1}{3}\%$ after 6 months and C withdrew 50% of his capital after 8 months, from the start of the business. If the total profit at the end of a year was ₹86,800, then the difference between the shares of A and C in the profit was:

HINTS

$$\begin{array}{l} A : B : C \\ 200 \times 4 + 300 \times 8 : 300 \times 6 + 400 \times 6 : 500 \times 8 + 250 \times 4 \\ 3200 : 4200 : 5000 \\ 16 : 21 : 25 = 62 \\ \text{Diff.} \quad \quad \quad \downarrow \times 1400 \\ 9 \text{ unit} = 12600 \quad 86800 \end{array}$$

Difference between profit of A and C = ₹12600

Based on Donation/Tax

Ex. Mayur and Aham enter into a partnership by making investments in the ratio 2 : 3. 10% of the total profit goes to charity. If Aham's share is ₹4,860, then what is the total profit (in ₹) earned?

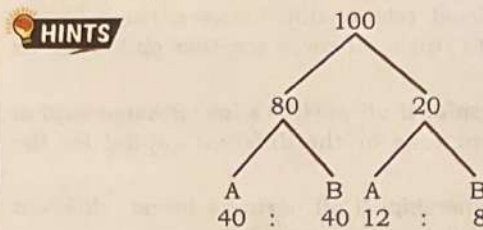
HINTS

$$\begin{array}{l} \text{Mayur : Aham} \\ \text{Investment} \rightarrow 2 : 3 \\ \text{Profit} \rightarrow 2 : 3 \\ \quad \quad \quad \text{sum} \\ \quad \quad \quad 5 \text{ unit} \\ \quad \quad \quad \downarrow \times 1620 \quad \downarrow \times 1620 \\ \quad \quad \quad 8100 \quad 4860 \end{array}$$

$$\therefore \text{Total Profit} = \frac{8100}{90} \times 100 = ₹9000$$

Based on Distribution

Ex. A & B invest ₹30,000 & ₹20,000 and 80% of total profit distributed equally among them. Rest profit distributed according to investment ratio. The difference between the shares of A & B is ₹4000. Find the total profit and find the share of A & B?



Ratio of investment of A and B = 3 : 2

Let Total Profit = 100 unit.

Share of A = 40 + 12 = 52 unit

Share of B = 40 + 8 = 48 unit

4 unit = ₹4000

$\therefore$ 1 unit = ₹1000

Total Profit = 100 × 1000 = ₹1,00,000

Share of A (52 unit) = ₹52,000

Share of B (48 unit) = ₹48,000

Ex. Keshav, Surjeet and Thomas started a business with investments in the ratio 2 : 3 : 4. The ratio of their period of investments is 5 : 6 : 9. 20% of the profit was spent on rent and maintenance of the office. Remaining profit was distributed among themselves. If the difference in the shares of profit of Keshav and Surjeet is ₹7264, then how much is the total profit (in ₹)?

HINTS Profit = Investment × Time

$$\begin{array}{l} K : S : T \\ \text{Investment} \rightarrow 2 : 3 : 4 \\ \text{Time} \rightarrow 5 : 6 : 9 \\ \text{Profit} \rightarrow 5 : 9 : 18 = 32 \text{ unit} \end{array}$$

80% of profit distributed in their respected ratio of profit.

Difference between profit of keshav and surjeet

$$= \frac{(9-5) \times 80}{100} = 3.2 \text{ units}$$

3.2 units → ₹7264

32 units → ₹72640

Alternatively

$$\begin{array}{l} K : S : T \\ \text{Investment} \rightarrow 2 : 3 : 4 \\ \text{Time} \rightarrow 5 : 6 : 9 \\ \text{Profit} \rightarrow 5 : 9 : 18 = 32 \text{ unit} \end{array}$$

Difference between profit of keshav and surjeet = 7264

4 unit = 7264

32 units = $\frac{7264}{4} \times 32 = ₹58112$ which is 80% of profit.

Total profit = $58112 \times \frac{100}{80} = ₹72640$

Partnership

Based on Management

Ex. Nasir is a working partner and Rohit is a sleeping partner in a business and Nasir puts in ₹5,000 and Rohit puts in ₹6,000. Nasir received 15% of the profit for managing the business and the rest is divided in proportion to their capitals. The amount received by Nasir out of the total profit of ₹880 is.

HINTS

Ratio of profit of Nasir and Rohit = 5 : 6

Total profit = ₹880

Money got by Nasir for managing the business

= 15% of ₹880 = ₹132

Remaining Amount = ₹880 - ₹132 = ₹748

Amount received by Nasir = ₹132 + ₹748 × $\frac{5}{11}$

= ₹132 + ₹340 = ₹472

Ex. A, B and C jointly started a business in which A invested ₹7,000 for 12 months, B invested ₹8,000 for 10 months and C invested ₹12,000 for 6 months. A was the working member for which he received 20% of the profits as his remuneration. The total profit earned at the end of one year was ₹14,750. What is C's share (in ₹) in the profit?

HINTS

	A	B	C
Investment →	7	8	12
Time →	12	10	6
Profit →	84	80	72
	21	20	18

100% = ₹14750

80% = ₹11800

Share of C = $\frac{18}{21+20+18} \times 11800$

= $\frac{18}{59} \times 11800 = ₹3600$

Based on Profit and Salary

Ex. P and Q start a shop with a capital of ₹1,50,000 and ₹4,50,000, respectively. After a year, out of the profit of ₹2,00,000, P gets his share of profit plus some money as his salary that is not a part of the profit. If P gets a total of ₹90,000, what is the amount of salary (in ₹) that he received?

HINTS

	P	Q
	150,000	450,000
Profit →	1	3

Share of P out of profit = $\frac{1}{4} \times 200,000$

= ₹50,000

But he get ₹90,000.

Salary = 90,000 - 50,000

= ₹40,000

CHAPTER

21

MIXTURE

A **mixture** is a material made up of two or more different substances.

Concentration of Mixture

Presence of a particular constituent in a mixture in percentage is called concentration of that constituent.

Addition or Removal of Some Quantity

Ex. 200 litres of a mixture contains milk and water in the ratio 17 : 3. After the addition of some more milk to it, the ratio of milk to water in the resulting mixture becomes 7 : 1. The quantity of milk added to it was:

HINTS [In this type of questions, the quantity of the parts which are not replaced will remain same]

In this question, quantity of water will be same.

$$\begin{array}{ccc} \text{M} & : & \text{W} \\ \text{Initial} & \rightarrow & (17 : 3)_{\times 1} \end{array}$$

$$\begin{array}{ccc} \text{M} & : & \text{W} \\ \text{Final} & \rightarrow & (7 : 1)_{\times 3} \end{array}$$

$$\begin{array}{ccc} \text{M} & : & \text{W} \\ \text{Initial} & \rightarrow & (17 : 3)_{\times 1} \end{array}$$

ATQ,

$$(17 + 3) \text{ unit} = 20 \text{ unit} \rightarrow 200 \text{ litre}$$

$$1 \text{ unit} \rightarrow 10 \text{ litre}$$

$$(21 - 17) \text{ unit} = 4 \text{ unit} \rightarrow 40 \text{ litre}$$

$$\therefore \text{Quantity of added milk} = 40 \text{ litre}$$

Ex. A vessel contains 32 litre solution of acid and water in which the ratio of acid and water is 5 : 3. If 12 litres of the solution is taken out and $7\frac{1}{2}$ litres of water is added

to it, then what is the ratio of acid and water in resulting solution?

HINTS $32 - 12 = 20 \text{ litre}$

20 L	
Acid	Water
5 : 3	
12.5L	7.5L
	+7.5
12.5	15
5	6

Ex. A beaker contains acid and water in the ratio 1 : x. When 50 ml of water is mixed in 300 ml of the mixture, the ratio of acid to water becomes 2 : 5. What is the value of 'x'?

HINTS

$$\text{A} : \text{W}$$

$$1 : x$$

$$2 : 5$$

$$\therefore (5 + 2) \text{ unit} = 7 \text{ unit} \rightarrow 300 + 50 = 350 \text{ ml}$$

$$1 \text{ unit} \rightarrow 50 \text{ ml}$$

$$2 \text{ unit} \rightarrow 100 \text{ ml}$$

$$\text{Initial quantity of water} = 300 - 100 = 200 \text{ ml}$$

$$\therefore x = 200$$

Ex. The ratio of milk and water in a vessel is 2 : 3. If we extract some water by chemical process, then the ratio becomes 5 : 7 and quantity of mixture reduced to 36 l. Find how much quantity of water was extracted.

HINTS

[In this type of questions, the quantity of the parts which are not replaced will remain same]

In this question, quantity of milk will be same.

$$\begin{array}{ccc} \text{M} & : & \text{W} \\ \text{Initial} & \rightarrow & (2 : 3)_{\times 5} \end{array}$$

$$\begin{array}{ccc} \text{M} & : & \text{W} \\ \text{Final} & \rightarrow & (5 : 7)_{\times 2} \end{array}$$

$$\text{ATQ,}$$

$$(10 + 14) \text{ unit} = 24 \text{ unit} \rightarrow 36 \text{ litre}$$

$$(15 - 14) \text{ unit} = 1 \text{ unit} \rightarrow 1.5 \text{ litre}$$

$$\therefore \text{Extracted amount of water} = 1.5 \text{ litre}$$

Ex. The ratio of milk and water in a mixture is 1 : 3. If we add 'x' litre milk to the mixture, then ratio becomes 7 : 15. If we add 50 litre water to the mixture, then ratio becomes 2 : 5. Find the value of 'x'.

HINTS

$$\begin{array}{ccc} \text{M} & : & \text{W} \\ 1 & : & 3 \end{array}$$

$$\begin{array}{ccc} \text{M} & : & \text{W} \\ 10 & : & 30 \end{array}$$

$$\begin{array}{ccc} \text{M} & : & \text{W} \\ 7_{\times 2} & : & 15_{\times 2} \end{array}$$

$$\begin{array}{ccc} \text{M} & : & \text{W} \\ 14 & : & 30 \end{array}$$

$$\text{ATQ,}$$

$$(35 - 30) \text{ unit} = 5 \text{ unit} \rightarrow 50 \text{ litre}$$

$$1 \text{ unit} \rightarrow 10 \text{ litre}$$

$$(14 - 10) \text{ unit} = 4 \text{ unit} \rightarrow 40 \text{ litre}$$

$$\therefore x = 40$$

Concept of Replacement

Based on this concept, 2 types of questions are asked.

Replacement of equal quantity

When amount 'x' of mixture of any substance 's' is replaced n-times with 'p' quantity of other substance, then quantity of substance 's' in replaced mixture

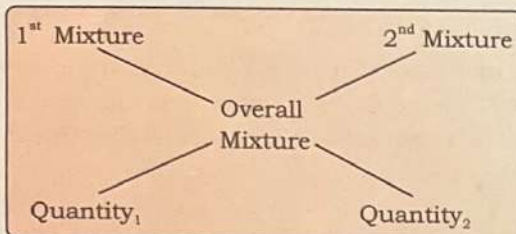
$$= x \left(1 - \frac{p}{x} \right)^n$$

ALLIGATION

- Applications of Alligation in different chapters.

- Mixture-based questions
- Percentage based questions
- Simple interest based questions
- Profit and loss based questions
- Discount based questions
- Average based questions
- Time and distance based questions
- Number system based questions

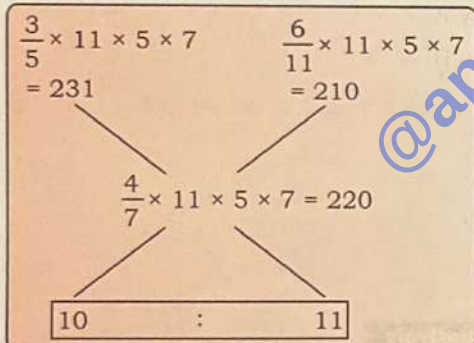
Mixture



- Ex.** Two containers have mixture of milk and water, respectively, in the ratio 3 : 2 and 6 : 5. In what ratio should the mixture in the two containers be mixed so that the ratio of milk to water in the final mixture is 4 : 3?



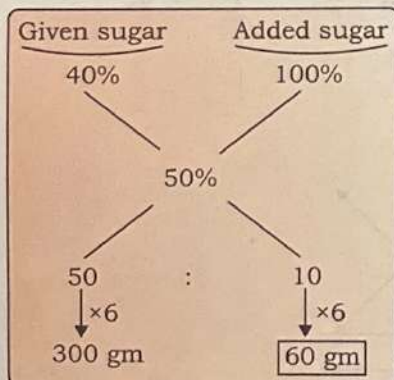
HINTS



- Ex.** 300 gm of sugar solution has 40% of sugar in it. How much sugar should be added to make it 50% in the solution?

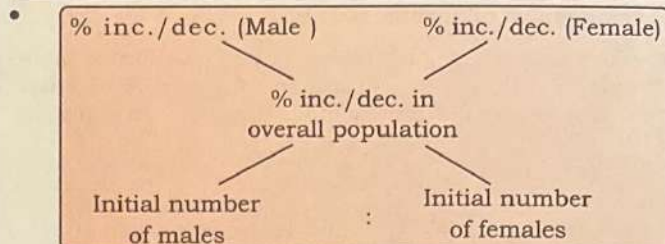


HINTS



∴ Amount of added sugar = 60 gram

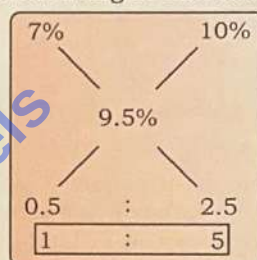
Percentage



- Ex.** Population of a town is 1,20,000. If the number of males increases by 7% and number of females increases by 10%, then the population of town would increase by 9.5%. Find the difference between male to female population living in that town.



HINTS



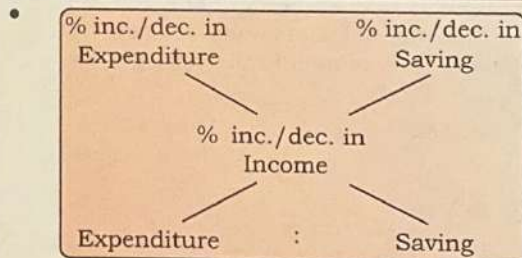
ATQ,

(5 + 1) unit = 6 unit → 1,20,000

(5 - 1) unit = 4 unit → 80,000

∴ Required difference = 80,000

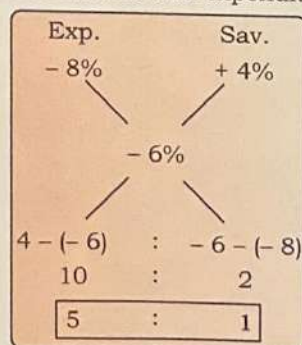
Income, Expenditure & Saving



- Ex.** If the income of a family decreases by 6% and expenditure decreases by 8% and its saving increases by 4%. Find the ratio of income and expenditure of that family.

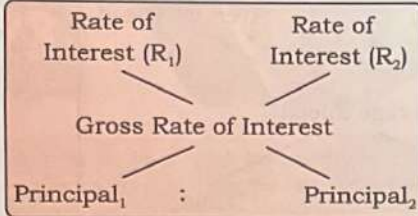


HINTS



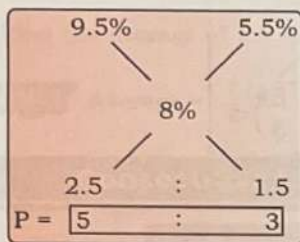
∴ Income : Expenditure = 6 : 5

Simple Interest



Ex. Harish is having ₹40,000. He deposited it in 2 banks. Interest given by 1st bank is 9.5% per annum and that of 2nd bank is 5.5% per annum. If at the end of a year, Harish got ₹3,200 as interest, then find the difference between the amount deposited in both banks.

HINTS $r = \frac{SI}{P} \times \frac{100}{t} = \frac{3200 \times 100}{40000 \times 1} = 8\%$



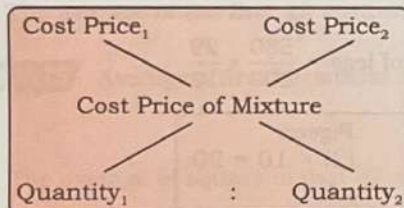
ATQ,

(5 + 3) unit = 8 unit → ₹40,000

(5 - 3) unit = 2 unit → ₹10,000

∴ Required difference = ₹10,000

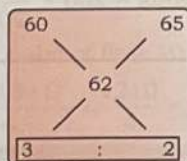
Profit and Loss



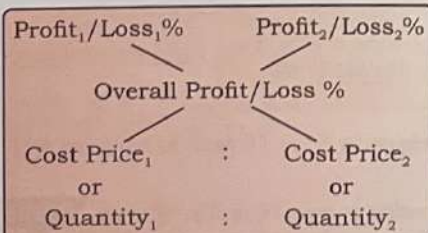
Ex. In what ratio must a grocer mix two varieties of tea worth ₹60 per kg and ₹65 per kg so that by selling the mixture at ₹68.20 per kg, he may gain 10%?

HINTS CP of Mixture = $\frac{68.20}{11} \times 10 = ₹62$

Now,

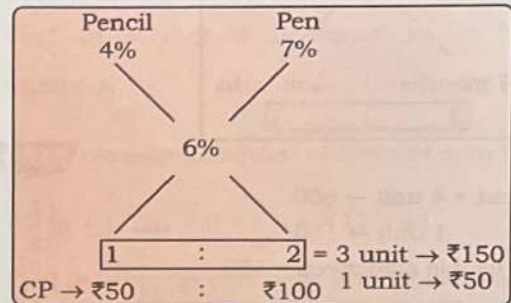


∴ Required ratio = 3 : 2



Ex. A student bought 10 pencils and 8 pens at ₹150. He sells pencil at 4% profit and pen at 7% profit. Overall he gains ₹9. Find the cost price of each pencil and pen.

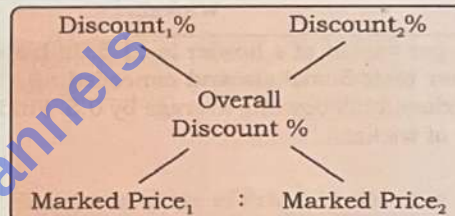
HINTS Overall profit % = $\frac{9}{150} \times 100\% = 6\%$



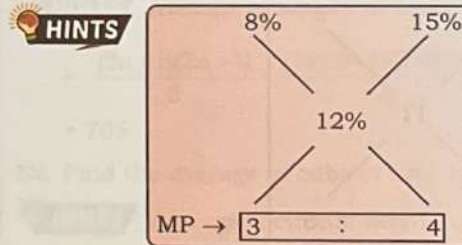
∴ CP of each pencil = $\frac{50}{10} = ₹5$

CP of each pen = $\frac{100}{8} = ₹12.5$

Discount



Ex. A seller sells 2 watches at 8% and 15% discount respectively. Total marked price of both the watches is ₹2,100. Find the difference between the marked price of both watches, if total discount is of 12%.



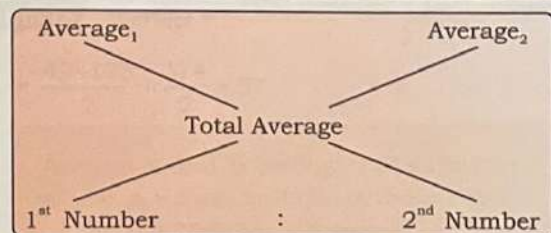
ATQ,

(3 + 4) unit = 7 unit → ₹2,100

(4 - 3) unit = 1 unit → ₹300

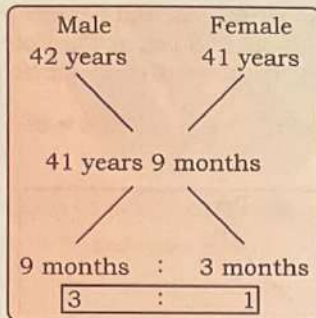
∴ Required difference = ₹300

Average



Ex. In a company with 600 employees, the average age of male employees is 42 years and that of female employees is 41 years. If the average age of all the employees in the company is 41 years and 9 months, then the number of female employees is:

HINTS

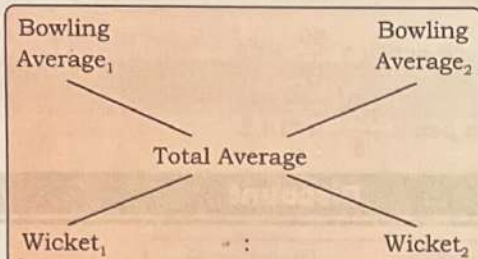


ATQ,

$$(3 + 1) \text{ unit} = 4 \text{ unit} \rightarrow 600$$

$$1 \text{ unit} \rightarrow 150$$

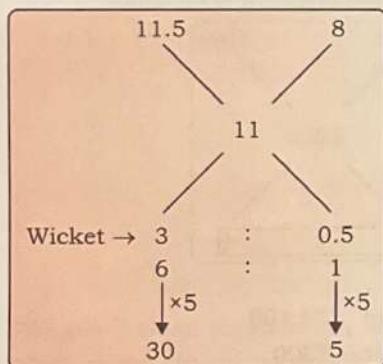
$$\therefore \text{No. of female employees} = 150$$



Ex. Average runs per wicket of a bowler is 11.5. In his next innings, bowler took 5 wickets and conceded 40 runs, there by he reduced his bowling average by 0.5. Find the total number of wickets.

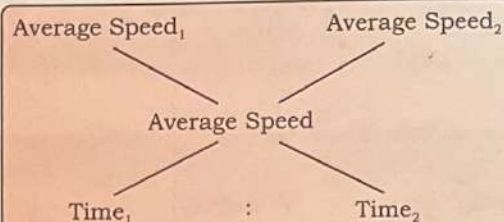
HINTS

$$\text{Bowling average} = \frac{\text{Total Runs}}{\text{Total Wickets}} = \frac{40}{5} = 8$$



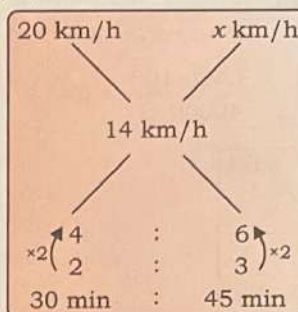
$$\therefore \text{Total number of wickets} = 30 + 5 = 35$$

Time and Distance



Ex. A car travels 20 km/h for 30 min and at 'x' km/h for 45 min. If the average speed of the car for entire journey is 14 km/h, then find the value of 'x'.

HINTS



$$\therefore x = 14 - 4 = 10 \text{ km/h}$$

Number System

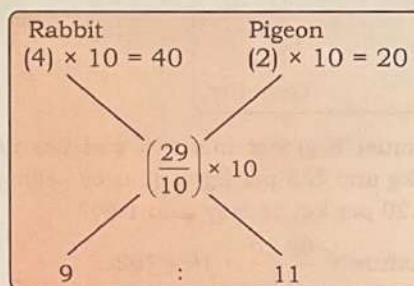
Ex. In a zoo, there are rabbits and pigeons. If heads are counted, there are 200 and if legs are counted, there are 580. How many pigeons are there?

HINTS

$$\text{Number of heads} = 200$$

$$\text{Number of legs} = 580$$

$$\text{Average number of legs} = \frac{580}{200} = \frac{29}{10}$$



$$\therefore \text{Number of Pigeons} = 200 \times \frac{11}{20} = 110$$

CHAPTER

23

AVERAGE

The result you get by adding two or more terms together and dividing by the number of terms is called **average**.

$$\text{Average} = \frac{\text{Sum of all terms}}{\text{Number of terms}}$$

Ex. The weight of the five friends is 67 kg, 85 kg, 70 kg, 90 kg and 103 kg. Find the average of their weight in kg.

HINTS Average = $\frac{67+85+70+90+103}{5} = \frac{415}{5} = 83$ kg

Ex. Find the average of 72, 81, 90, 68, 61.

HINTS Average = $\frac{72+81+90+68+61}{5} = \frac{372}{5} = 74.4$

Consecutive Natural Numbers

Consecutive Natural Numbers

Average of first 'n' numbers = $\frac{(n+1)}{2}$

Average of square of first 'n' numbers = $\frac{(n+1)(2n+1)}{6}$

Average of cube of first 'n' numbers = $\frac{n(n+1)^2}{4}$

Ex. The average of the first 10 natural numbers is _____.

HINTS Average of first 10 natural numbers = $\frac{n+1}{2} = \frac{10+1}{2} = 5.5$

Ex. The average of square of first 47 natural numbers is:

HINTS Average of square of first 47 natural no.

$$\begin{aligned} &= \frac{(n+1)(2n+1)}{6} \\ &= \frac{(47+1)(2 \times 47+1)}{6} = \frac{48 \times 95}{6} = 95 \times 8 = 760 \end{aligned}$$

Ex. Find the average of cube of first 10 natural numbers.

HINTS Average of cube of first 10 natural numbers

$$= \frac{n(n+1)^2}{4} = \frac{10 \times (10+1)^2}{4} = \frac{1210}{4} = 302.5$$

Consecutive Even Numbers

Average of first 'n' numbers = $(n+1)$

Average of square of first 'n' numbers = $\frac{2(n+1)(2n+1)}{3}$

Average of cube of first 'n' numbers = $2n(n+1)^2$

Ex. The average of first 91 even numbers is:

HINTS Average of first 91 even numbers = $(n+1) = 91+1 = 92$

Ex. Find the average of the square of the first five even numbers.

HINTS Average of square of first five even numbers

$$= \frac{2}{3} (n+1)(2n+1) = \frac{2}{3} \times 6 \times 11 = 44$$

Ex. Find the average of cube of first 15 even numbers.

HINTS Average of cube of first 15 even numbers

$$\begin{aligned} &= 2n(n+1)^2 \\ &= 2 \times 15 \times (15+1)^2 = 30 \times 16 \times 16 = 7680 \end{aligned}$$

Consecutive Odd Numbers

Average of first 'n' numbers = n

Average of square of first 'n' numbers = $\frac{(2n+1)(2n-1)}{3}$

Average of cube of first 'n' numbers = $n(2n^2-1)$

Ex. Find the average of first 125 odd numbers.

HINTS Average of first 125 odd numbers = $n = 125$

Ex. Find the average of square of first 23 odd numbers.

HINTS Average of square of first 23 odd numbers

$$\begin{aligned} &= \frac{(2n+1)(2n-1)}{3} = \frac{(2 \times 23+1)(2 \times 23-1)}{3} = \frac{47 \times 45}{3} \\ &= 705 \end{aligned}$$

Ex. Find the average of cube of first 10 odd numbers.

HINTS Average of cube of first 10 odd numbers

$$\begin{aligned} &= n(2n^2-1) \\ &= 10(2 \times 10^2-1) = 1990 \end{aligned}$$

Ex. What is the average of all the natural numbers from 49 to 125?

HINTS Average = $\frac{\text{First number} + \text{Last number}}{2}$

$$= \frac{49+125}{2} = \frac{174}{2} = 87$$

Average of first 'n' multiples of a number = $a \times \frac{(n+1)}{2}$
where, a = First multiple of the number
n = Number of terms

Ex. Find the average of first 20 multiples of 7.

HINTS Average = $a \times \frac{(n+1)}{2} = 7 \times \frac{(20+1)}{2} = 7 \times 10.5 = 73.5$

Finding Average using Deviation method

Step I: Assume any term as average in observation.

Step II: Calculate the difference between the observations and assumed average. (These differences are known as **Deviation**).

Step III: Add all deviations calculated in Step-II and divide it by total number of terms given in the observations. (This is known as **net deviation** in the average).

Step IV: To get the actual average, add or subtract the net deviation in the assumed average (as per the result of Step-III).

Ex. Find the average of the following observations:

48, 52, 40, 60, 53, 54, 49

HINTS **Step I:** Let, average = 50

Step II: Calculating the difference between the observations and the assumed average.

48 52 40 60 53 54 49
 Let, average = 50

Deviation → -2 +2 -10 +10 +3 +4 -1 = 6

Step III: Net deviation = $\frac{+6}{7} = +0.857$

Step IV: Actual average = $50 + 0.857 = 50.857$

Based on Weighted Average

If the average of ' n_1 ' observations is ' a_1 ', the average of ' n_2 ' observations is ' a_2 ' and so on, then:

$$\text{Average of all the observations} = \frac{n_1 a_1 + n_2 a_2 + \dots}{n_1 + n_2 + \dots}$$

Ex. John buys 20 kg of wheat at ₹30 per kg and 30 kg of wheat at ₹50 per kg. Find its average price.

HINTS Quantity → 20 : 30
 Rate → 30 : 50
 Total price → 60 : 150

$$\therefore \text{Average price} = \frac{60 + 150}{2 + 3} = \frac{210}{5} = ₹42/\text{kg}$$

Ex. In a primary school, there are 60 boys of age 12 years each, 40 of age 13 years each, 50 of age 14 years each and 50 of age 15 years each. The average age (in years) of all the boys in the school is:

HINTS Let, average = 12

No. of boys → 60 40 50 50

Ratio → 6 : 4 : 5 : 5

Avg. age → 12 13 14 15

Let, Avg. = 12

Deviation → 0 +1 +2 +3

$$\text{Net deviation} = \frac{4 \times 1 + 5 \times 2 + 5 \times 3}{6 + 4 + 5 + 5} = \frac{29}{20} = 1.45$$

$$\therefore \text{Average} = 12 + 1.45 = 13.45 \text{ years}$$

Based on Inclusion

If a new person is added in the group, then age/income/weight/height of added the person = Initial average $\pm$ Average increment/decrement $\times$ Final no. of persons

Ex. The mean weight of 34 students of a school is 42 kg. If the weight of the teacher is also included, the mean rises by 400 grams. Find the weight of the teacher (in kg).

HINTS Weight of teacher = Initial average + Average increment $\times$ Final no. of persons
 $= 42 + 0.4 \times 35 = 42 + 14 = 56 \text{ kg}$

If more than one person is added in the group:

Ex. The average weight of 20 students of a class was 30.5 kg. After 6 new students joined the class, its average weight increased by 2 kg. Find the average weight of the 6 new students who joined the class (rounded off to one decimal place).

HINTS Average weight of 6 new students
 $= 30.5 + \frac{(26 \times 2)}{6} = 30.5 + 8.7 = 39.2 \text{ kg}$

When initial number of persons is asked:

Ex. The average weight of some students in a class was 58.4 kg. When 5 students having the average weight 62.8 kg joined the class, the average weight of all students in the class increased by 0.55 kg. The number of students initially in the class was:

HINTS Total deviation = $5 \times 4.4 = +22$
 No. of students now = $\frac{22}{0.55} = 40$
 $\therefore$ Initial no. of students = $40 - 5 = 35$

Ex. In a class with a certain number of students if one student weighing 30 kg is added, then the average weight of the class increases by 1 kg. If one more student weighing 30 kg is added, then the average weight of the class increases by 1.5 kg over the original average. What is the original average weight (in kg) of the class?

HINTS Let, no. of students = n

Initial average = x

Total weight of students = nx

Case I:

$$\frac{nx + 30}{n + 1} = x + 1$$

$$\Rightarrow nx + 30 = nx + x + n + 1$$

$$\Rightarrow x + n = 29 \quad \dots\dots\dots(i)$$

Case II:

$$\frac{nx + 30 + 30}{n + 2} = x + 1.5$$

$$\Rightarrow nx + 60 = nx + 2x + 1.5n + 3$$

$$\Rightarrow 2x + 1.5n = 57 \quad \dots\dots\dots(ii)$$

Average

From (i) and (ii):

$$57 = 2(29 - n) + 1.5n$$

$$\Rightarrow 57 = 58 - 2n + 1.5n$$

$$\Rightarrow -1 = -0.5n \Rightarrow n = 2$$

From eqⁿ. (i):

$$x + n = 29 \Rightarrow x = 27$$

$\therefore$ Original average weight of the class = 27 kg

Alternatively

Let, total no. of students initially be 'n'.

Case I:

$$\text{Original avg.} = 30 - 1(n + 1) \dots\dots\dots(i)$$

Case II:

$$1 + \text{Original avg.} = 30 - \frac{1}{2}(n + 2)$$

$$\Rightarrow \text{Original avg.} = 30 - \frac{1}{2}(n + 2) - 1 \dots\dots\dots(ii)$$

From eqⁿ. (i) and (ii):

$$30 - (n + 1) = 30 - \frac{1}{2}(n + 2) - 1 \Rightarrow n = 2$$

$\therefore$ Original average weight = $30 - (2 + 1) = 27$ kg

Based on Exclusion

If a person leaves the group then age/ income/ height/ weight of the left person = Initial average $\pm$ Average increment / decrement $\times$ Final no. of persons

Ex. The average age of 24 workers and their supervisor is 41 years. If the supervisor's age is excluded, the average reduces by 0.75 years. The age of the supervisor is:

HINTS Age of supervisor = $41 + 24 \times 0.75 = 41 + 18$
 $= 59$ yrs

When more than one person leaves the group:

Ex. The average height of 25 boys is 1.4 m. When 5 boys leave the group, then the average height increase by 0.15 m. What is the average height of the 5 boys who leave?

HINTS Average height of the boys who left
 $= 1.4 - \frac{0.15 \times 20}{5} = 1.4 - 0.6 = 0.8$ m

Based on Replacement

If a person is replaced by a new person then age/height/weight/income of the new person = Age of replaced person $\pm$ Average increment/decrement $\times$ No. of persons

Ex. There are 50 students in a class. One of them weighing 50 kg goes away and a new student joins. By this the average weight of the class is increased by $\frac{1}{2}$ kg. The weight of the new student is:

HINTS Weight of new student = Weight of the student who left + Average increment $\times$ Initial no. of students
 $= 50 + \frac{1}{2} \times 50 = 50 + 25 = 75$ kg

When more than one person is replaced by equal number of persons:

Ex. The average weight of 49 students in a class is 39 kg. Seven of them whose average weight is 40 kg leave the class and other seven students whose average weight is 54 kg join the class. What is the new average weight (in kg) of the class?

HINTS Net deviation = $(-40 + 39) \times 7 + (54 - 39) \times 7$
 $= -7 + 105 = 98$

$$\therefore \text{New avg. weight} = 39 + \frac{98}{49} = 39 + 2 = 41 \text{ kg}$$

When number of persons is not replaced by equal number of persons:

Ex. The average score in an English exam of a class of 45 students in a coaching centre is 52. A group of 6 boys having an average score of 40 leave the class and another group of eight boys having an average score of 43 joins the class. What is the new average score of the class?

	Initially Left	Join
No. of students	45	6
Average	52	40

$$\therefore \text{Avg. score of class} = 52 - (52 - 40) \times 6 + (52 - 43) \times 8$$

$$= 52 - 72 + 72 = 52$$

Ex. There are some children in a camp and their average weight is 40 kg. If 5 children with average weight 36 kg joined the camp or if 5 children with average weight 43.2 kg leave the camp, the average weight of children in both cases is equal. How many children are there in the camp initially?

HINTS ATQ,

Let, total no. of students initially = x

$$\Rightarrow \frac{x+5}{5 \times 4} = \frac{x-5}{3.2 \times 5}$$

$$\Rightarrow \frac{x+5}{20} = \frac{x-5}{16}$$

$$\Rightarrow x = 45$$

$\therefore$ There are 45 children in the camp initially.

Ex. The average weight of a group of 12 girls is 48 kg. If 2 girls, R and S replace P and Q, the new average weight becomes 46 kg. The weight of P = The weight of Q. The weight of R = The weight of S. Another girl, T is included in the group and new average weight becomes 46 kg. The weight of T = The weight of R. What is the weight of P? (in kg)

HINTS $10G + (P + Q) = 48 \times 12 = 576$

$$10G + (R + S) = 46 \times 12 = 552$$

$$2P - 2R = 24$$

$$P - R = 12$$

$$10G + (R + S) + T = 46 \times 13$$

$$T = 598 - 552 = 46$$

$$T = R = 46 \text{ kg}$$

$$P - 46 = 12 \Rightarrow P = 58 \text{ kg}$$

$$\therefore \text{Weight of P} = 58 \text{ kg}$$

Based on Wrongly Entered Data

- (A) If average of 'n' numbers is 'm' but later on it was found that a number 'a' was misread as 'b', then:

$$\text{Correct average} = m + \frac{(a-b)}{n}$$

- Ex.** While finding the average of 10 given numbers, a student, by mistake, wrote 64 in place of a number 46 and got his average 50. The correct average of the given numbers is:

HINTS

$$\text{Correct average} = 50 + \frac{(46-64)}{10} = 50 + \frac{(-18)}{10} = 48.2$$

- (B) If the average of 'n' numbers is 'm' but later on it was found that the two numbers 'a' and 'b' were misread as 'p' and 'q', respectively, then:

$$\text{Correct average} = m + \frac{(a+b-p-q)}{n}$$

- Ex.** The average marks of 40 students was found to be 68 when the marks of two students were incorrectly entered as 48 and 64 instead of 84 and 46 respectively, then what is the correct average?

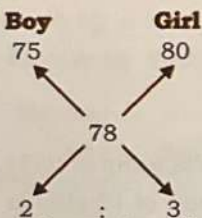
HINTS

$$\text{Correct average} = 68 + \frac{(84+46-48-64)}{40} = 68.45$$

Based on Alligation

- Ex.** The average marks (out of 100) of boys and girls in an examination are 75 and 80, respectively. If the average marks of all the students in that examination are 78. Find the ratio of the number of boys to the number of girls.

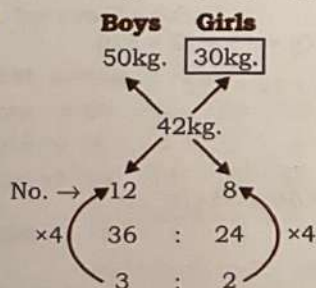
HINTS



$\therefore$ Ratio of the number of boys to the number of girls = 2 : 3

- Ex.** In a class, there are 60 students. The average weight of the entire class is 42 kg. If there are 36 boys in the class and the average weight of all the boys is 50 kg, then what is the average weight (in kg) of all the girls?

HINTS



$\therefore$ Average weight of girls = 30kg

Based on Cricket

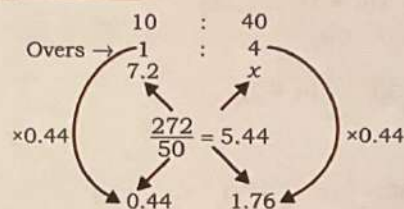
- Ex.** In the first 10 overs of a cricket game, the run rate was only 7.2. What should be the average run rate in the remaining 40 overs to reach the target of 272 runs?

HINTS

	10 overs	40 overs
Average $\rightarrow$	7.2	5
Total runs $\rightarrow$	72	200
	Total 272	

$\therefore$ Average run rate = 5

Alternatively



$$x = 5.44 - 0.44 = 5$$

$\therefore$ Average run rate = 5

- Ex.** A cricket batsman had a certain average of runs for his 11 innings. In the 12th innings, he made a score of 90 runs after that his average run is decreased by 5. Find his average runs after 12th innings.

HINTS

Let the initial average = x

New average

$$= \frac{(\text{Initial innings}) \times \text{Average} + \text{Recent runs}}{\text{Total innings}}$$

$$\Rightarrow x - 5 = \frac{11 \times x + 90}{12} \Rightarrow x = 150$$

$\therefore$ New average = 150 - 5 = 145

Alternatively

Let, average for 11 innings be 'x'.

N = Initial average $\pm$ Average increment/decrement $\times$ Final no. of innings

$$\Rightarrow 90 = x - 5 \times 12 \Rightarrow x = 150$$

$\therefore$ New average = 150 - 5 = 145

- Ex.** Average of 64 innings of a player is 62 runs, difference between his maximum and minimum score is 180 runs. If these two innings are excluded then the average of remaining innings is 60 runs. Then his maximum score is:

HINTS

$$64 \times 62 - H - L = 62 \times 60$$

$$\Rightarrow 248 = H + L \quad \dots\dots(i)$$

Given that:

$$H - L = 180 \quad \dots\dots(ii)$$

On adding (i) and (ii), we get:

$$248 = H + L$$

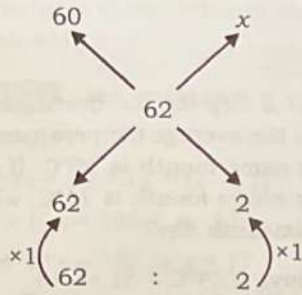
$$180 = H - L$$

$$428 = 2H$$

$$\Rightarrow H = 214$$

$\therefore$ Maximum score = 214

Alternatively



$$x = 62 + 62 = 124$$

$$\text{Highest} + \text{Lowest} = 2 \times 124 = 248$$

$$248 = H + L$$

$$180 = H - L$$

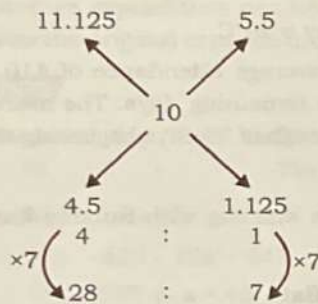
$$428 = 2H$$

$$\Rightarrow H = 214$$

$$\therefore \text{Maximum score} = 214$$

Ex. A cricketer whose bowling average is 11.125 runs per wicket. He takes 7 wickets for 38.5 runs and thereby decrease his average by 1.125. The number of wickets taken by him till last match was:

HINTS Bowling Average = $\frac{\text{Total runs}}{\text{Total wickets}} = \frac{38.5}{7} = 5.5$



$$\text{So, } \frac{7}{1.125} \times 4.5 = 28$$

$$\therefore \text{Number of wickets till the last Match} = 28 + 7 = 35$$

Based on Age

Ex. 5 years ago, the average age of A, B, C and D were 45 years. With the joining of E, the average age of all the five is 48 years. How old is E?

HINTS Total present age of (A, B, C, D) = $(45 + 5) \times 4 = 200$ years

$$\text{Total present age of (A, B, C, D, E)} = 48 \times 5 = 240 \text{ years}$$

$$\text{Present Age of E} = 240 - 200 = 40 \text{ years}$$

Alternatively

$$\text{Total present Average of (A, B, C, D)} = 45 + 5 = 50$$

$$\text{Total present Average of (A, B, C, D, E)} = 48$$

$$\therefore \text{Age of E} = \text{Initial average} \pm \text{Average increment/decrement} \times \text{Final members}$$

$$= 50 - (2 \times 5) = 40 \text{ years}$$

Ex. The average age of Ravi, his wife and their two children is 23 years. His wife is just 4 years younger than Ravi and his wife was 24 years old when his daughter was born. He was 32 years old when his son was born. The average age of Ravi and his daughter is:

HINTS Total Sum = $23 \times 4 = 92$

(R)	(W)	(D)	(S)	
32	28	4	0	= 64
↓ +x	↓ +x	↓ +x	↓ +x	

$$\Rightarrow 64 + 4x = 92$$

$$\Rightarrow x = 7$$

$$\text{Age of Ravi} = 32 + 7 = 39 \text{ years}$$

$$\text{Age of Daughter} = 4 + 7 = 11 \text{ years}$$

So, average age of Ravi and his daughter

$$= \frac{(39 + 11)}{2} = \frac{50}{2} = 25 \text{ years}$$

Based on Numbers

Ex. Average of 25 natural numbers is 67. 1st number is increased by 1, 2nd number is increased by 3 and 3rd number is increased by 5 and so on. Find the new average.

HINTS We know, average of n-odd numbers = n

$$\text{So, average of 25 odd no.} = 25$$

$$\therefore \text{New average} = 67 + 25 = 92$$

Ex. The average of 15 results is 21. The average of the first 7 of those is 21 and the average of the last 7 is 20. What is the 8th result?

HINTS 8th Result = $15 \times 21 - (7 \times 21 + 7 \times 20) = 315 - 287 = 28$

Alternatively

	First	Last	8 th
Result →	7	7	1
Average →	21	20	x
	↑		
	21		

$$\text{Net deviation} = (-1) \times 7 = -7$$

$$\therefore x = 21 + 7 = 28$$

Ex. The average of 21 numbers is 44. The average of first 11 numbers is 48 and that of last 11 numbers is 42. If 11th number is excluded, what is the average of the remaining numbers?

HINTS

No. →	21	11	11
Average →	44	48	42
Deviation →		↑ +4	↑ -2

$$\text{Net deviation} = 11 \times 4 - 11 \times 2$$

$$= 44 - 22 = +22$$

$$11^{\text{th}} \text{ number} = 44 + 22 = 66$$

$$\therefore \text{New average after removing 11th number}$$

$$= \frac{21 \times 44 - 66}{20}$$

$$= 42.9$$

Average

Ex. The average of 15 numbers is 45. The average of the first six numbers is 42 and that of last six numbers is 43. The 7th number is two times the 8th number but 5 more than the 9th number. The average of the 7th and the 9th number is :

HINTS

	All	First	Last
No. →	15	6	6
Average →	45	42	43
Deviation →		-3	-2

$$\text{Net deviation} = (-3 \times 6) + (-2 \times 6) = -30$$

Let, 8th number = x

7 th	8 th	9 th
$2x$	x	$2x - 5$

$$\text{Now, } 5x - 5 = 45 \times 3 + 30$$

$$\Rightarrow 5x = 170 \Rightarrow x = 34$$

$$\therefore \text{Average of 7th and 9th number} = \frac{68 + 63}{2} = 65.5$$

Ex. The average of 21 data is 36, out of which the first 12 data have an average of 15. The average of the rest 9 data is:

HINTS

Average of remaining 9 numbers

$$= \frac{(\text{Total sum} - \text{Sum of first 12 data})}{9}$$

$$= \frac{(21 \times 36 - 12 \times 15)}{9} = \frac{576}{9} = 64$$

Alternatively

	First	Rem.
Data →	12	9
Ratio →	4	3
Average →	15	x
Deviation →		-21

$$\text{Net deviation} = \frac{-4 \times 21}{3} = -28$$

$$\therefore \text{Average of remaining data } (x) = 36 + 28 = 64$$

Miscellaneous

If there are 3 natural numbers and average of any two numbers when added with third number gives a , b , c , then:

$$\text{Sum of numbers} = \frac{a + b + c}{2} = k$$

$$\text{First number} = 2a - k$$

$$\text{Second number} = 2b - k$$

$$\text{Third number} = 2c - k$$

Ex. Three numbers are such that if the average of any two of them is added to the third number, the sums obtained are 168, 174 and 180 respectively. What is the average of the original three numbers?

HINTS

$$\text{Sum} = \frac{168 + 174 + 180}{2} = \frac{522}{2} = 261$$

$$\text{Average} = \frac{261}{3} = 87$$

Ex. The average temperature of a city for the first sixteen days of January is 22°C and the average temperature for the last sixteen days of the same month is 26°C. If the average temperature for the entire month is 24°C, what is the temperature on the sixteenth day?

HINTS

$$\text{Temperature of 31 days} = 24^\circ\text{C} \times 31 = 744$$

$$\text{Temperature of first 16 days} = 22^\circ\text{C} \times 16 = 352$$

$$\text{Temperature of last 16 days} = 26^\circ\text{C} \times 16 = 416$$

$$\text{Temperature of 16th day} = (352 + 416) - 744$$

$$= 768 - 744 = 24^\circ\text{C}$$

Alternatively

	First	Last
Days →	16	16
Average →	22°C	26°C
Deviation →		24°C

$$\text{Net deviation} = (24 - 22) \times 16 + (24 - 26) \times 16$$

$$= 32 - 32 = 0$$

$$\therefore \text{Temperature on 16th day} = 24^\circ\text{C}$$

Ex. A public library has an average attendance of 410 on Sundays and 230 for the remaining days. The average attendance per day of a month of 30 days beginning with Sunday would be:

HINTS

Month of 30 days starting with Sunday has 5 Sundays.

	Sun	Mon-Sat
Day →	5	25
Average →	410	230

$$\text{Let, average} = 230$$

$$\text{Net deviation} = (410 - 230) \times 5 = 900$$

$$\therefore \text{Average attendance of whole month} = 230 + \frac{900}{30}$$

$$= 230 + 30 = 260$$

Alternatively

Month of 30 days starting with Sunday has 5 Sundays.

Day →	5	25
	1	5
	↓ × 410	↓ × 230

$$\therefore \text{Average} = \frac{410 \times 1 + 230 \times 5}{6} = 260$$

Ex. The average of 27 numbers is zero. Out of them, how many may be greater than zero, at the most?

HINTS

Maximum 26 numbers may be greater than zero and 27th number will be of opposite sign of the sum of 26 numbers.

Average

Ex. Eleven friends spent ₹18 each on a tour and the 12th friend spent ₹11 less than the average expenditure of all twelve of them. What is the total money spent by twelve friends?

HINTS Let, average = x

ATQ,

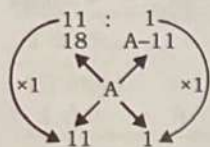
$$12x = 18 \times 11 + (x - 11)$$

$$\Rightarrow 12x = 198 + x - 11$$

$$\Rightarrow 11x = 187 \Rightarrow x = 17$$

$$\therefore \text{Total money} = 12 \times 17 = ₹204$$

Alternatively



$$A = 18 - 1 = 17$$

$$\therefore \text{Total money spent} = 17 \times 12 = ₹204$$

Ex. There are 75 students in the hostel. Due to new admissions, 15 new students join the mess and the daily expenses of the mess increased by ₹240, while the average expenditure per head diminished by ₹5. What was the original expenditure (in ₹) of the mess?

HINTS

Students Exp./Head Daily expenses

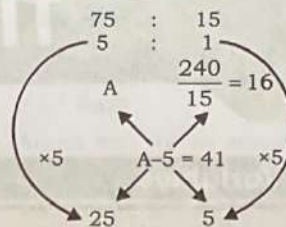
75	x	$75x$
90	$(x - 5)$	$90x - 450$

$$\Rightarrow 90x - 450 - 75x = 240$$

$$\Rightarrow 15x = 690 \Rightarrow x = 46$$

$$\therefore \text{Original expenditure} = 75 \times 46 = ₹3450$$

Alternatively



$$A - 5 = 41 \Rightarrow A = 46$$

$$\text{Original expenditure} = 75 \times 46 = ₹3450$$

Ex. The average weight of a group of 3 people A, B and C is 70 kg. When D joins this group, the average becomes 60 kg. One man E, whose weight is 5 kg more than that of D, replace A and the average weight of B, C, D and E now becomes 59 kg. A's weight (in kg) is:

HINTS $A + B + C = 70 \times 3 = 210$ kg

$$A + B + C + D = 60 \times 4 = 240$$
 kg

$$D = 240 - 210 = 30$$
 kg

$$E = 30 + 5 = 35$$
 kg

$$B + C + D + E = 59 \times 4 = 236$$
 kg

$$B + C = 236 - 30 - 35 = 171$$
 kg

$$A + B + C = 210$$
 kg

$$A = 210 - 171 = 39$$
 kg

Alternatively

$$D = (70 - 60) \times 3 = 30$$
 kg

$$E = 30 + 5 = 35$$
 kg

$$A - E = (60 - 59) \times 4$$

$$\Rightarrow A - E = 4$$
 kg

$$\Rightarrow A - 35 = 4$$
 kg

$$\Rightarrow A = 39$$
 kg

CHAPTER

24

TIME & WORK

Based on Two Variables

Ex. If A completes a piece of work in 'x' days and B completes the same work in 'y' days, then -

Total time taken by both A and B to complete the work

$$= \left(\frac{xy}{x+y} \right)$$

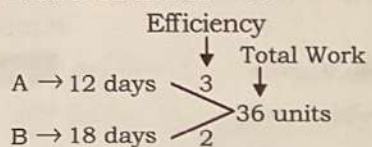
Ex. A can do a piece of work in 12 days whereas B can do the same work in 18 days. In how many days can A and B together complete the same work?

HINTS Time taken by A and B together to complete the

$$\text{work} = \frac{18 \times 12}{30} = 7 \frac{1}{5} \text{ days}$$

Alternatively

LCM of 12 and 18 = 36



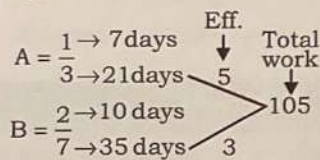
$$\therefore T_{(A+B)} = \frac{T.W.}{T.E.} = \frac{36}{5} = 7 \frac{1}{5} \text{ days}$$

Ex. If A can finish $\frac{m}{n}$ part of the work in D days. Then, total time taken to finish the work by A = $\frac{n}{m} \times D$ days

Ex. A can complete $\frac{1}{3}$ of a work in 7 days and B can complete $\frac{2}{7}$ of the same work in 10 days. In how many days can

both A and B together complete the work?

HINTS



$$\text{Time taken by (A + B)} = \frac{105}{8} = 13 \frac{1}{8} \text{ days}$$

Based on Three Variables

Ex. If A can do a work in 'x' days, B can do the same work in 'y' days and C can do the same work in 'z' days, then total time taken by A, B and C to complete the work together

$$= \frac{1}{\frac{1}{x} + \frac{1}{y} + \frac{1}{z}} = \frac{xyz}{xy + yz + zx}$$

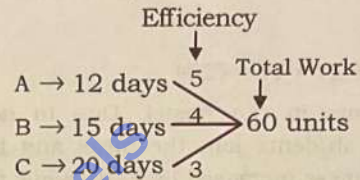
Ex. A, B and C can do a work in 12, 15 and 20 days respectively. In how many days will they complete the same work together?

HINTS Time taken by A, B and C together to complete the work

$$= \frac{12 \times 15 \times 20}{(12 \times 15) + (15 \times 20) + (20 \times 12)} = \frac{12 \times 15 \times 20}{720} = 5 \text{ days}$$

Alternatively

LCM of 12, 15 and 20 = 60



$$T_{(A+B+C)} = \frac{\text{Total Work}}{E_A + E_B + E_C} = \frac{60}{12} = 5 \text{ days}$$

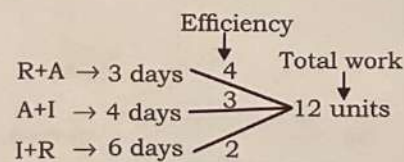
Ex. If A and B can do a work in 'x' days. B and C can do the same work in 'y' days. C and A can do the same work in 'z' days. Then total time taken when A, B and C work together = $\frac{2xyz}{xy + yz + zx}$ days

Ex. Raghav and Aditya can complete a work in 3 days. Aditya and Ishan can complete the same work in 4 days. Ishan and Raghav can complete it in 6 days. How many days will it take for Raghav, Aditya and Ishan to complete the same work, working together?

HINTS Time taken by Raghav, Aditya and Ishan to

$$\begin{aligned} \text{complete the work} &= \frac{2 \times 3 \times 4 \times 6}{(3 \times 4) + (4 \times 6) + (6 \times 3)} = \frac{2 \times 3 \times 4 \times 6}{54} \\ &= 2 \frac{2}{3} \text{ days} \end{aligned}$$

Alternatively



$$2(I+R+A) = 9 \text{ units}$$

Efficiency of (I+R+A) = $\frac{9}{2}$ unit

$$\therefore T_{I+R+A} = \frac{T.W.}{T.E.} = \frac{12 \times 2}{9} = \frac{8}{3} = 2 \frac{2}{3} \text{ days}$$

Concept of Alternate Days

Ex. A and B can complete a work in 10 days and 20 days respectively. A started the work and they work in alternate days. In how many days will the total work be completed?

HINTS

		Efficiency	Total Work
A → 10 days	2	↓	20 units
B → 20 days	1	↓	
Days	Man	Work	
1 st	A	2 units	
2 nd	B	1 unit	
2 days _{×6}	→	3 units _{×6}	
12 days	→	18 units	
1 day → A	→	2 units	
13 days	→	20 units	

Hence, work will be completed in 13 days.

Ex. If Mohit, Nitu and Soma can complete a piece of work in 5 days, 6 days and 8 days respectively, then in how many days will the work be completed if they work on alternate days starting with Mohit on first day, Nitu on second day, Soma on third day, Mohit on fourth day and so on?

HINTS LCM of 5, 6 and 8 = 120

		Efficiency	Total work
Mohit → 5 days	24	↓	120 units
Nitu → 6 days	20	↓	
Soma → 8 days	15	↓	
Days	Man	Work	
1 st	M	→ 24	
2 nd	N	→ 20	
3 rd	S	→ 15	
3 days _{×2}	→	59 units _{×2}	
6 days	→	118 units	
$\frac{1}{12}$ day	M	→ 2 units	
$6\frac{1}{12}$ days	→	120 units	

Hence, they will complete the work in $6\frac{1}{12}$ days.

Ex. Ratnesh and Vineet worked on alternate days, starting with Ratnesh. The entire work got completed in $54\frac{3}{4}$ days. If Ratnesh alone can finish the work in 48 days, then in how many days can Vineet alone do the work?

HINTS Total number of days = $54\frac{3}{4}$ days

Ratnesh	Vineet
$27\frac{3}{4}$ days	27 days

Let, efficiency of Ratnesh and Vineet be R & V respectively.

$$\therefore \text{Total work} = 48R = R \times \frac{111}{4} + 27V$$

$$\Rightarrow 81R = 108V$$

$$\Rightarrow \frac{R}{V} = \frac{4}{3} \text{ (Efficiency)}$$

$$\text{Total work} = 48 \times 4 = 192 \text{ unit}$$

$$\therefore \text{Time taken by Vineet alone to do the work} = \frac{192}{3}$$

$$= 64 \text{ days}$$

Ex. Aman and Rajan are working at a construction site. In some engineering experiment, Aman is constructing a wall while Rajan is demolishing that wall. Aman can completely build the wall in 15 days while Rajan will take 20 days to completely demolish it. In how many days will the complete wall be built for the first time if they work on alternate days, with Aman working on the 1st day?

HINTS LCM of 15 and 20 = 60

		Efficiency	Total Work
Aman → 15 days	4	↓	60 units
Rajan → 20 days	3	↓	
Days	Man	Work	
1 st	A	→ +4	
2 nd	R	→ -3	
2 days _{×6}	→	1 unit _{×6}	
12 days	→	56 units	
1 day → A	→	4 units	
13 days	→	60 units	

$\therefore$ Wall will be built for the first time in 13 days.

Ex. A and B are equally efficient and each could individually complete a piece of work in 30 days, if none took any holiday. A and B started working together on this piece of work but A took a day off after every four days of work while B took a day off after every five days of work. If the duo had started work on 01 August 2022, on which date was the work completed?

HINTS

$$\begin{matrix} A : B \\ \text{Efficiency} \rightarrow 1 : 1 \end{matrix}$$

$$\text{Total work} = 1 \times 30 = 30 \text{ units}$$

Days	Man	Work
4 days	A+B	8 units
5 th	B	1 unit
6 th	A	1 unit

6 days _{×3}	→	10 units _{×3}
18 days	→	30 units

So, work will be completed on 18 August 2022.

Concept of leaving the work

Leaving the work from forward

If A can do a piece of work in 'x' days and B can do the same work in 'y' days. If they work together for 'd' days, after that A leaves the work then the time taken by B to do the remaining work

$$= \frac{xy - (x + y) \times d}{x} \text{ days}$$

Time & Work

Ex. A can finish a piece of the work in 16 days and B can finish it in 12 days. They worked together for 4 days and then A left. B finished the remaining work. For how many days did B work to finish the work completely?

HINTS Time taken by B to finish the remaining work

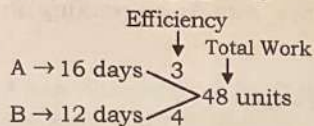
$$= \frac{16 \times 12 - (16 + 12) \times 4}{16} = \frac{192 - 112}{16} = \frac{80}{16} = 5 \text{ days}$$

∴ Total number of days in which B worked

$$= 4 + 5 = 9 \text{ days}$$

Alternatively

LCM of 16 and 12 = 48



Here A and B worked for 4 days and after that B had completed the remaining work. Let the number of days in which B worked alone be 't'.

$$\text{So, } 4(A+B) + B \times t = \text{T.W.}$$

$$\Rightarrow 28 + 4t = 48 \Rightarrow 4t = 20 \Rightarrow t = 5 \text{ days}$$

∴ Total number of days in which B worked = 4 + 5 = 9 days

Lefting the work from backward

A can do a piece of work in 'p' days and B can do the same work in 'q' days. They start the work together but A leaves 'd' days before the work is finished.

$$\text{Required time to complete the work} = \frac{pq + qd}{p + q} \text{ days}$$

Ex. A and B working alone can complete a work in 8 days and 12 days, respectively. They started working together but A left 2 days before completion of the work. In how many days was the work completed?

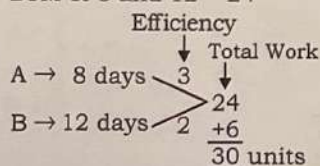
HINTS Required time to complete the work

$$= \frac{8 \times 12 + 12 \times 2}{(8 + 12)} = \frac{120}{20} = 6 \text{ days}$$

Alternatively

Tips: Add the work of left days (before completion of the work) in total work and solve using basic concept.

LCM of 8 and 12 = 24



Required time to complete the work

$$= \frac{\text{T.W.}}{\text{T.E.}} = \frac{30}{5} = 6 \text{ days}$$

Lefting the work from both forward and backward

A can do a piece of work in 'x' days, B in 'y' days and C in 'z' days. If all three of them start the work together but B leaves the work 'd' days before its completion and C leaves the work after 'D' days of starting, then required time to complete the work

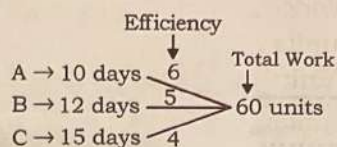
$$= \frac{xyz + zx \times d - xy \times D}{yz + zx} \text{ days}$$

Ex. A, B and C can do a work in 10, 12 and 15 days respectively. They begin the work together but after 3 days, C left the work and 4 days before the completion of work, B left the work. In how many days will the total work be completed?

HINTS Required time to complete the work

$$= \frac{10 \times 12 \times 15 + 15 \times 10 \times 4 - 10 \times 12 \times 3}{12 \times 15 + 15 \times 10} = \frac{2040}{330} = \frac{68}{11} \text{ days}$$

Alternatively



∴ Required time to complete the work

$$= \frac{\text{T.W.}}{\text{T.E.}} = \frac{60 - 3C + 4B}{A + B} = \frac{60 - 12 + 20}{11} = \frac{68}{11} \text{ days}$$

Concept of Efficiency

Efficiency of a machine or a person is the rate of work done in per unit time.

$$\text{Efficiency (E)} = \frac{\text{Work done}}{\text{Time taken}}$$

$$\text{Also, Efficiency (E)} \propto \frac{1}{\text{Time taken}}$$

Ex. If the efficiency of A is twice the efficiency of B, then ratio of time taken by A and B = 1 : 2

Ex. Tim is thrice as good a workman as Joya and they together finish a piece of work in 75 days. In how many days will Tim alone finish the work?

HINTS Tim : Joya
Efficiency → 3 : 1

$$\text{Total Work} = (E_T + E_J) \times \text{Time taken} = (3 + 1) \times 75 = 300 \text{ unit}$$

∴ Time taken by Tim alone to finish the work

$$= \frac{\text{Total Work}}{E_T} = \frac{300}{3} = 100 \text{ days}$$

MDH Concept

$$\frac{M_1 D_1 H_1}{W_1} = \frac{M_2 D_2 H_2}{W_2}$$

where,

M → Man, D → Days

H → Hours, W → Work/Wages

Ex. If 35 men can finish a work in 6 days, then in how many days can 7 men do half of the same work?

HINTS $\frac{M_1 D_1 H_1}{W_1} = \frac{M_2 D_2 H_2}{W_2}$

$$\Rightarrow \frac{35 \times 6}{2} = \frac{7 \times x}{1}$$

$$\Rightarrow x = 15 \text{ days}$$

Ex. 18 persons working 8 hours a day can complete 3 units of work in 10 days. How many days are required by 25 persons to complete 5 units of work working 6 hours a day?

HINTS $\frac{M_1 D_1 H_1}{W_1} = \frac{M_2 D_2 H_2}{W_2}$

$$\Rightarrow \frac{18 \times 10 \times 8}{3} = \frac{25 \times x \times 6}{5} \Rightarrow x = 16 \text{ days}$$

Ex. 64 men working 8 hours a day, plan to complete a piece of work in 9 days. However, 5 days later, they found that they had completed only 40% of the work. They now wanted to finish the remaining work in 4 days. How many hours per day should they need to work in order to achieve the target?

HINTS $\frac{M_1 D_1 H_1}{W_1} = \frac{M_2 D_2 H_2}{W_2} \Rightarrow \frac{64 \times 5 \times 8}{40\%} = \frac{64 \times 4 \times x}{60\%}$

$$\Rightarrow x = 15 \text{ hours}$$

Ex. A hostel had food provision for 300 students for a month. After 20 days, 50 students left the hostel. How long would the remaining food last? (1 month = 30 days)

HINTS Remaining time after 20 days = 30 - 20 = 10 days

Remaining students after 20 days = 300 - 50 = 250

$$\therefore M_1 D_1 = M_2 D_2$$

$$\Rightarrow 300 \times 10 = 250 \times D_2 \Rightarrow D_2 = 12$$

$\therefore$ Remaining food would last for 12 days.

Ex. (N + 18) persons, each working for 7.5 hours a day, can complete 48% of a work in 20 days. (N + 12) persons can complete the remaining work in 30 days if each of them works for 6.5 hours per day. Determine the value of N.

HINTS ATQ,

$$\frac{(N+18) \times 7.5 \times 20}{48\%} = \frac{(N+12) \times 6.5 \times 30}{52\%}$$

$$\Rightarrow (N+18)5 = 6(N+12) \Rightarrow 5N+90 = 6N+72$$

$$\Rightarrow N = 18$$

MWC Concept

Ex. 6 men or 2 women or 4 boys can finish a work in 77 days. How many days will 1 man, 1 woman and 1 boy together take to finish the same work?

HINTS Total Work $\Rightarrow 6M \times 77 = 2W \times 77 = 4B \times 77$

$$\Rightarrow 6M = 2W = 4B$$

$$M : W : B$$

$$2 : 6 : 3$$

Let, required number of days be 'x'.

$$\therefore (1M + 1W + 1B)x = 6M \times 77$$

$$\Rightarrow 11x = 6 \times 2 \times 77 \Rightarrow x = 84 \text{ days}$$

Ex. 3 men can finish a work in 10 days, 4 women can finish it in 12 days and 10 qualified workers can finish it in 6 days. In how many days is the work finished by 4 men, 4 women and 4 qualified workers, working together everyday?

HINTS ATQ,

$$\text{Total work} \Rightarrow 3M \times 10 = 4W \times 12 = 10L \times 6$$

$$\Rightarrow 5M = 8W = 10L$$

$$M : W : L$$

$$8 : 5 : 4$$

$$\text{Total work} = 30M = 30 \times 8 = 240 \text{ unit}$$

$$\therefore \text{Time taken by } 4M + 4W + 4L = \frac{240}{4 \times 17} = \frac{60}{17} \text{ days}$$

Concept of contractor

Ex. A contractor decided to complete a work in 80 days and employed 60 men at the beginning and 20 men additionally after 20 days and got the work completed as per schedule. If he had not employed the additional men, how many extra days would he have needed to complete the work (round off to the nearest integer)?

HINTS 60 Men $\rightarrow$ 20 days

80 Men $\rightarrow$ 60 days

$$\text{Total work} = (60 \times 20) + (80 \times 60) = 6000$$

$$\therefore \text{Required days} = \frac{6000}{60} = 100 \text{ days}$$

$$\text{Additional days} = 100 - 80 = 20 \text{ days}$$

Alternatively

$$60M \times 20 + 80M \times 60 = 60M \times n$$

$$\Rightarrow 6000 = 60n \Rightarrow n = 100$$

$$\therefore \text{Additional days} = 100 - 80 = 20 \text{ days}$$

Concept of AP in time and work

Ex. A group of men decided to complete a job in 4 days. However, since 10 men dropped out everyday, the job got completed at the end of the 7th day. How many men were there in the beginning?

HINTS Total work $\Rightarrow (x) + (x-10) + \dots + (x-60) = 4x$

$$\Rightarrow 7x - 210 = 4x \Rightarrow 3x = 210$$

$$\Rightarrow x = 70$$

$\therefore$ There were 70 men in the beginning.

Ex. Each one of the 5 men independently can complete a work in 20 days. The work is started by one person. Next day, one more person joins and every next day, one more person joins. From the fifth day, five persons continued working as a team. In how many days will the work be completed?

HINTS Let the efficiency of each man be 1 unit.

Total work = 20 units

Day $\rightarrow$ 1st 2nd 3rd 4th 5th 6th

Men $\rightarrow$ 1 2 3 4 5 5

15 men = 15 units

Remaining work = 20 - 15 = 5 unit

Time in which remaining work is completed

$$= \frac{5}{5} = 1 \text{ day}$$

$$\therefore \text{Total time} = 5 + 1 = 6 \text{ days}$$

Miscellaneous

If A working alone takes 'x' days more than A & B and B working alone takes 'y' days more than A & B, then the number of days taken by A & B working together to complete the same job = $\sqrt{xy}$ days

Ex. A and B working together can complete a work in 'd' days. Working alone, A takes (8+d) days and B takes (18+d) days to complete the same work. Find the value of 'd'.

HINTS $d = \sqrt{18 \times 8} = \sqrt{144} = 12$ days

Ex. When A alone does a piece of work, he takes 25 days more than the time taken by A and B to do the work together. On the other hand, B alone takes 16 days more than the time taken by A and B to do the work together. How many days will A and B, working together, take to do the work?

HINTS

Required time in which A and B together complete the work = $\sqrt{25 \times 16} = 20$ days

Ex. In an office, 75 employees can finish a project in 60 days. After few days, 15 employees left the job and the project was completed in 65 days. After how many days did 15 employees leave?

HINTS Let, 15 employees left after 'n' days.

ATQ,

$$75 \times 60 = (75 \times n) + 60 \times (65 - n)$$

$$\Rightarrow 4500 = 75n + 3900 - 60n \Rightarrow 600 = 15n$$

$$\Rightarrow n = 40 \text{ days}$$

Ex. A work was finished by L, P and Q together. L and P together finished 50 percent of the work and P and Q together finished 60 percent of the work. Who among the three is the most efficient?

HINTS Let, total work = 100 unit

$\therefore$ L and P together finished 50% work.

$\therefore Q = 100 - 50 = 50$ unit

$\therefore$ P and Q together finished 60% work.

$\therefore L = 100 - 60 = 40$ unit

So, $P = 100 - (50 + 40) = 10$ unit

$\therefore Q$ is the most efficient among all.

Ex. Four persons, P, Q, R, S were engaged for doing a task with the condition that P; Q; R; S work respectively on (Monday, Thursday); (Tuesday, Friday); (Wednesday, Saturday); (Sunday). The task was began on Monday and got completed on the 15th day which was also a Monday. If the efficiencies of P, Q, R, S in respect of doing this task were in the proportion 1 : 2 : 3 : 4, then in how many days could R complete the task, working alone without break?

HINTS Given,

Efficiencies of P, Q, R, S = 1 : 2 : 3 : 4

Total work of 7 days = $1 + 2 + 3 + 1 + 2 + 3 + 4 = 16$ unit

Total work of 15 days = $16 \times 2 + 1 = 33$ unit

$\therefore$ Time taken by R = $\frac{33}{3} = 11$ days

Work and Wages

Ex. Total wages = Per day wage $\times$ Number of working days
Ex. Wage is shared among the persons in the ratio of their efficiencies.

Ex. A and B worked together and received a total of Rs. 18,000 for 15 days. A's efficiency in the work was 5 times that of B's. The daily wage of A (in Rs.) was:

HINTS Ratio of efficiency of A and B = 5 : 1

$$\text{Per day wages of A and B} = \frac{\text{Total sum}}{\text{Time taken}}$$

$$= \frac{18000}{15} = \text{Rs. } 1200$$

$$\text{Daily wage of A} = 1200 \times \frac{5}{6}$$

$$= \text{Rs. } 1000$$

Ex. A and B undertake a project worth Rs. 36,000. A alone can do the work in 25 days. They worked together for 5 days. For the next five days, B worked alone. After that, A substituted B and completed the remaining work in 5 days. The share of A in the earnings is:

HINTS ATQ,

$$5(A + B) + 5B + 5A = 36000 \Rightarrow 10(A + B) = 36000$$

$\therefore$ A can do a work in 25 days.

$$\therefore \text{Amount received by A in 1 day} = \text{Rs. } \frac{36000}{25}$$

Hence, amount received by A in 10 days

$$= \frac{36000}{25} \times 10 = 36 \times 4 \times 100$$

$$= \text{Rs. } 14400$$

Alternatively

ATQ,

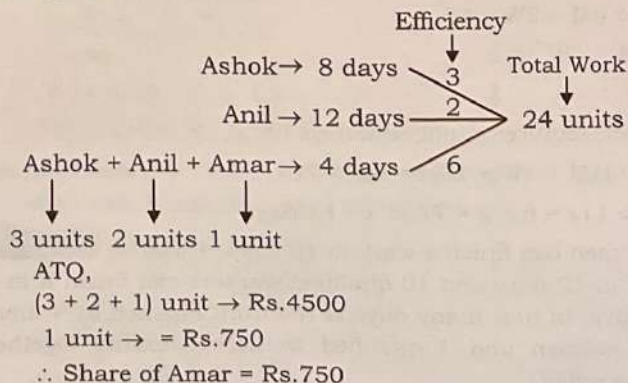
$$\text{Total work} \Rightarrow 5(A + B) + 5B + 5A = 25A$$

$$\Rightarrow 10A + 10B = 25A \Rightarrow \frac{A}{B} = \frac{2}{3}$$

$$\therefore \text{A's share} = \frac{3600}{5} \times 2 = \text{Rs. } 14400$$

Ex. Ashok and Anil undertake to do a piece of work for Rs. 4,500. Ashok alone could do the work in 8 days and Anil in 12 days. With the assistance of Amar, they finished the work in 4 days. What is the share of Amar?

HINTS



CHAPTER

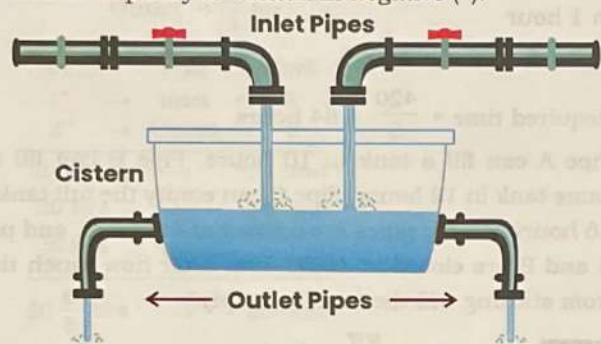
25

PIPE & CISTERN

Amount of water released or filled = Rate $\times$ time.

Inlet: The pipe that fills the tank is called inlet and its capacity is written in positive (+).

Outlet: The pipe that empties the tank is called outlet and its capacity is written in negative (-).



Completely Filling or Emptying

Two taps 'A' and 'B' can fill or empty a tank in 'x' hours and 'y' hours respectively. If both the taps are opened simultaneously, how much time will it take to fill the tank?

$$\text{Required time} = \left(\frac{xy}{x+y} \right) \text{ hrs}$$

Ex. Two pipes X and Y can fill a tank in 14 hours and 21 hours respectively. Both pipes are opened simultaneously to fill the tank. In how many hours will the empty tank be filled?

HINTS Required time = $\frac{14 \times 21}{14 + 21} = \frac{42}{5} = 8 \frac{2}{5}$ hours

Alternatively

X $\rightarrow$ 14 hours $\downarrow$ 3
 Y $\rightarrow$ 21 hours $\downarrow$ 2
 Total capacity $\downarrow$ 42
 Time required to fill the tank
 $= \frac{42}{3+2} = \frac{42}{5} = 8 \frac{2}{5}$ hours

Ex. Pipe P can empty a tank alone in 15 hours. Pipe Q can empty the same tank alone in 10 hours. In how much time can they together empty the tank?

HINTS
 P $\rightarrow$ 15 hrs $\downarrow$ 2
 Q $\rightarrow$ 10 hrs $\downarrow$ 3
 Total capacity $\downarrow$ 30
 Required time = $\frac{30}{2+3} = 6$

Ex. Pipes A, B and C can fill a tank in 20, 30 and 60 hours respectively. Pipes A, B and C are opened at 7 a.m., 8 a.m. & 9 a.m. respectively on the same day. When will the tank be full?

HINTS
 7AM A $\rightarrow$ 20 hrs $\downarrow$ 3
 8AM B $\rightarrow$ 30 hrs $\downarrow$ 2
 9AM C $\rightarrow$ 60 hrs $\downarrow$ 1
 Tank capacity $\downarrow$ 60

Tank filled by A till 9 a.m. = $2 \times 3 = 6$ unit

Tank filled by B till 9 a.m. = $2 \times 1 = 2$ unit

Remaining capacity = $60 - (6 + 2) = 52$ unit

Required time = $\frac{52}{6} = 8 \frac{2}{3}$ hrs = 8 hrs 40 min

Time = 9 a.m. + 8 hrs 40 min = 5 : 40 p.m.

Filling and Emptying Simultaneously

Ex. Tap 'A' can fill a tank in 'x' hours and tap 'B' can empty the tank in 'y' hours. How much time will it take to fill the tank when both the taps are opened simultaneously.

$$\text{Required time} = \left(\frac{xy}{y-x} \right) \text{ hrs } [y > x]$$

Ex. A tap can fill a cistern in 10 minutes and another tap can empty it in 12 minutes. If both the taps are open, the time (in hours) taken to fill the tank will be:

HINTS Required time = $\frac{12 \times 10}{12 - 10} = \frac{60}{1}$
 $= 60 \text{ min.} = 1 \text{ hr}$

Alternatively

P' $\rightarrow$ 10 min $\downarrow$ 6
 Q' $\rightarrow$ 12 min $\downarrow$ -5
 Total capacity $\downarrow$ 60
 Eff. of P and Q = $6 - 5 = 1$

Time taken by P and Q to fill the tank = $\frac{60}{1} = 60 \text{ min.} = 1 \text{ hrs.}$



Positive result shows that the tank is filling and Negative result shows that the tank is getting empty.

Ex. Time taken to empty the tank when both are opened.

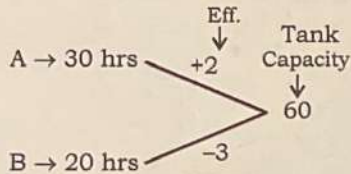
$$= \left(\frac{xy}{x-y} \right) \text{ hrs } [x > y]$$

Pipe & Cistern

Ex. Pipe A can fill an empty tank completely in 30 hours. Pipe B can empty this fully filled tank in 20 hours. If both the pipes are opened simultaneously, in what time will the full tank be emptied?

HINTS Time taken to empty the tank = $\frac{30 \times 20}{30 - 20} = \frac{600}{10} = 60 \text{ hrs}$

Alternatively



Time taken to empty the tank = $\frac{60}{1} = 60 \text{ hrs}$

Ex. If x, y, z,..... all taps are opened together, then the time required to fill/empty the tank will be:

$$\frac{1}{x} \pm \frac{1}{y} \pm \frac{1}{z} \pm \dots = \frac{1}{T}$$

where T, is the required time.

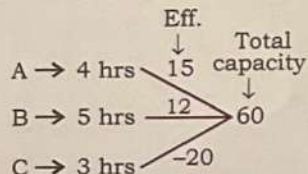
Ex. Pipe A can fill a cistern in 4 hours and Pipe B can fill the same cistern in 5 hours. Pipe C can empty a full cistern in 3 hours. If all three pipes are opened together, then the time (in minutes) taken to fill the cistern is: (round off to the nearest integer)

HINTS $\frac{1}{T} = \frac{1}{4} + \frac{1}{5} - \frac{1}{3} = \frac{15 + 12 - 20}{60} = \frac{7}{60} \Rightarrow T = \frac{60}{7}$

Time taken by all three pipe to fill the tank = $\frac{60}{7} \text{ hrs}$

= $\frac{60 \times 60}{7} \sim 514 \text{ min}$

Alternatively

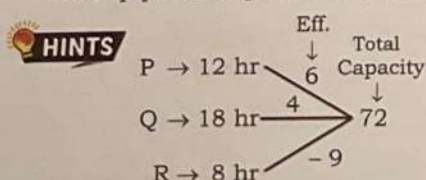


A + B + C = 15 + 12 - 20 = 7

Time taken by all three pipe to fill the tank = $\frac{60}{7} \text{ hrs}$

= $\frac{60 \times 60}{7} \sim 514 \text{ min}$

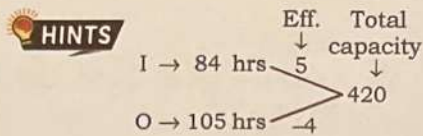
Ex. Two pipes can fill a tank in 12 hours and 18 hours, respectively, while a third pipe can empty it in 8 hours. How long (in hours) will it take to fill the empty tank if all three pipes are opened simultaneously?



Time taken by them to fill the empty tank = $\frac{72}{6 + 4 - 9}$

= 72 hrs.

Ex. Each inlet pipe can fill an empty cistern in 84 hours while each drain pipe can empty the same cistern from a filled condition in 105 hours. When the cistern is empty, 9 inlet pipes and 10 outlet pipes are simultaneously opened. After how many hours will the cistern be completely filled?

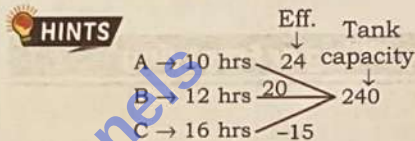


Part of the tank filled by 9 Inlet pipe and 10 Outlet pipe in 1 hour

= $[9 \times 5 + 10 \times (-4)] = 5$

Required time = $\frac{420}{5} = 84 \text{ hours}$

Ex. Pipe A can fill a tank in 10 hours. Pipe B can fill the same tank in 12 hours. Pipe C can empty the full tank in 16 hours. All the pipes are opened at 8:00 a.m. and pipe A and B are closed at 10:00 a.m. After how much time from starting will the tank be empty?



All pipe are opened at 8:00 a.m. and pipe (A and B) are closed at 10:00 a.m.

So, $(24 + 20 - 15) \times 2 \text{ hr} = (44 - 15) \times 2$

= $29 \times 2 = 58 \text{ unit}$

Time taken by C to empty the tank

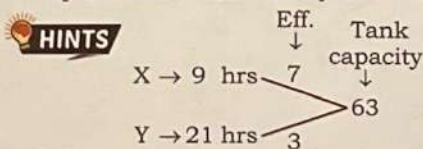
= $\frac{58}{15} \text{ hr} = 3 \text{ hr } \frac{13}{15} \times 60 \text{ min}$

= 3 hr 52 min

So, Total Time = (2 hr + 3 hr 52 min) = 5 hr 52 min

When Taps Opened Alternately

Ex. Pipe X can fill a tank in 9 hours and Pipe Y can fill it in 21 hours. If they are opened on alternate hours and Pipe X is opened first, in how many hours shall the tank be full?



Starting from 'X'

Hour	Pipe	Work
1 st	→ X	→ 7
2 nd	→ Y	→ 3

2 hr. × 6 → 10 × 6 unit

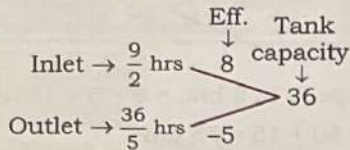
12 hr. → 60 unit

$\frac{3}{7}$ hr. → X → 3 unit

$12 \frac{3}{7}$ hr. → 63 unit

∴ Total time = $12 \frac{3}{7}$ hours.

Ex. An inlet pipe can fill an empty tank in $4\frac{1}{2}$ hours while an outlet pipe drains a completely filled tank in $7\frac{1}{5}$ hours. The tank is initially empty, and the two pipes are alternately opened for an hour each, till the tank is completely filled, starting with the inlet pipe. Then in how much time the tank will be full.

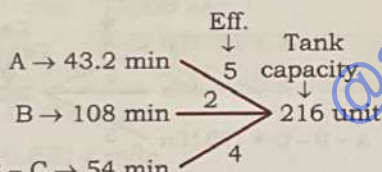
HINTS

Hours	Pipe	Work
1 st	$\rightarrow$ Inlet	$\rightarrow$ 8
2 nd	$\rightarrow$ Outlet	$\rightarrow$ -5
2 hrs $\times 10$	$\rightarrow$	3×10 unit
20 hrs	$\rightarrow$	30 unit
$\frac{6}{8}$ hrs	$\rightarrow$ Inlet	$\rightarrow$ 6 unit
$20\frac{6}{8}$ hrs	$\rightarrow$	36 unit

Time taken to fill the tank completely = $20\frac{6}{8} = 20\frac{3}{4}$ hrs

To find the capacity of tank

Ex. Pipe A and B fill a tank in 43.2 minutes and 108 minutes respectively, Pipe C can empty it at 3 litres/minutes. When all the three pipes are opened together, they will fill the tank in 54 minutes. The capacity (in litres) of the tank is:

HINTS

$\therefore$ Efficiency of C = $5 + 2 - 4 = 3$ unit/min
 i.e., C empty 3 units in 1 min.

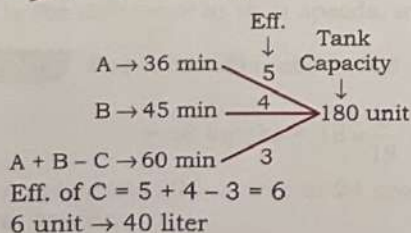
3 unit = 3 litre

Therefore capacity of tank = $216 \times 1 = 216$ liter

Ex. Two pipes can fill a tank separately in 36 minutes and 45 minutes respectively. A drain pipe installed in the tank can remove 40 liters of water per minute. If all three pipes are opened together, the tank is filled in one hour. Find the capacity/holding (in litres) of the tank.

HINTS

Let drain pipe is C.



Eff. of C = $5 + 4 - 3 = 6$

6 unit $\rightarrow$ 40 liter

180 unit $\rightarrow \frac{40}{6} \times 180 = 1200$ liter

$\therefore$ Capacity of tank = 1200 liter

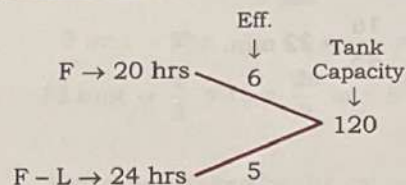
Leakage in tank

If a pipe fills a tank in 'x' hours but due to leakage it takes 't' hours more to fill tank. If tank is filled completely, then in how many hours it will be empty? [due to leakage outlet]

$$\text{Required time} = \frac{x(x+t)}{t} \text{ hours}$$

Ex. If a pipe fills a tank in 20 hours but it takes 4 hours more to fill it due to leakage in the tank. If the tank is filled completely, then in how many hours it will be empty?

$$\text{Required time} = \frac{20 \times (20+4)}{4} = 120 \text{ hrs}$$

Alternatively

Efficiency of leakage (L) = $6 - 5 = 1$

Time taken to empty the tank due to leakage

$$= \frac{120}{1} = 120 \text{ hrs}$$

Ex. Pipe A can fill a tank in 12 hours and pipe B takes 18 hours to fill it. Both pipes were opened together and a leak was spotted which increased the filling-up time by 48 minutes. Find how many hours will it take for the leak to empty a full tank.

$$\text{HINTS } t_{(A+B)} = \frac{12 \times 18}{12+18} = 7.2 = 7 \text{ hrs. } 12 \text{ min.}$$

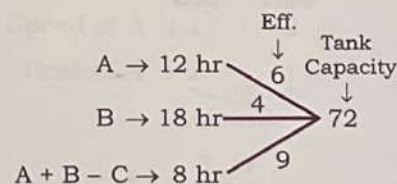
$$\text{Required time} = \frac{x(x+t)}{t}$$

$$= \frac{7.2 \left(7.2 + \frac{48}{60} \right)}{\frac{48}{60}} = \frac{7.2 \times 8}{0.8} = 72 \text{ hr.}$$

Alternatively

$$t_{(A+B)} = \frac{12 \times 18}{12+18} = 7.2 = 7 \text{ hrs. } 12 \text{ min.}$$

$$A + B - C = 7 \text{ hrs } 12 \text{ min} + 48 \text{ min} = 8 \text{ hrs}$$



Efficiency of leak = $6 + 4 - 9 = 1$

$$\text{Required time} = \frac{72}{1} = 72 \text{ hrs.}$$

When 1 or 2 taps are closed during filling or emptying a tank

Ex. Two taps A and B can fill a tank in x hours and y hours respectively. If both the pipes are opened together, then the time after which pipe B should be closed so that the tank is full in t hours

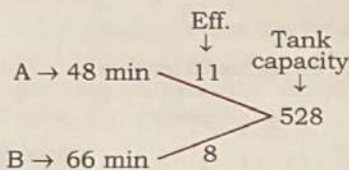
$$\text{Required time} = \left[y \left(1 - \frac{t}{x} \right) \right] \text{ hours}$$

Ex. Two pipes A and B can fill a tank in 48 minutes and 66 minutes, respectively. If both the pipes are opened simultaneously, then after how many minutes should pipe B be closed so that the tank gets filled in 32 minutes?

HINTS Required time = $y \left(1 - \frac{t}{x} \right)$

$$= 66 \left(1 - \frac{32}{48} \right) = 66 \times \frac{16}{48} = 22 \text{ min.}$$

Alternatively



Tank filled by pipe A in 32 min = $32 \times 11 = 352$ unit
Remaining part = $528 - 352 = 176$ unit

$$\therefore \text{Time taken by pipe B} = \frac{176}{8} = 22 \text{ min}$$

Then, after 22 min pipe B should be closed.

Ex. Two taps A and B can fill a tank in x hours and y hours respectively. If both the taps are opened together and tap A is closed before z hours of filling the tank, then in how much time the tank will be filled.

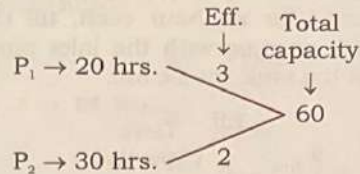
$$\text{Required time} = \frac{(x+z)y}{(x+y)}$$

• If tap B is closed before z hours of filling the tank then

$$\text{Required time} = \frac{(y+z)x}{(x+y)}$$

Ex. Two inlet pipes, P_1 and P_2 , can fill a cistern in 20 hours and 30 hours, respectively. They were opened at the same time, but pipe P_1 had to be closed 5 hours before the cistern was full. How many hours in total did it take for the two pipes to fill the cistern?

HINTS



Tank filled by pipe P_1 in 5 hrs. = $3 \times 5 = 15$ unit

Total capacity = $60 + 15 = 75$ unit

$$\text{Time taken by them to fill the cistern} = \frac{75}{3+2} = \frac{75}{5} = 15 \text{ hrs.}$$

Alternatively

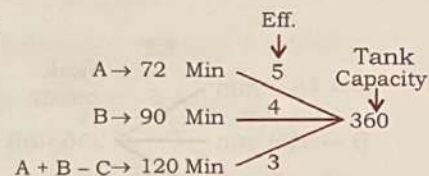
$$\text{Required time} = \frac{(x+z)y}{(x+y)}$$

$$= \frac{(20+5) \times 30}{20+30} = \frac{25 \times 30}{50}$$

= 15 hrs

Ex. Pipe A and B are filling pipes while C is an emptying pipe. A and B can fill a tank in 72 and 90 minutes respectively. When all the three pipes are opened together, the tank gets filled in 2 hours. A and B are opened together for 12 minutes, then closed and C is opened. The tank will be empty after.

HINTS



Efficiency of C = $(5 + 4 - 3) = 6$

Tank filled by pipe A and B in 12 minutes = $(5 + 4) \times 12 = 108$ units

$$\text{Time taken by pipe C to empty tank} = \frac{108}{6} = 18 \text{ min}$$

CHAPTER

26

TIME & DISTANCE

Basic Concepts

- Speed = $\frac{\text{Distance}}{\text{Time}}$
- When distance is constant, then:

$$\text{Speed} \propto \frac{1}{\text{Time}}$$

- When time is constant, then:
- When speed is constant, then:

$$\text{Distance} \propto \text{Speed}$$

$$\text{Distance} \propto \text{Time}$$

Units of Measurement

- Time is usually measured in second (sec), minute (min) or hour (hr).
- Distance is usually measured in meter (m), kilometer (km), mile, yard or feet.
- Speed is usually measured in km/hr, mile/hr or m/sec.

S.No.	Conversion of Units
(i)	1 hr = 60 min = 3600 sec
(ii)	1 km = 1000 m
(iii)	1 mile = 1.609 km
	1 km = 0.6214 mile
	5 mile = 8 km
(iv)	1 yard = 3 feet
(v)	$a \text{ km/hr} = a \times \frac{1000}{60 \times 60} \text{ m/sec} = a \times \frac{5}{18} \text{ m/sec}$
(vi)	$a \text{ m/sec} = a \times \frac{60 \times 60}{1000} \text{ km/hr} = a \times \frac{18}{5} \text{ km/hr}$

Ex. Dimple is travelling at a speed of 90 km/hr on a highway, while Sachin is travelling at a speed of 108 km/hr. What is the difference in their speeds, in m/sec?

HINTS Required difference = $(108 - 90) \text{ km/hr}$
 $= 18 \text{ km/hr} = 18 \times \frac{5}{18} \text{ m/sec} = 5 \text{ m/sec}$

Ex. A man runs 200 metres in 24 seconds. Find his speed in km/hr.

HINTS Speed = $\frac{200}{24} = \frac{25}{3} \text{ m/s}$
 $= \frac{25}{3} \times \frac{18}{5} \text{ km/hr} = 30 \text{ km/hr}$

When speed is increased/decreased

Ex. A car when increases its usual speed by 30%, it would reach certain destination 2 hours early. Find the usual time taken by the car to cover the same distance.

HINTS

	Old	:	New
Speed →	10	:	13
Time →	13	:	10
	Diff. 3 unit		

∴ 3 unit → 2hr

$$13 \text{ unit} \rightarrow \frac{2}{3} \times 13 = \frac{26}{3} \text{ hr} = 8 \frac{2}{3} \text{ hr}$$

∴ Usual time taken by the car = $8 \frac{2}{3} \text{ hr}$

$$= 8 \text{ hr} + \frac{2}{3} \times 60 \text{ min} = 8 \text{ hr } 40 \text{ min}$$

Ex. A vehicle is going at one-fourth of its usual speed and it takes an extra 45 minutes to reach its destination. Find the usual time taken by the vehicle to cover the same distance.

HINTS

	Old	:	New
Speed →	4	:	1
Time →	1	:	4
	Diff. 3 unit		

∴ 3 unit → 45 min

1 unit → 15 min

∴ Usual time taken by the vehicle = 15 min

Ex. A takes 2 hours 30 minutes more than B to walk 40 km. If A doubles his speed, then he can make it in 1 hour less than B. How much time (in hours) does A require for walking a 40 km distance?

HINTS

	Time _B	
	+2hr 30min	-1hr
	= $\frac{5}{2} \text{ hr}$	
	Time _A (old)	Time _A (new)

	Old	:	New
Speed of A →	1	:	2
Time of A →	2	:	1
	Diff. 1 unit		

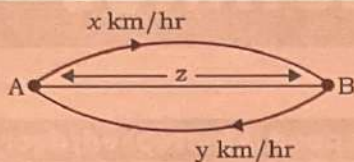
$$\therefore 1 \text{ unit} \rightarrow 1 + \frac{5}{2} = \frac{7}{2} \text{ hr}$$

$$2 \text{ unit} \rightarrow 2 \times \frac{7}{2} = 7 \text{ hr}$$

∴ Time taken by A to cover the given distance = 7hr

Time & Distance

When distance is constant and total time is given



A man travels a distance 'z' from point A to point B with a speed of x km/hr and comes back from point B to point A with a speed of y km/hr. If he takes 't' hours to go and come back, then the distance 'z' travelled by him:

$$\text{Distance 'z'} = \left(\frac{xy}{x+y} \right) \times t$$

Ex. A student goes from home to school at a speed of $5\frac{1}{2}$ km/hr and returns to home at a speed of 4 km/hr. If he takes $4\frac{3}{4}$ hours for the entire journey, then find the distance between home and school (in km).

HINTS Distance = $\left(\frac{\frac{11}{2} \times 4}{\frac{11}{2} + 4} \right) \times \frac{19}{4} = \frac{44}{19} \times \frac{19}{4} = 11$ km

Alternatively

Old	New
Speed $\rightarrow 11$	: 8
Time $\rightarrow 8$	: 11
Sum	
19 unit	

$$\therefore 19 \text{ unit} \rightarrow \frac{19}{4} \text{ hr}$$

$$1 \text{ unit} \rightarrow \frac{1}{4} \text{ hr}$$

$$8 \text{ unit} \rightarrow 2 \text{ hr}$$

$\therefore$ Distance between home and school

$$= \frac{11}{2} \times 2 = 11 \text{ km}$$

Ex. A person travels from one place to another at 60 km/hr and returns at 75 km/hr. If the total time taken is 6 hours, then find the total distance travelled.

HINTS Let, Total distance = $2d$

ATQ,

$$6 = \frac{d}{75} + \frac{d}{60} \Rightarrow 6 = \frac{9d}{300} \Rightarrow d = 200$$

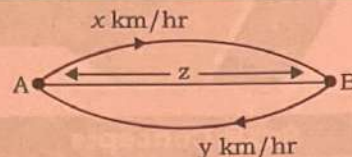
$$\therefore \text{Total distance} = 2 \times 200 = 400 \text{ km}$$

Alternatively

$$\text{Total Distance} = \frac{2 \times S_1 \times S_2}{S_1 + S_2} \times t$$

$$\text{Total Distance} = \frac{2 \times 60 \times 75}{135} \times 6 = 400 \text{ km}$$

When distance is constant and difference of time is given



A man travels a distance 'z' with a speed of x km/hr from point A to point B and comes back to point A with a speed of y km/hr. If he takes 't' hours more to come back from B to A than to travel from A to B, then the distance 'z':

$$\text{Distance 'z'} = \left(\frac{xy}{x-y} \right) \times (\text{difference between time})$$

The difference between time can be solved by the following tricks:

'Early-early' case	- (Subtraction)
'Late-late' case	- (Subtraction)
'Early-late' case	+ (Addition)
'Late-early' case	+ (Addition)

Ex. Ramesh drives from his home at a speed of 40 km/hr and reaches his college 25 minutes late. The next day, he increases his speed by 10 km/hr, yet he is late by 10 minutes. How far is his college from his home?

HINTS Distance = $\frac{40 \times 50}{50 - 40} \times \left(\frac{25 - 10}{60} \right) = \frac{40 \times 50}{10} \times \frac{15}{60} = 50$ km

Alternatively

Old	New
Speed $\rightarrow 4$	: 5
Time $\rightarrow 5$	: 4
Diff.	
1 unit	

$$\therefore 1 \text{ unit} \rightarrow (25 - 10) \text{ min} = 15 \text{ min}$$

$$5 \text{ unit} \rightarrow 75 \text{ min}$$

$$\therefore \text{Distance} = 40 \times \frac{75}{60} = 50 \text{ km}$$

Ex. A train covers a certain distance at a certain speed. If the speed of the train is 6 km/hr more, then to cover the same distance, it takes 6 hours less. But, if the speed of the train is 6 km/hr less, then it would take 10 hours more to cover the same distance. Find the distance covered by the train.

HINTS Let, original speed of train = s

ATQ,

$$\frac{s(s+6)6}{6} = \frac{s(s-6)}{6} \times 10$$

$$\Rightarrow 3s + 18 = 5s - 30 \Rightarrow 2s = 48$$

$$\Rightarrow s = 24 \text{ km/h}$$

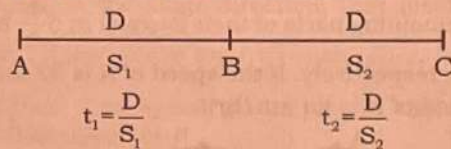
$$\therefore \text{Total distance covered by train} = \frac{24 \times 30}{6} \times 6 = 720 \text{ km}$$

Average speed

$$\text{Average speed} = \frac{\text{Total distance covered}}{\text{Total time taken}}$$

Cases for Average Speed:

1. When distance is constant.
2. When distance is different.
3. When distance is given in fraction.

Case-I: When distance is constant

$$\text{Average speed} = \frac{D + D}{\frac{D}{S_1} + \frac{D}{S_2}} = \frac{2D}{D \left(\frac{1}{S_1} + \frac{1}{S_2} \right)}$$

$$= \frac{2}{\left(\frac{1}{S_1} + \frac{1}{S_2} \right)} = \frac{2S_1S_2}{S_1 + S_2} \quad (\text{when two distance is equal})$$

- If 'n' distance is equal, then:

$$\text{Average speed} = \frac{n}{\left(\frac{1}{S_1} + \frac{1}{S_2} + \frac{1}{S_3} + \dots + \frac{1}{S_n} \right)}$$

From above explanation, it is clear that when distance is constant, then average speed is free from distance. It means that there is no need of distance or question can be solved through any supposed distance.

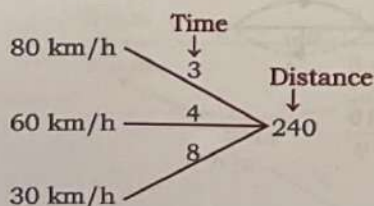
Ex. A car covers a certain distance travelling at a speed of 60 km/hr and returns to the starting point at a speed of 40 km/hr. The average speed of the car for the entire journey is:

HINTS Average speed of car = $\frac{2 \times 60 \times 40}{60 + 40} = 48 \text{ km/hr}$

Ex. A person divides his total route of journey into three equal parts and decides to travel the three parts at the speed of 80 km/hr, 60 km/hr and 30 km/hr, respectively. What is the average speed during the journey?

HINTS

$$\text{Average speed} = \frac{3}{\frac{1}{80} + \frac{1}{60} + \frac{1}{30}} = \frac{3 \times 240}{3 + 4 + 8} = \frac{3 \times 240}{15} = 48 \text{ km/hr}$$

Alternatively

$$\therefore \text{Average speed} = \frac{240 \times 3}{15} = 48 \text{ km/hr}$$

Case-II: When distance is different

Ex. A person completes 80 km of a journey at a speed of 10 km/hr and the remaining 70 km in 7 hours. His average speed for the whole journey is:

HINTS

$$\text{Average Speed} = \frac{\text{Total distance}}{\text{Total time}} = \frac{80 + 70}{\frac{80}{10} + 7} = \frac{150}{15} = 10 \text{ km/hr}$$

Ex. A person covers 11 km with a speed of 7 km/hr, 25 km with a speed of 10 km/hr and 30 km with a speed of 20 km/hr. What is the average speed of the person?

HINTS

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}} = \frac{11 + 25 + 30}{\frac{11}{7} + \frac{25}{10} + \frac{30}{20}} = \frac{66}{\frac{220 + 350 + 210}{140}} = \frac{66 \times 140}{780} = 11 \frac{11}{13} \text{ km/hr}$$

Case-III: When distance is given in fraction

If distance is in fraction $\left(\frac{1}{x}, \frac{1}{y}, \frac{1}{z}, \dots \right)$ and speed is $S_1, S_2, S_3, \dots$, then:

$$\text{Average speed} = \frac{1}{\left(\frac{1}{xS_1} + \frac{1}{yS_2} + \frac{1}{zS_3} + \dots \right)}$$

Ex. A person covers $\frac{1}{4}$ of his journey at the speed of 25 km/hr, $\frac{1}{2}$ of the journey at 40 km/hr and the remaining at the speed of 50 km/hr. Find his average speed for the whole journey (in km/hr).

HINTS

$$\text{Average speed} = \frac{1}{\frac{1}{4 \times 25} + \frac{1}{2 \times 40} + \frac{1}{4 \times 50}} = \frac{400}{4 + 5 + 2} = \frac{400}{11} = 36 \frac{4}{11} \text{ km/hr}$$

Ex. Vikas covered a certain distance by bike. If he covers 40% of the distance at 40 km/hr, 50% of the distance at 25 km/hr and the remaining 10% distance at 10 km/hr. Find his average speed over the whole distance.

HINTS

$$40\% = \frac{2}{5}, 50\% = \frac{1}{2}, 10\% = \frac{1}{10}$$

$$\text{Average speed} = \frac{1}{\frac{2}{5 \times 40} + \frac{1}{2 \times 25} + \frac{1}{10 \times 10}} = \frac{100}{1 + 2 + 1} = 25 \text{ km/hr}$$

Alternatively

Let, total distance = 100 km

$$\therefore \text{Average speed} = \frac{100}{\frac{40}{40} + \frac{50}{25} + \frac{10}{10}} = 25 \text{ km/hr}$$

Average speed with stoppage

When stoppage time is taken into consideration while calculating average speed, it is said **average speed with stoppage**.

Excluding stoppage, the average speed of a train is 'u' and with stoppage, its average speed is 'v', then:

$$\text{Stoppage time per hour} = \frac{\text{Diff. between their average speed}}{\text{Speed without stoppage}} = \frac{u-v}{u}$$

where, $u > v$ and $u, v \neq 0$

Ex. Without any stoppage, speed of the bus is 54 km/hr and with stoppage, its speed is 45 km/hr. The bus stops for how many minutes per hour?

HINTS Stoppage time per hour = $\frac{54-45}{54} \times 60$ minutes
 $= \frac{9}{54} \times 60 = 10$ minutes

Ex. The speed of a bus without stoppages is 40 km/h and with stoppages is 32 km/h. How many minutes per hour does the bus stop?

HINTS Stoppage time
 $= \frac{40-32}{40} \times 60 \text{ min} = \frac{1}{5} \times 60 = 12$ minutes

Ex. Speed of a car without stoppage is 48 km/hr and that of with stoppage is 36 km/hr. If car stops each time for 5 min, then find the number of stoppage in 1 hr 20 min.

HINTS 1 hr 20 min = 80 minutes

$$\begin{aligned} \text{Stoppage time in 1 hr 20 min} &= \frac{48-36}{48} \times 80 \text{ min} \\ &= \frac{12}{48} \times 80 = 20 \text{ min} \end{aligned}$$

$$\therefore \text{Number of stops} = \frac{20}{5} = 4$$

Radha concept



On this concept, three types of questions are asked in the exams and they are based on the given formulae:

(a) $T = \sqrt{T_1 \times T_2}$

(b) $\frac{S_1}{S_2} = \sqrt{\frac{T_2}{T_1}}$

(c) $D = S_1 T_1 + S_2 T_2$

$S_1 \rightarrow$ Speed of object starting from A

$S_2 \rightarrow$ Speed of object starting from B

$T \rightarrow$ Time after which they meet each other.

$T_1 \rightarrow$ Time taken by the object 1 to reach at its destination after crossing object 2.

$T_2 \rightarrow$ Time taken by the object 2 to reach at its destination after crossing object 1.

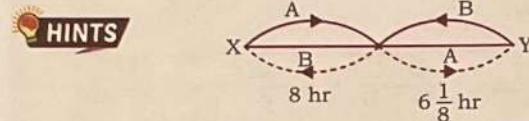
$D \rightarrow$ Total distance from A to B.

Ex. Shyam starts walking from A towards B at 10 a.m. and Radha starts walking from B towards A at 10 a.m. and after meeting at C, they both reach their destination in 24 minutes and 54 minutes respectively. Find out the time at which they meet at C.

HINTS $T = \sqrt{24 \times 54} = 36$ minutes

$\therefore$ Meeting time at C = 10 : 36 a.m.

Ex. A and B started their journey from X to Y and Y to X, respectively. After crossing each other, A and B completed remaining parts of their journey in $6\frac{1}{8}$ hours and 8 hours, respectively. If the speed of A is 32 km/hr, then the speed of B is (in km/hr):



Let, speed of B = x

$$\therefore \frac{x}{32} = \sqrt{\frac{49}{64}} \Rightarrow \frac{x}{32} = \frac{7}{8}$$

$$\Rightarrow x = 28 \text{ km/hr}$$

Ex. A and B start moving towards each other from places X and Y, respectively, at the same time on the same day. The speed of A is 20% more than that of B. After meeting on the way, A and B take p hours and q hours, respectively, to reach Y and X, respectively. What is the value of p ?

HINTS $A : B$
 Speed 6 : 5

Here,

$$S_A = 6, \quad S_B = 5$$

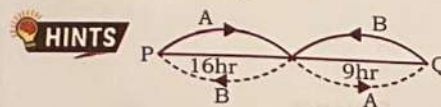
$$t_A = p, \quad t_B = \frac{36}{5}$$

$$\frac{S_A}{S_B} = \sqrt{\frac{t_B}{t_A}} \Rightarrow \frac{6}{5} = \sqrt{\frac{36}{5p}}$$

Square both side

$$\Rightarrow \frac{5p}{36} = \frac{25}{36} \Rightarrow p = 5 \text{ hours}$$

Ex. Two trains, A and B started travelling towards each other at the same time, from places P to Q and Q to P, respectively. After crossing each other, A and B took 9 hours and 16 hours respectively, to reach Q and P. If the speed of A was 56 km/hr, then what was the distance between P and Q?



$$\frac{S_A}{S_B} = \sqrt{\frac{t_B}{t_A}} \Rightarrow \frac{56}{S_B} = \sqrt{\frac{16}{9}}$$

$$\Rightarrow S_B = 42 \text{ km/hr}$$

$$\begin{aligned} \therefore \text{Distance between P and Q} &= S_A t_A + S_B t_B \\ &= 56 \times 9 + 42 \times 16 = 504 + 672 = 1176 \text{ km} \end{aligned}$$

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Relative Speed

Relative speed refers to the speed of one moving body in relation to other.

- Relative speed in same direction = $S_1 - S_2$
- Relative speed in opposite direction = $S_1 + S_2$

where, S_1 = Speed of first object

S_2 = Speed of second object

Ex. Place A and B are 144 km apart. Two cars start simultaneously, one from A and the other from B. If they move in the same direction, they meet after 12 hours, but if they move towards each other, they meet after $\frac{9}{8}$ hours. The speed (in km/hr) of the car moving at a faster speed is:

HINTS Let, speed of faster and slower car = x and y

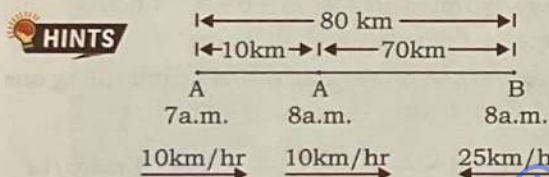
$$x - y = \frac{144}{12} = 12 \text{ km/hr} \quad \dots(i)$$

$$x + y = \frac{144 \times 8}{9} = 128 \text{ km/hr} \quad \dots(ii)$$

Solving equation (i) and (ii):

$x = 70 \text{ km/hr} \Rightarrow$ Speed of faster car = 70 km/hr

Ex. The distance between two cities A and B is 80 km. A motorcycle rider starts from A towards B at 7 a.m. at a speed of 10 km/hr. Another motorcyclist starts from B towards A at 8 a.m. at a speed of 25 km/hr. At what time will they cross each other?



Distance travelled by motorcycle A in 1 hr at the speed of 10 km/hr = $1 \times 10 = 10 \text{ km}$

Remaining distance = $80 - 10 = 70 \text{ km}$

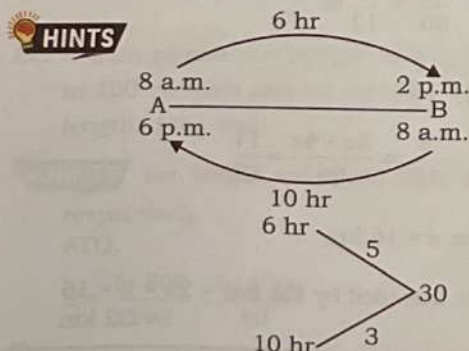
Relative speed = $25 + 10 = 35 \text{ km/hr}$

$$\text{Meeting time} = \frac{70}{35} = 2 \text{ hours}$$

$\therefore$ Time at which they will cross each other

= 8 a.m. + 2 hours = 10 a.m.

Ex. Ram starts from point A at 8 a.m. and reaches point B at 2 p.m. on the same day. On the same day, Raju starts from point B at 8 a.m. and reaches point A at 6 p.m. on the same day. Both points A and B are separated by only a straight line track. At what time they both meet?



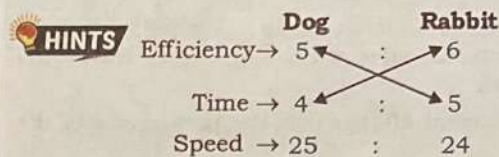
$$\text{Total time} = \frac{30}{5+3} \times 60 \text{ minute} = 3 \text{ hr } 45 \text{ minute}$$

Time at which they meet each other

= 8 am + 3 hr 45 minute

= 11 : 45 a.m.

Ex. A dog takes 5 leaps for every 6 leaps of a rabbit, but 4 leaps of the dog are equal to 5 leaps of the rabbit. What is the ratio of the dog's speed to the rabbit's speed?

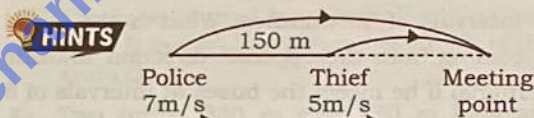


$\therefore$ Ratio of the dog's speed to the rabbit's speed = 25 : 24

Based on police and thief

Ex. A policeman sees a thief at a distance of 150 m. He starts chasing the thief who is running at a speed of 5 m/sec, while the policeman chases with a speed of 7 m/sec.

- What is the time taken by the police to chase the thief?
- Find the distance covered by the thief when he is caught.
- Find the distance covered by the policeman when he caught the thief.



- Relative speed = $7 - 5 = 2 \text{ m/sec}$

$$\therefore \text{Time} = \frac{150}{2} = 75 \text{ sec}$$

- Distance (D_t) = $75 \times 5 = 375 \text{ m}$

- Distance (D_p) = $75 \times 7 = 525 \text{ m}$

Ex. A thief was noticed by a policeman from a distance of 80 metres. The speed of the policeman is 21 km/hr and the thief is 20 km/hr. What is the distance between them after 18 seconds?

HINTS Relative speed = $21 - 20 = 1 \text{ km/hr}$

$\therefore$ Distance between thief and police after 18 sec

$$= 80 - 1 \times \frac{5}{18} \times 18 = 80 - 5 = 75 \text{ metres}$$

Ex. A thief running at a speed of ' x ' km/hr is chased by a policeman running at a speed of 10 km/hr. If the thief is ahead by 100 metres, the policeman catches the thief after 3 minutes. At what speed is the thief running (' x ' being the unknown speed)?

HINTS ATQ,

$$3 \times 60 = \frac{100}{(10 - x) \times \frac{5}{18}} \Rightarrow 9 = \frac{18}{10 - x} \Rightarrow x = 8 \text{ km/hr}$$

$\therefore$ Speed of thief = 8 km/hr

Calculate speed, based on regular interval

- Buses are leaving a bus terminal after every 10 minutes (T_1). A person who is moving towards the terminal, meets the buses after every 8 minutes (T_2), then:

$$\frac{\text{Speed of man}}{\text{Speed of bus}} = \frac{|T_1 - T_2|}{T_2}$$

where,

T_1 = Time interval after which the buses leaves the terminal.

T_2 = Time interval after which the person meets the buses.

- Two guns are fired from the same place at an interval of 13 minutes (T_1). A person in a train approaching the place hears the second firing sound 12 minutes 30 seconds (T_2) after the first, then:

$$\frac{\text{Speed of Train}}{\text{Speed of Sound}} = \frac{|T_1 - T_2|}{T_2}$$

where,

T_1 = Time interval after which the second gun is fired after the first.

T_2 = Time interval between hearing the first sound and the second sound by the person.

Ex. Buses start from a bus terminal with a speed of 20 km/hr at intervals of 10 minutes. What is the speed of a man coming from the opposite direction towards the bus terminal if he meets the buses at intervals of 8 minutes?

HINTS Let, speed of man = x

$$\Rightarrow \frac{x}{20} = \frac{10 - 8}{8} \Rightarrow \frac{x}{20} = \frac{2}{8} \Rightarrow x = 5 \text{ km/hr}$$

$\therefore$ Speed of man = 5 km/hr

Ex. Two guns are fired from the same place at an interval of 13 minutes. A man in a train approaching the place hears the second firing sound 12 minutes 30 seconds after the first. Find the speed of train (Suppose that the sound travels at a speed of 300 m/sec).

HINTS Let, speed of train = x

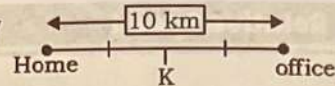
$$\Rightarrow \frac{x}{300} = \frac{13 - 12.5}{12.5} \Rightarrow \frac{x}{300} = \frac{0.5}{12.5} \Rightarrow x = 12 \text{ m/sec}$$

$\therefore$ Speed of train = 12 m/sec

Miscellaneous

Ex. R jogs at twice the speed of walking and runs at twice the speed of jogging. From his home to office, he covers half of the distance by walking and the rest by jogging. From his office to home, he covers half the distance by jogging and the rest by running. What is his average speed (in km/hr) in a complete round from his home to office and back home if the distance between his office and home is 10 km and he walks at the speed of 5 km/hr?

HINTS



Walking : Jogging : Running

1 : 2 : 4

Speed $\rightarrow$ 5 km/hr 10 km/hr 20 km/hr

$$T_1 = \frac{5}{5} = 1 \text{ hr}, T_2 = \frac{5}{10} = \frac{1}{2} \text{ hr}$$

$$T_3 = \frac{5}{10} = \frac{1}{2} \text{ hr}, T_4 = \frac{5}{20} = \frac{1}{4} \text{ hr}$$

$$\therefore \text{Average speed} = \frac{\text{Total distance}}{\text{Total time}}$$

$$= \frac{20}{1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{4}} = \frac{20}{2 + \frac{1}{4}} = \frac{80}{9} \text{ km/hr}$$

Ex. A man takes 6 hours to go to a place and come back by walking both the ways. He could have saved 2 hours by riding both the ways. The distance covered in the whole journey is 20 miles. What is the average speed for the whole journey if he goes by walking and comes back by riding?

HINTS Total distance = 20 miles

Time taken by walking both sides = 6 hours

Time taken by walking one side = 3 hours

Time taken by riding both ways = 6 - 2 = 4 hours

Time taken by riding one way = 2 hours

$\therefore$ Total time taken on walking one side and riding one side = 3 + 2 = 5 hours

$$\therefore \text{Average Speed} = \frac{\text{Total distance}}{\text{Total time}} = \frac{20}{5} = 4 \text{ miles/hr}$$

Ex. A delivery boy started from his office at 10 a.m. to deliver an article. He rides his scooter at a speed of 32 km/hr. He delivered the article and waited for 15 minutes to get the payment. After the payment was made, he reached his office at 11:25 a.m., travelling at a speed of 24 km/hr. Find the total distance travelled by the boy.

HINTS Let, distance between the office and delivering destination = x km

Start time = 10:00 a.m.

Final time = 11:25 a.m.

$$\text{Total time} = 1 + \frac{25}{60} = \frac{17}{12} \text{ hr}$$

$$\Rightarrow \frac{x}{32} + \frac{15}{60} + \frac{x}{24} = \frac{17}{12}$$

$$\Rightarrow \frac{x}{32} + \frac{x}{24} = \frac{17}{12} - \frac{3}{12} \Rightarrow \frac{3x + 4x}{96} = \frac{14}{12}$$

$$\Rightarrow x = \frac{14 \times 96}{12 \times 7} \Rightarrow x = 16 \text{ km}$$

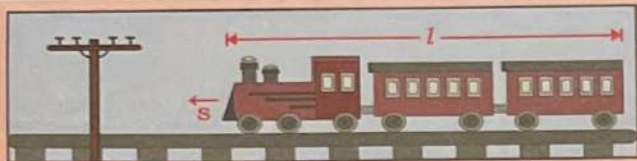
$\therefore$ Total distance travelled by the boy = $2x = 2 \times 16 = 32 \text{ km}$

CHAPTER

27

TRAIN

Train Passing a Telegraph Pole



Time taken by a train of length l m, moving with a speed of s m/s, to pass a pole or a stationary man:

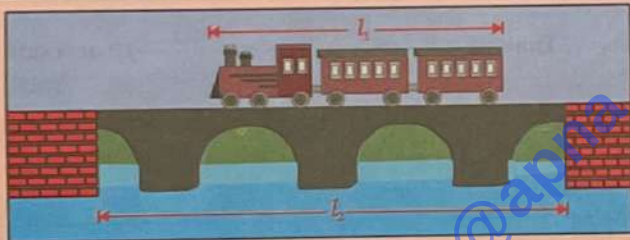
$$\text{Time} = \frac{\text{Length of train } (l)}{\text{Speed of train } (s)}$$

Ex Distance covered by a train while passing a pole or a stationary man is equal to the length of the train.

Ex. What is the time taken by a 180 m long train running at a speed of 15 m/s to cross a man standing on the platform?

HINTS $\text{Time} = \frac{\text{Length of train}}{\text{Speed of train}} = \frac{180}{15} = 12 \text{ seconds}$

Train passing a bridge or a platform



Time taken by a train of length l_1 m, moving with a speed of s m/s, to pass a bridge/platform of length l_2 m:

$$\text{Time} = \frac{\text{Length of train } (l_1) + \text{Length of bridge } (l_2)}{\text{Speed of train } (s)}$$

Ex Distance covered by a train while passing a bridge is equal to the sum of length of the train and the bridge.

Ex. A 900 m long train is running at a speed of 30 m/s. How much time will it take to cross a 900 m long platform?

HINTS $\text{Time} = \frac{\text{Length of train } (l_1) + \text{Length of platform } (l_2)}{\text{Speed of train } (s)}$

$$= \frac{900 + 900}{30} = \frac{1800}{30} = 60 \text{ seconds}$$

Ex. A train passes two bridges of length 800 m and 400 m in 100 seconds and 60 seconds, respectively. Find the length of the train.

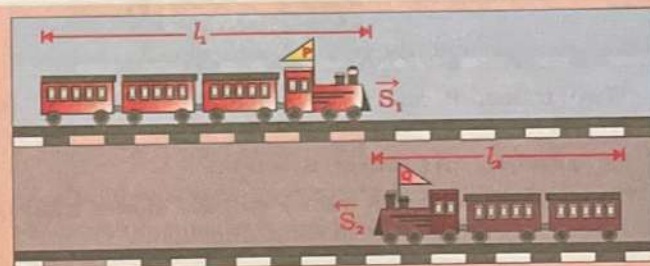
HINTS Let, length and speed of train be l m and s m/s, respectively.

ATQ,

$$s = \frac{l + 800}{100} = \frac{l + 400}{60}$$

$$\Rightarrow 3l + 2400 = 5l + 2000 \Rightarrow 2l = 400 \Rightarrow l = 200 \text{ m}$$

Two Trains Crossing Each other in Opposite Direction



Two trains, P and Q of length l_1 m and l_2 m respectively, are moving with respective speed of s_1 m/s and s_2 m/s.

Time taken by train P and train Q to pass each other when they are moving in opposite direction:

$$\text{Time} = \frac{l_1 + l_2}{s_1 + s_2}$$

Ex Distance covered by a train P while crossing another train Q is equal to the sum of length of the train P and train Q.

Ex. Two trains 230 m and 270 m long are running in opposite direction at the speed of 42 km/h and 48 km/h, respectively. How much time will they take to cross each other?

HINTS $l_1 = 230 \text{ m}, l_2 = 270 \text{ m}$

$$s_1 = 42 \text{ km/h}, s_2 = 48 \text{ km/h}$$

$$\therefore \text{Time} = \frac{l_1 + l_2}{s_1 + s_2}$$

$$= \frac{230 + 270}{(42 + 48) \times \frac{5}{18}} = \frac{500}{90 \times \frac{5}{18}} = 20 \text{ seconds}$$

Ex. Two trains, P and Q, are running on parallel tracks in opposite directions with lengths of 110 m and 140 m, respectively. Train P is running at a speed of 70 km/h and took 5 seconds to cross train Q. The speed of train Q is:

HINTS $\therefore s_1 + s_2 = \frac{l_1 + l_2}{\text{Time}}$

$$= \frac{110 + 140}{5} = \frac{250}{5} = 50 \text{ m/s} = 50 \times \frac{18}{5} = 180 \text{ km/h}$$

$$\text{Speed of train P} = 70 \text{ km/h}$$

$$\text{Speed of train Q} = 180 - 70$$

$$= 110 \text{ km/h}$$